RF & Microwave Engineering

An Interactive Textbook

with MRI Physics & Coil Engineering

10 Parts ▪ 64 Chapters ▪ 119 figures

with companion interactive calculators

RF Toolbox

rftoolbox.ca

Compiled from the RF Toolbox reference library.

Preface

This book collects the RF Toolbox reference library — sixty-four illustrated articles on radio-frequency engineering, microwave design, and magnetic-resonance imaging — into a single, sequenced volume. Each topic began as a standalone web article at rftoolbox.ca; here they are re-ordered into a foundations-to-advanced progression, grouped into ten Parts, and typeset with their original diagrams.

The material spans classical electromagnetism, transmission-line theory and impedance matching, guided-wave structures, passive components and filters, noise and amplifier design, signal generation and frequency conversion, antennas, complete RF systems, electromagnetic compatibility, and a full treatment of MRI physics and RF coil engineering. Every chapter is paired online with an interactive calculator that lets you put its equations to work with your own numbers.

How to Use This Book

The ten Parts are arranged so that each builds on the last, but chapters are also written to stand alone for reference. If you are new to RF engineering, read Parts I and II in order — waves, decibels, transmission lines, VSWR, the Smith chart, and matching are the vocabulary everything else uses. Practitioners can jump directly to the chapter they need.

Displayed equations are boxed set-off results; symbols follow the conventions in the Notation section below. Wherever this book shows a formula, the online edition offers a live calculator for it: for example, the VSWR / return-loss converter, the Smith chart, the cascaded noise-figure and IP3 tools, the antenna calculators, and the MRI coil and SAR estimators. Visit rftoolbox.ca to use them.

Notation

Z_0	Characteristic impedance (ohms), typically 50 or 75 Ω.
Γ	Voltage reflection coefficient, $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$.
VSWR	Voltage standing-wave ratio, $(1 + \|\Gamma\|)/(1 - \|\Gamma\|)$.
λ	Wavelength; $\beta = 2\pi/\lambda$ is the phase constant.
$f,\ \omega$	Frequency (Hz) and angular frequency $\omega = 2\pi f$.
S_{ij}	Scattering parameters (S-parameters).
NF, F	Noise figure (dB) and noise factor (linear).
$B_0,\ B_1$	MRI static field and RF transmit field.
γ	Gyromagnetic ratio (MRI); also used for propagation constant in line theory.

Decibel quantities: dBm is power referenced to 1 milliwatt; dBc is relative to a carrier; dBi is antenna gain relative to an isotropic radiator.

Contents

Part I

Electromagnetic Foundations

Chapter 1

Maxwell's Equations

Maxwell's equations are the four partial differential equations that completely describe classical electromagnetism. They were formulated by James Clerk Maxwell in 1865 and unified electricity, magnetism, and optics into a single theoretical framework.

1.1 The Four Equations

$\nabla \cdot \mathbf{E} = \dfrac{\rho}{\varepsilon_0}$ — *Gauss's Law: electric field diverges from charges.*
$\nabla \cdot \mathbf{B} = 0$ — *Gauss's Law for Magnetism: no magnetic monopoles.*
$\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}$ — *Faraday's Law: a changing magnetic field induces an electric field.*
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \dfrac{\partial \mathbf{E}}{\partial t}$ — *Ampère–Maxwell Law: currents and changing electric fields create magnetic fields.*

1.2 Displacement Current

Maxwell's key insight was adding the displacement current term $\varepsilon_0 \partial \mathbf{E}/\partial t$ to Ampère's law. This predicts that a time-varying electric field produces a magnetic field even in the absence of physical current — allowing electromagnetic waves to propagate through a vacuum.

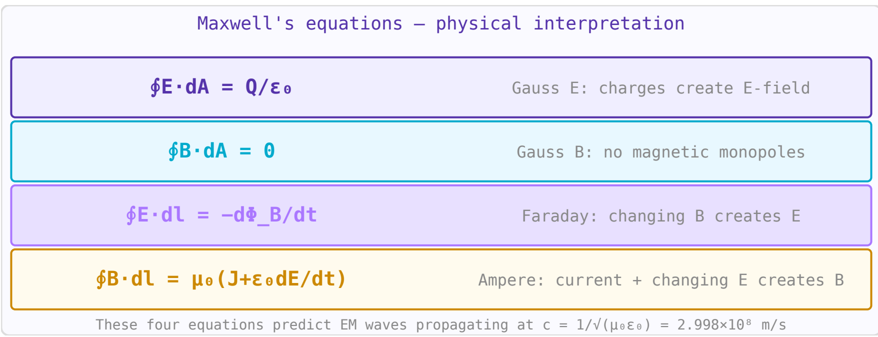

Figure 1.1: Maxwell's equations in integral form. Together they are the complete foundation of classical electromagnetism.

1.3 Electromagnetic Wave Speed

Taking the curl of Faraday's law and substituting Ampère–Maxwell gives a wave equation. The wave speed is:

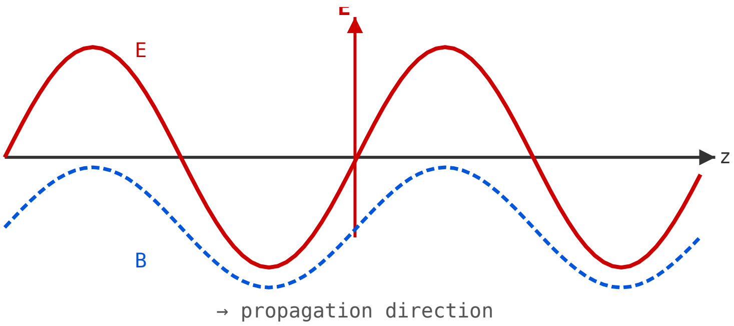

Figure 1.2: Transverse EM wave: E (red) and B (blue dashed) oscillate perpendicular to each other and to the propagation direction z.

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \times 10^8 \text{ m/s}$$

Maxwell recognised this as the speed of light, demonstrating that light is an electromagnetic wave.

1.4 In Materials

In a dielectric medium with relative permittivity ε_r and relative permeability μ_r, the wave speed becomes $v = c/\sqrt{\varepsilon_r \mu_r}$. This slowing is why wavelengths in PCB substrates are shorter than in free space.

Chapter 2

Electromagnetic Waves

An electromagnetic wave is a self-propagating oscillation of electric and magnetic fields, perpendicular to each other and to the direction of travel. No medium is required — EM waves propagate in vacuum at the speed of light.

2.1 Wave Properties

A plane wave travelling in the z-direction has:

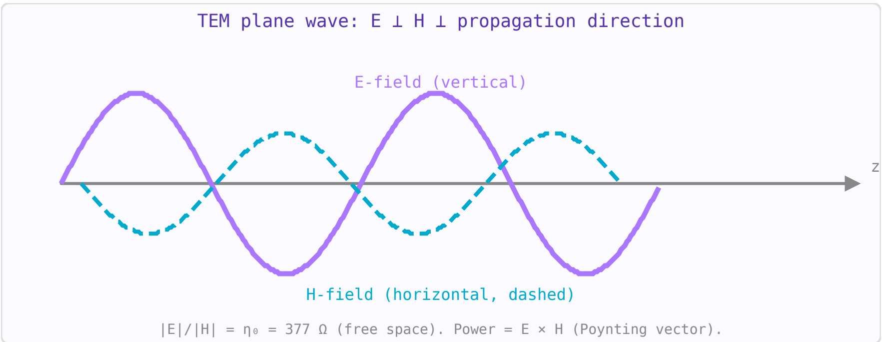

Figure 2.1: TEM plane wave: E and H oscillate perpendicular to each other and to the propagation direction z.

$$\mathbf{E}(z,t) = E_0 \cos(\omega t - kz)\,\hat{x} \quad \mathbf{B}(z,t) = \frac{E_0}{c}\cos(\omega t - kz)\,\hat{y}$$

The wavenumber $k = 2\pi/\lambda = \omega/c$, angular frequency $\omega = 2\pi f$, wavelength $\lambda = c/f$.

2.2 The EM Spectrum

- ELF / VLF < 30 kHz — submarine communications, power lines
- MF / HF 300 kHz–30 MHz — AM radio, shortwave
- VHF / UHF 30 MHz–3 GHz — FM radio, TV, WiFi, cellular
- Microwave 3–300 GHz — radar, satellite, 5G, microwave ovens, MRI
- Infrared / Visible / UV above 300 GHz
- X-ray / Gamma very high frequencies — ionising radiation

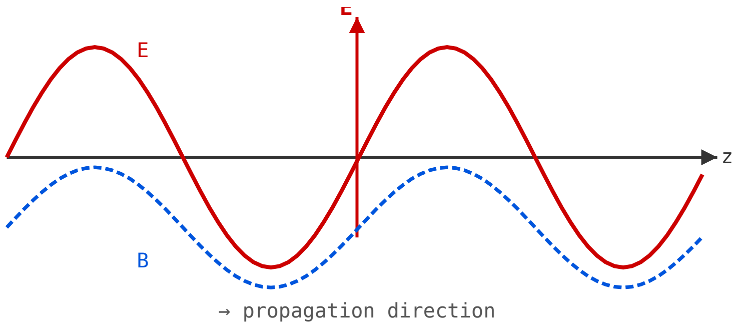

Figure 2.2: Linearly polarised plane wave propagating in the z-direction.

2.3 Polarisation

The polarisation of an EM wave describes the orientation of the electric field vector. Linear polarisation: E oscillates in one plane. Circular polarisation: E rotates, tracing a helix. In MRI, circular (quadrature) polarisation of the B1 field is used to improve SNR by 40%.

2.4 Poynting Vector

Power flow per unit area is given by the Poynting vector: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m^2). Time-averaged power: $\langle S \rangle = E_0^2/(2\eta)$ where $\eta = \sqrt{\mu/\varepsilon}$ is the wave impedance (377 Ω in free space).

Chapter 3

Dielectrics and Permittivity

A dielectric is an insulating material that, when placed in an electric field, stores energy by polarising its molecules. The key RF properties of a dielectric are its relative permittivity (dielectric constant) and loss tangent.

3.1 Relative Permittivity (ε_r)

The dielectric constant ε_r relates the permittivity of the material to that of free space: $\varepsilon = \varepsilon_r \varepsilon_0$. It determines how much wavelengths are shortened in the material: $\lambda = \lambda_0/\sqrt{\varepsilon_r}$.

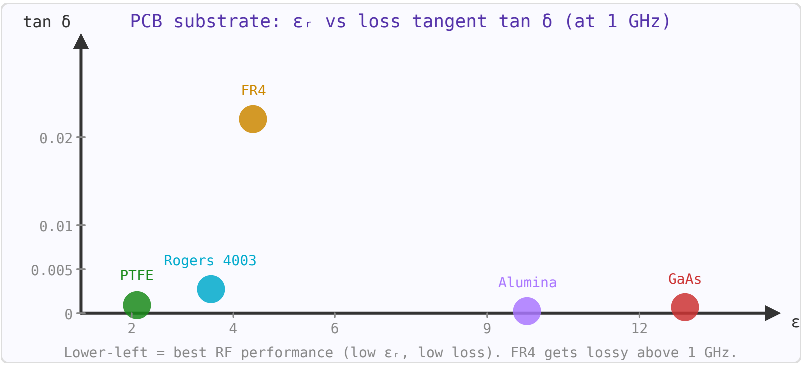

Figure 3.1: PCB substrate map. FR4 is cheapest but lossy; Rogers and PTFE offer much lower $\tan\delta$ for microwave applications.

3.2 Loss Tangent ($\tan\delta$)

A real dielectric also dissipates energy. The loss tangent represents the ratio of the imaginary to real part of the complex permittivity:

$$\tan\delta = \frac{\varepsilon_r''}{\varepsilon_r'}$$

Lower $\tan\delta$ means lower dielectric loss. Attenuation in a transmission line increases with loss tangent and frequency.

3.3 Common PCB Substrate Materials

- FR4 — $\varepsilon_r \approx 4.4$, $\tan\delta \approx 0.02$. Standard PCB material, adequate to ~5 GHz.

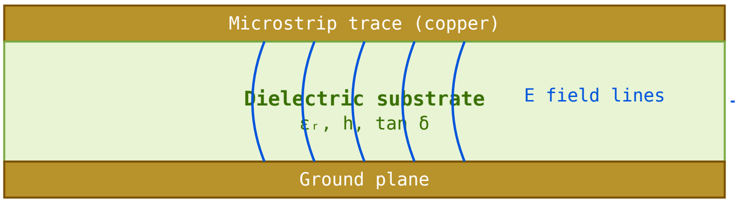

Figure 3.2: Microstrip cross-section. Field lines extend partly into air above the trace, giving εeff < εr.

- Rogers 4003C — $\varepsilon_r = 3.55$, $\tan\delta = 0.0027$. Low loss, stable with temperature, suitable to 77 GHz.
- PTFE (Teflon) — $\varepsilon_r \approx 2.1$, \(\tan\delta < 0.001\). Excellent high-frequency performance, used in coaxial cable dielectrics.
- Alumina (Al_2O_3) — $\varepsilon_r \approx 9.8$, $\tan\delta \approx 0.001$. Used in MMIC substrates.

3.4 Frequency Dependence

Both ε_r and $\tan\delta$ vary with frequency. For PCB design above 1 GHz, always use substrate data measured at the operating frequency rather than the low-frequency value printed on datasheets.

Chapter 4

Magnetic Materials

Magnetic materials are used in RF and power electronics to make inductors, transformers, ferrite beads, and circulators. Understanding their properties is essential for choosing the right core material.

4.1 Relative Permeability (μ_r)

Permeability μ_r relates the magnetic flux density to the applied field: $B = \mu_r \mu_0 H$. High μ_r increases inductance but also limits useful frequency range (permeability drops at high frequencies).

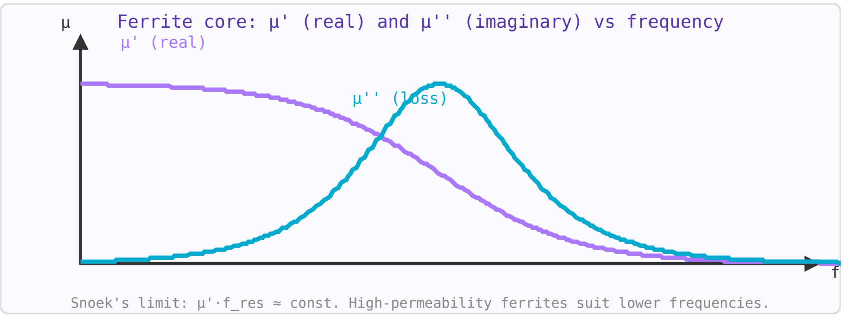

Figure 4.1: Ferrite permeability: μ' falls above resonance; μ'' (loss) peaks at resonance. Use below resonance for low-loss performance.

4.2 Ferrites

Ferrites are ceramic magnetic materials with high resistivity (minimising eddy current losses at RF). They are used in:

- **Ferrite beads** — suppress high-frequency noise on power lines and signal traces by presenting high impedance above a target frequency.
- **Ferrite cores** — increase inductance of RF chokes and RF transformers, particularly below 50 MHz.
- **Circulators and isolators** — exploit the non-reciprocal properties of magnetised ferrite (Faraday rotation) to route signals directionally.

4.3 Core Saturation

At high flux densities, magnetic cores saturate — μ_r collapses, inducing harmonics and power loss. Always verify that peak flux density is below the saturation flux density B_{sat} of the chosen material under worst-case conditions.

4.4 Core Loss

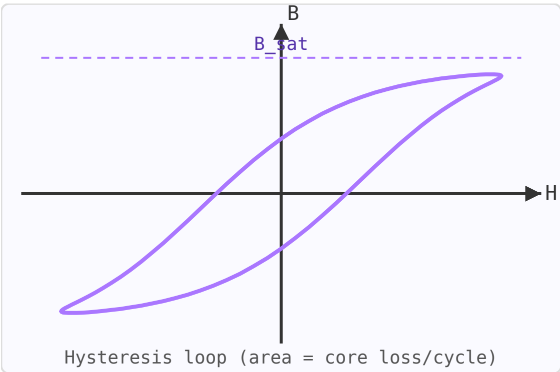

Figure 4.2: Hysteresis loop. Area enclosed = energy dissipated as heat per cycle.

Core loss (hysteresis + eddy current losses) is characterised by the complex permeability $\mu = \mu' - j\mu''$. The loss factor μ''/μ' (analogous to $\tan\delta$) increases with frequency. Ferrite materials are categorised by their intended frequency range (e.g. MnZn ferrites for <5 MHz, NiZn for 1–100 MHz).

Chapter 5

Skin Effect

The skin effect is the tendency of alternating electric current to flow primarily near the surface of a conductor. At higher frequencies, the current density decreases exponentially with depth from the surface, concentrating in a thin layer called the skin depth.

5.1 Skin Depth Formula

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{1}{\pi f \mu\sigma}}$$

where $\omega = 2\pi f$ is angular frequency, μ is permeability, and σ is conductivity. Current density falls to $1/e \approx 37\%$ of its surface value at depth δ.

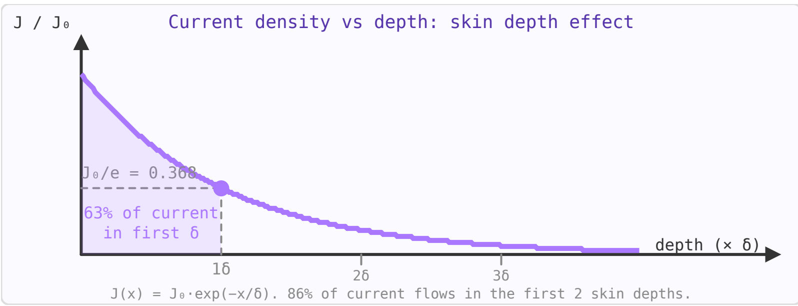

Figure 5.1: Current density decays exponentially with depth. At depth $= \delta$, J has fallen to 1/e $\approx$ 37% of its surface value.

5.2 Values for Common Conductors at 100 MHz

- Copper ($\sigma = 5.8 \times 10^7$ S/m): $\delta \approx 6.6$ μm
- Aluminium ($\sigma = 3.8 \times 10^7$ S/m): $\delta \approx 8.2$ μm
- Silver ($\sigma = 6.3 \times 10^7$ S/m): $\delta \approx 6.3$ μm

5.3 Impact on RF Resistance

Because current only flows in a thin annulus near the surface, the effective cross-sectional area of the conductor is reduced, increasing resistance compared to DC. The AC resistance of a round wire of radius $r \gg \delta$ is approximately:

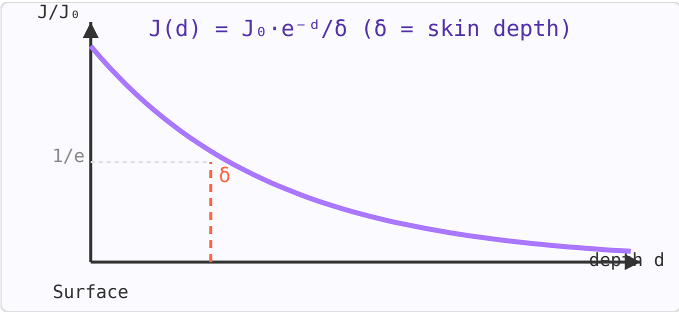

Figure 5.2: Current density J vs depth d. At one skin depth δ, J falls to 1/e $\approx$ 37% of its surface value.

$$R_{AC} \approx R_{DC} \cdot \frac{r}{2\delta}$$

For a 1 mm diameter copper wire at 100 MHz, $R_{AC}/R_{DC} \approx 38$. This is why multi-strand (Litz) wire is used in high-Q inductors at lower RF frequencies — many thin strands, each smaller than the skin depth, provide more effective conductor area.

5.4 Design Implications

- Conductor thickness in PCBs should be at least 3–5 skin depths to minimise resistive loss.
- Silver plating adds only marginal benefit over copper above ~100 MHz (skin depth ~6 μm) as long as the copper is thicker than a few skin depths.
- In MRI coils at 128 MHz (3 T), copper tubing (thick wall) is preferred over solid wire for high-Q resonators.

Part II

Transmission Lines and Impedance

Chapter 6

Decibels, dBm and RF Power Units

The decibel (dB) is the most important unit in RF engineering. It turns multiplication into addition, making cascade analysis practical. Once you're fluent in dB arithmetic, working with gain, loss, and power levels becomes fast and intuitive.

6.1 Why Logarithmic Scales?

A receiver handles signals ranging from 10^{-15} W (noise floor) to 10^{-3} W (nearby transmitter) — a range of 10^{12}. Writing and plotting these on a linear scale is impractical. Logarithms compress that 10^{12} ratio to a 120 dB range that fits on a graph. More importantly, a cascade of gain and loss stages multiplies powers together — in dB you just add.

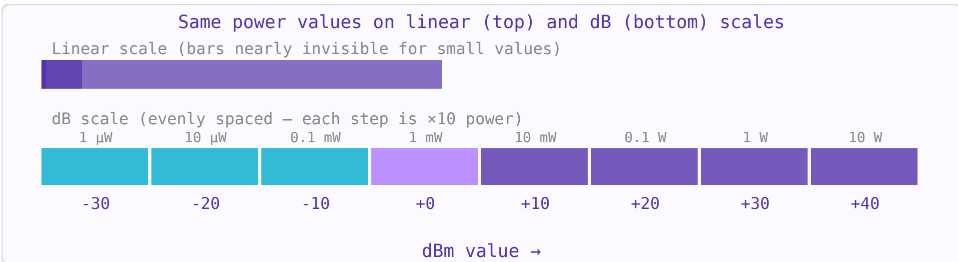

Figure 6.1: The dB scale compresses a million-fold power range into readable 10 dB steps.

6.2 The Decibel (dB)

The decibel expresses a *ratio* between two power values:

$$\text{ratio (dB)} = 10\log_{10}\left(\frac{P_2}{P_1}\right)$$

For voltage or current ratios (into the same impedance), the factor is 20 instead of 10 because power is proportional to voltage squared:

$$\text{ratio (dB)} = 20\log_{10}\left(\frac{V_2}{V_1}\right)$$

Key values to memorise:

Power ratio	Voltage ratio	dB	Memory aid

10×	3.16×	+10 dB	One order of magnitude
2×	1.41×	+3 dB	Double the power
1.26×	1.12×	+1 dB	Smallest audible difference
1×	1×	0 dB	No change
0.5×	0.71×	−3 dB	Half the power
0.1×	0.316×	−10 dB	One tenth
0.001×	0.0316×	−30 dB	One thousandth

6.3 Absolute Power: dBm and dBW

dB alone is a ratio — it needs a reference to become an absolute level. **dBm** uses 1 milliwatt as the reference:

$$P_{\text{dBm}} = 10\log_{10}\left(\frac{P_{\text{watts}}}{1\ \text{mW}}\right)$$

dBW uses 1 watt. The conversion: P(dBm) = P(dBW) + 30.

Power	dBm	dBW	Context
1 kW	+60 dBm	+30 dBW	AM broadcast transmitter
10 W	+40 dBm	+10 dBW	Handheld radio TX
1 W	+30 dBm	0 dBW	Bluetooth TX max
100 mW	+20 dBm	−10 dBW	Wi-Fi typical
1 mW	0 dBm	−30 dBW	Reference level
1 μW	−30 dBm	−60 dBW	Strong receiver input
1 nW	−60 dBm	−90 dBW	Typical receiver input
−100 dBm	−100 dBm	−130 dBW	Near noise floor
10^{-15} W = 1 fW	−120 dBm	−150 dBW	Thermal noise floor, 1 MHz BW, room temperature

6.4 dB Arithmetic in Cascades

This is where dB pays off. A system with a 20 dB LNA, a 7 dB cable loss, a 10 dB mixer loss, and 30 dB of IF gain, driven by a −90 dBm signal:

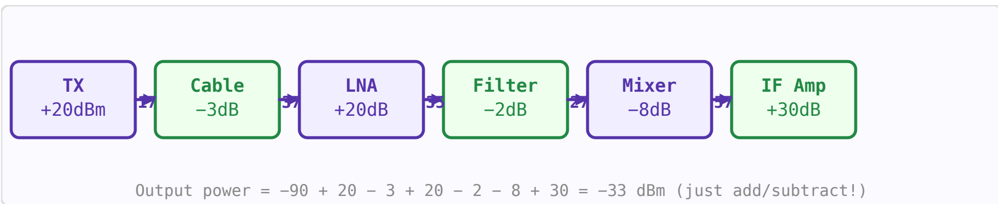

Figure 6.2: In dB arithmetic, cascade gain/loss is simple addition. No need to multiply or divide linear values.

$$P_{out} = -90 + 20 - 7 - 10 + 30 = -57\ \text{dBm}$$

Compare doing this in linear: $10^{-12} \times 100 \times 0.2 \times 0.1 \times 1000 = 2{\times}10^{-12}$ W. Much harder to get right mentally.

6.5 Other dB Variants

Unit	**Reference**	**Use**
dBm	1 mW	Absolute RF power, transmitters and receivers
dBW	1 W	Satellite EIRP specifications
dBc	Carrier power	Relative: spurious, harmonics, phase noise
dBi	Isotropic antenna	Antenna gain (absolute)
dBd	Half-wave dipole	Antenna gain (relative to dipole). 0 dBd = 2.15 dBi
dBμV	1 μV	EMC and receiver sensitivity specifications in Europe
dBμV/m	1 μV/m	Field strength in EMC measurements
dBFS	Full scale	Digital audio and ADC levels (always $\leq$ 0 dBFS)

6.6 Noise Floor and kTB

Thermal noise power available from a resistor at temperature T in bandwidth B:

$$P_{noise} = kTB \approx -174 \text{ dBm/Hz at 290 K}$$

In a bandwidth B (Hz): $P_{noise} = -174 + 10\log_{10}(B)$ dBm. In 1 MHz bandwidth: $-174 + 60 = -114$ dBm. In 20 MHz (Wi-Fi): $-174 + 73 = -101$ dBm. Add the receiver noise figure to get the actual noise floor.

6.7 Converting Back to Linear

$$P_{\text{watts}} = 10^{P_{dBm}/10} \times 10^{-3} \text{ W}$$

Shortcut: 0 dBm = 1 mW, every +10 dB multiplies by 10, every +3 dB doubles.

Chapter 7

Complex Impedance and Reactance

At DC, resistance is the only opposition to current flow. At RF, inductors and capacitors also oppose current — but in a frequency-dependent and phase-shifting way. The complete description requires complex numbers: impedance $Z = R + jX$, where the imaginary part X is the reactance.

7.1 Phasors and Sinusoidal Steady State

In a circuit driven by a sinusoidal source at frequency f, all voltages and currents are sinusoids at the same frequency. We represent them as phasors — complex numbers whose magnitude gives amplitude and whose angle gives phase. Multiplying a phasor by j rotates it 90° (a quarter cycle), which is exactly what an inductor does: it makes current lag voltage by 90°.

7.2 Reactance

Reactance X is the imaginary part of impedance. It stores and releases energy without dissipating it:

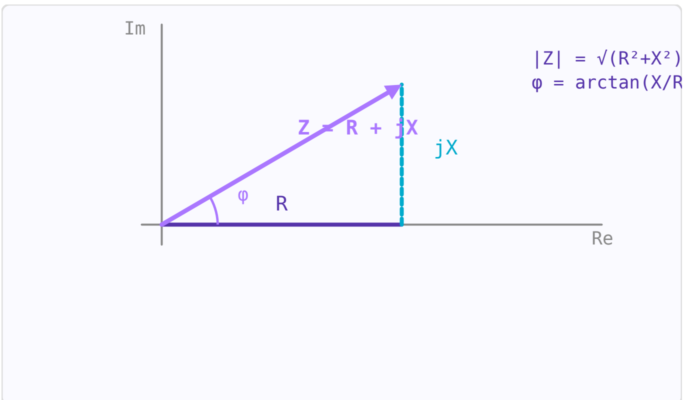

Figure 7.1: Impedance $Z = R + jX$ as a phasor in the complex plane. R is resistance, X is reactance.

Component	Impedance Z	Reactance X	Phase
Resistor R	R	0	V and I in phase
Inductor L	$j\omega L$	$+\omega L = +2\pi f L$	I lags V by 90°
Capacitor C	$1/(j\omega C)$	$-1/(\omega C) = -1/(2\pi f C)$	I leads V by 90°

At resonance (ωL = 1/ωC), the inductive and capacitive reactances cancel and the impedance is purely resistive.

7.3 Admittance Y = 1/Z

For parallel circuit analysis, admittance Y = G + jB is more convenient than impedance. G is the conductance (1/R) and B is the susceptance (−1/X for an inductor, +1/X for a capacitor — note the sign flip). On a Smith Chart, the admittance chart is the impedance chart rotated 180°.

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2} = G + jB$$

7.4 Series vs Parallel Equivalents

A real inductor has both inductance L and series resistance R_s. It can be represented equivalently as a parallel combination of L_p and R_p, where the two representations are equivalent at one frequency. This conversion matters when designing matching networks:

$$Q = \frac{X_s}{R_s} = \frac{R_p}{X_p} \qquad R_p = R_s(1 + Q^2) \qquad X_p = X_s\frac{Q^2 + 1}{Q^2}$$

7.5 Quality Factor Q

The Q factor of a reactive element describes how "ideal" it is — the ratio of energy stored to energy lost per cycle:

$$Q = \frac{|X|}{R} = \frac{\omega_0 L}{R_s} = \frac{1}{\omega_0 C R_s}$$

High Q means low loss. A typical SMD inductor at 100 MHz has Q = 30–80. A quartz crystal resonator has Q = 10,000–100,000. Q also sets bandwidth: $BW = f_0/Q$.

7.6 Frequency Dependence

At higher frequencies, parasitics dominate. A real capacitor has series inductance (ESL) that causes it to become inductive above its self-resonant frequency (SRF). A real inductor has inter-winding capacitance that causes it to self-resonate. Always check the SRF of a component against your operating frequency:

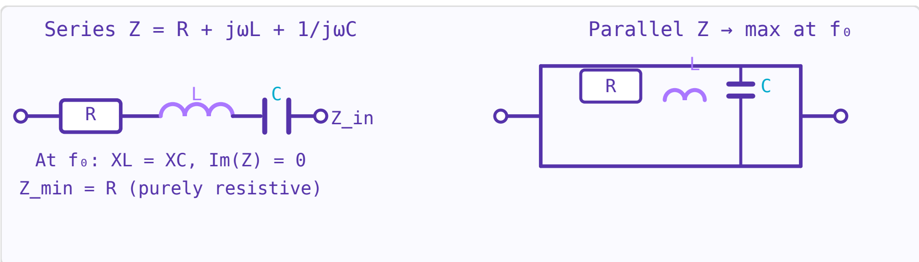

Figure 7.2: Series RLC: minimum impedance at resonance. Parallel RLC: maximum impedance at resonance.

Component value	Typical SRF	Notes
10 μF electrolytic	500 kHz	Useless for RF bypassing
100 nF ceramic (0402)	50 MHz	Good for audio, limited at RF
10 nF ceramic (0402)	200 MHz	OK for HF/VHF bypassing
100 pF ceramic (0402)	2 GHz	Good RF bypass to ~1 GHz
10 pF ceramic (0402)	8 GHz	Good for microwave matching
1 pF ceramic (0402)	20+ GHz	Usable at mmWave

7.7 Impedance vs Frequency: What to Expect

A 50 Ω transmission line port expects 50 + j0 Ω. Impedance that deviates from this causes reflections. The Smith Chart provides an intuitive visual representation of complex impedance across frequency — a single-frequency impedance is a point, and an impedance varying with frequency traces a curve. Understanding complex impedance is the prerequisite for reading and using the Smith Chart.

Chapter 8

Transmission Line Theory

A transmission line is a structure that guides electromagnetic waves from one point to another. At RF frequencies, ordinary wire connections behave as transmission lines — understanding this is essential for signal integrity and impedance matching.

8.1 Telegrapher's Equations

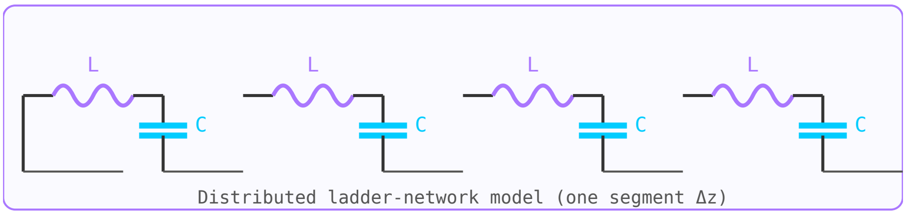

Figure 8.1: Distributed-element model: series L, shunt C per unit length Δz.

A transmission line is modelled as a ladder network of distributed resistance R, inductance L, capacitance C, and conductance G per unit length. The resulting wave equations are:

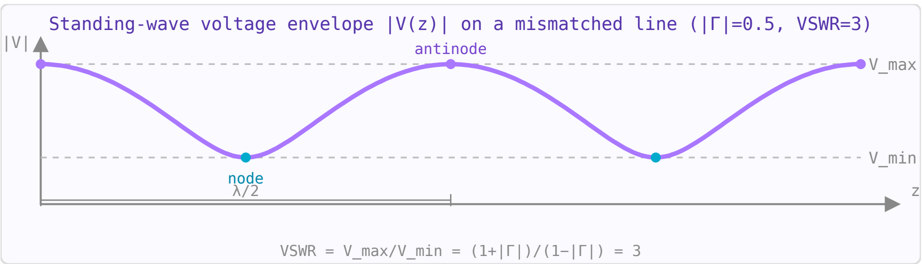

Figure 8.2: Standing wave on a mismatched line: V_max/V_min = VSWR = 3 for $|\Gamma|$=0.5.

$$\frac{\partial V}{\partial z} = -(R + j\omega L)I \qquad \frac{\partial I}{\partial z} = -(G + j\omega C)V$$

8.2 Characteristic Impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \xrightarrow{\text{lossless}} \sqrt{\frac{L}{C}}$$

Coax with 50 Ω characteristic impedance: $Z_0 = (60/\sqrt{\varepsilon_r})\ln(D/d)$.

8.3 Reflection Coefficient

When a wave hits a load impedance $Z_L \neq Z_0$, some power is reflected. The voltage reflection coefficient is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Return loss $= -20 \log_{10} |\Gamma|$ dB. VSWR $= (1 + |\Gamma|)/(1 - |\Gamma|)$.

8.4 Standing Waves

The superposition of forward and reflected waves creates a standing wave pattern on the line. The voltage varies with position; the ratio of maximum to minimum voltage is the VSWR. A matched load ($\Gamma = 0$) has VSWR = 1 and no standing waves.

8.5 Input Impedance

A transmission line of length ℓ terminated in Z_L presents input impedance:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta\ell)}{Z_0 + jZ_L \tan(\beta\ell)}$$

Special cases: $\ell = \lambda/4 \to Z_{in} = Z_0^2/Z_L$ (quarter-wave transformer); $\ell = \lambda/2 \to Z_{in} = Z_L$ (half-wave section is transparent).

Chapter 9

Transmission Line Loss and Attenuation

Every real transmission line attenuates signals as they travel. Two independent mechanisms cause this loss: **conductor loss** (resistive heating in the metal) and **dielectric loss** (power absorbed in the substrate or insulator). Total attenuation $\alpha = \alpha c + \alpha d$ in Np/m; multiply by 8.686 to convert to dB/m.

9.1 Conductor Loss & Skin Effect

At RF frequencies current crowds into a thin surface layer of thickness $\delta s = \surd(1/\pi f \mu_0 \sigma)$ — the skin depth. AC resistance rises as $\surd f$. The surface resistance (resistance per square) is:

$$R_s = \sqrt{\frac{\pi f \mu_0}{\sigma}} \;\; [\Omega/\square]$$

For a **coaxial line** with inner conductor radius a and outer radius b:

$$\alpha_c = \frac{R_s}{4\pi Z_0}\left(\frac{1}{a} + \frac{1}{b}\right) \;\; [\text{Np/m}]$$

For **microstrip** the conductor loss depends on the effective strip width w and involves a correction for current crowding at the edges; simplified for $w/h > 1$:

$$\alpha_c \approx \frac{R_s}{Z_0\, w} \;\; [\text{Np/m}]$$

9.2 Dielectric Loss

The loss tangent $\tan\delta$ of the substrate causes power absorption proportional to frequency. For **microstrip** (Pozar, Microwave Engineering eq. 3.30):

$$\alpha_d = \frac{k_0\, \varepsilon_r\, (\varepsilon_{\text{eff}} - 1)\, \tan\delta}{2\sqrt{\varepsilon_{\text{eff}}}\, (\varepsilon_r - 1)} \;\; [\text{Np/m}]$$

where $k_0 = 2\pi f/c$ and εeff is the effective permittivity. For **coaxial** line:

$$\alpha_d = \frac{\pi f \sqrt{\varepsilon_r}}{c}\, \tan\delta \;\; [\text{Np/m}]$$

Dielectric loss is proportional to frequency and often dominates conductor loss at GHz frequencies in lossy substrates such as FR4 ($\tan\delta \approx 0.02$).

9.3 Total Loss & Unit Conversion

$$\alpha_{\text{total}} = \alpha_c + \alpha_d \ \ [\text{Np/m}] \qquad \alpha_{\text{dB}} = 8.686\,\alpha_{\text{total}} \ \ [\text{dB/m}]$$

Insertion loss over a physical length L is simply α_dB × L dB. Both frequency and length increase loss linearly (for dB/m at fixed f, or for fixed length scaling with f).

9.4 Propagation Delay

Signals travel at the phase velocity vph = c/$\sqrt{\varepsilon}$eff, giving a time delay:

$$t_d = \frac{L\sqrt{\varepsilon_{\text{eff}}}}{c} \ \ [\text{s}]$$

On FR4 microstrip (εeff ≈ 3.5), vph ≈ 0.53c — roughly 6 ps/mm of delay. Delay is independent of frequency for a non-dispersive TEM line.

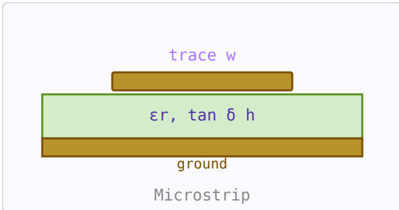

Figure 9.1: Microstrip cross-section showing substrate height h, trace width w.

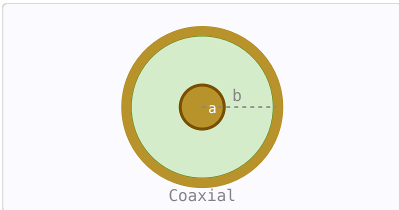

Figure 9.2: Coax cross-section. Conductor loss from inner (a) and outer (b) conductors.

9.5 Practical Values

- FR4 microstrip 50 Ω, 1.6 mm substrate: ≈ 0.5 dB/cm at 10 GHz
- Rogers RO4003C microstrip 50 Ω: ≈ 0.09 dB/cm at 10 GHz (tan δ = 0.0027)
- RG-58 coax: ≈ 0.22 dB/m at 100 MHz; ≈ 1.1 dB/m at 2.4 GHz
- LMR-400 coax: ≈ 0.068 dB/m at 1 GHz; ≈ 0.12 dB/m at 2.4 GHz

Chapter 10

VSWR, Return Loss and Reflection Coefficient

VSWR, return loss, reflection coefficient, and mismatch loss are four equivalent ways to describe the same physical phenomenon: how well two impedances are matched. They are all simple algebraic transformations of each other. Understanding all four is essential for RF work because different instruments and standards use different conventions.

10.1 Reflection Coefficient Γ

When a travelling wave encounters a discontinuity — a change in impedance — some energy is reflected. The complex voltage reflection coefficient at the load is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where Z_0 is the reference (usually 50 Ω) and Z_L is the load impedance. $|\Gamma|$ ranges from 0 (perfect match) to 1 (total reflection). The angle of Γ gives the phase of the reflection.

10.2 Return Loss (RL)

Return loss is the magnitude of the reflection expressed in dB, always as a positive number for a passive load:

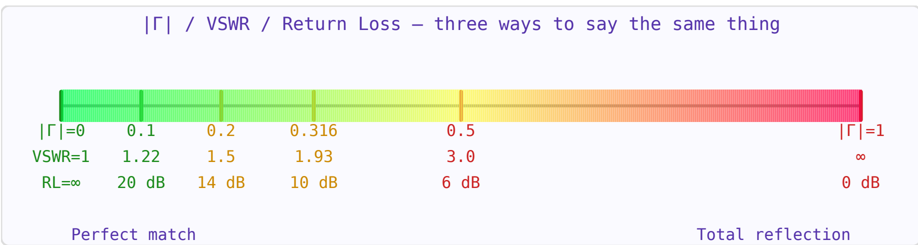

Figure 10.1: The match quality spectrum from perfect (green) to total reflection (red).

$$RL = -20 \log_{10} |\Gamma| \text{ dB}$$

Higher return loss means a better match. Typical design targets:

Return Loss	$\|\Gamma\|$	VSWR	Mismatch Loss	Context

∞ dB	0	1.00:1	0 dB	Perfect match (theoretical)
20 dB	0.100	1.22:1	0.04 dB	Good — typical antenna target
14 dB	0.200	1.50:1	0.18 dB	Acceptable for many systems
10 dB	0.316	1.93:1	0.46 dB	Minimum acceptable for most RF
6 dB	0.500	3.00:1	1.25 dB	Poor — significant power loss
0 dB	1.000	∞	∞ dB	Total reflection (open or short)

10.3 VSWR — Voltage Standing Wave Ratio

When forward and reflected waves coexist on a transmission line, they superimpose to create a standing wave pattern. The ratio of the maximum to minimum voltage amplitude along the line is the VSWR:

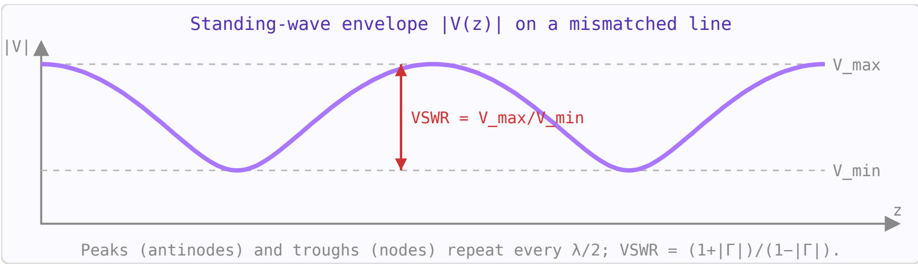

Figure 10.2: Voltage amplitude along a mismatched line. VSWR = ratio of maximum to minimum voltage.

$$\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

A VSWR of 1 means no standing waves (perfect match). VSWR is measured with a slotted line or a VNA. It is always ≥ 1.

10.4 Mismatch Loss

Even without any dissipative loss, a mismatch causes a reduction in delivered power because some fraction of the incident power is reflected and not absorbed by the load:

$$ML = -10\log_{10}(1-|\Gamma|^2) \text{ dB}$$

This is also called the *transmission coefficient* $T = 1 - |\Gamma|^2$. At RL = 10 dB, mismatch loss is only 0.46 dB — often acceptable. But at RL = 6 dB ($|\Gamma| = 0.5$), 25% of the power is reflected and mismatch loss reaches 1.25 dB.

10.5 The S_{11} Connection

On a VNA, S_{11} is the complex reflection coefficient Γ measured at port 1. Its magnitude in dB equals the negative of the return loss: $|S_{11}|$_dB $= -$RL. A Smith Chart plots Γ directly in the complex plane.

10.6 Physical Interpretation

The standing wave on a mismatched line creates voltage maxima and minima that can exceed the incident amplitude. At VSWR = 2 the peak voltage is twice the matched case — this matters for power handling, dielectric breakdown, and cable ratings. High-power transmitters specify maximum allowed VSWR to protect their output stages.

Chapter 11

The Smith Chart

The Smith Chart is a polar plot of the reflection coefficient Γ with constant-resistance circles and constant-reactance arcs overlaid. Every point on the chart represents a unique normalised impedance $z = Z/Z_0$. Once you understand its structure, the chart makes impedance matching design immediate and visual.

11.1 Chart Structure

The chart is a circle of radius 1 (the $|\Gamma| = 1$ boundary, representing total reflection). The centre is the perfect match point ($z = 1$, $\Gamma = 0$). The right-most point is an open circuit ($z = \infty$). The left-most point is a short circuit ($z = 0$).

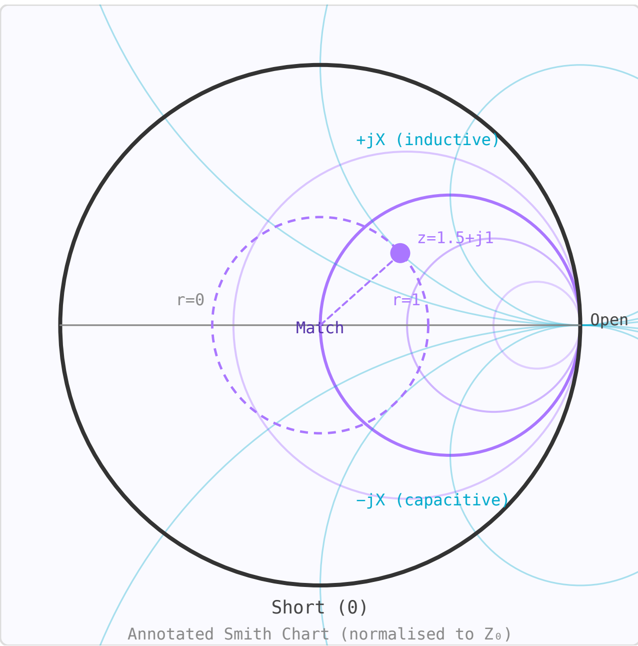

Figure 11.1: Smith Chart: constant-resistance circles (purple) and constant-reactance arcs (teal). Upper half = inductive; lower half = capacitive.

- **Constant-resistance circles** (drawn in purple on the RF Toolbox chart): Circles tangent to the right side. The $r = 1$ circle passes through the centre and represents all impedances with real part equal to Z_0.
- **Constant-reactance arcs** (drawn in teal): Arcs emanating from the right-most point.

Upper half = inductive (positive X), lower half = capacitive (negative X). The real axis (X = 0) represents purely resistive impedances.

11.2 Reading an Impedance

To plot Z = 75 + j50 Ω in a 50 Ω system, first normalise: z = (75 + j50)/50 = 1.5 + j1.0. Find the intersection of the r = 1.5 circle and the x = +1.0 arc. That point is in the upper half of the chart (inductive).

11.3 Moving Along a Transmission Line

Moving along a *lossless* transmission line toward the generator rotates the impedance point *clockwise* around the centre of the chart. The radius stays constant (|Γ| is constant because the line has no loss). One full revolution = λ/2 of travel. Moving λ/4 toward the generator rotates 180°.

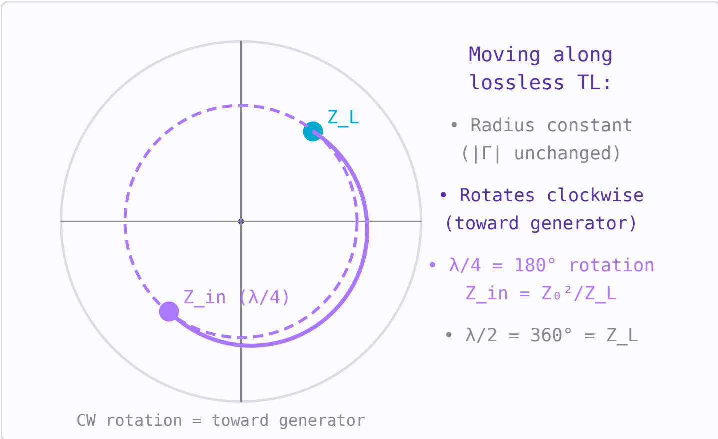

Figure 11.2: Transmission line moves impedance clockwise on a constant-|Γ| circle. A λ/4 section rotates 180°.

$$\Gamma_{in} = \Gamma_L \cdot e^{-j2\beta\ell}$$

This is the basis of the quarter-wave transformer: rotate 180°, and impedance Z_L becomes Z_0^2/Z_L at the input.

11.4 Reading |Γ|, Return Loss, and VSWR

The radius of the point from the chart centre equals |Γ|. The dashed constant-|Γ| circle that passes through the operating point is the "VSWR circle" — the VSWR equals (1 + |Γ|)/(1 − |Γ|). Return loss = −20 $\log_{10}$|Γ| dB. A point right at the centre has |Γ| = 0, VSWR = 1, RL = ∞ dB (perfect match).

11.5 Admittance: The Rotated Chart

The admittance Smith Chart is identical to the impedance chart but rotated 180° around the centre. On a combined Z/Y chart, the admittance of any impedance point is found by rotating that point 180° through the centre. This is useful for adding shunt (parallel) elements — on the admittance chart, adding a shunt susceptance moves the point along a constant-conductance circle.

11.6 Matching Network Design — Step by Step

To match Z_L = 25 − j30 Ω to 50 Ω using an L-network:

1. Normalise: z_L = (25 − j30)/50 = 0.5 − j0.6. Plot this point.
2. Adding a series inductor moves the point clockwise along the constant r = 0.5 circle (toward more positive X). Add series L until you reach the r = 1 circle.
3. The new point is on r = 1 with some positive X. Now add a shunt capacitor: switch to the admittance chart (rotate 180°), move along constant g = 1 circle to the centre by adding susceptance. Read off the capacitor value.
4. Alternatively, use the RF Toolbox L-Network calculator to get exact values, then verify on the Smith Chart.

11.7 Stub Matching

A shunt stub matching network uses a transmission line stub to add the right susceptance to cancel the load's reactive part, bringing the point to the real axis, then a series line section to rotate it to the centre. On the Smith Chart this is done graphically: move along the constant-|Γ| circle until you hit the g = 1 circle (for shunt matching), then add stub susceptance to reach the centre.

11.8 Practical Tips

- An impedance that sweeps a clockwise circle on the Smith Chart as frequency increases is dominated by transmission line behaviour.
- A small anticlockwise hook in the sweep indicates a series resonance.
- The upper half of the chart = inductive = store energy in magnetic field. Lower half = capacitive = store energy in electric field.
- The outer rim (|Γ| = 1) represents lossless reactive terminations: shorts, opens, and purely reactive loads.

Chapter 12

S-Parameters

S-parameters (scattering parameters) describe how RF signals are scattered by a network. Unlike Z or Y parameters, S-parameters are measured with all ports terminated in a reference impedance Z_0 (typically 50 Ω), making them practical to measure with a vector network analyser (VNA) at microwave frequencies.

12.1 Definition

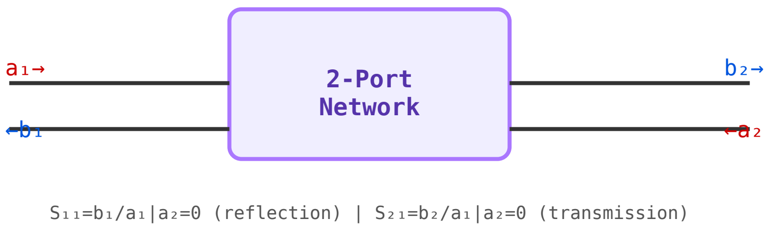

Figure 12.1: 2-port S-parameter definitions. All ports terminated in Z_0 during measurement.

For a 2-port network, S-parameters relate incident (a) and reflected/transmitted (b) wave amplitudes:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

- S_{11} — input reflection coefficient (return loss at port 1)
- S_{21} — forward transmission (insertion loss or gain)
- S_{12} — reverse isolation
- S_{22} — output reflection coefficient

12.2 Smith Chart

The Smith Chart maps the complex reflection coefficient Γ onto a plot of normalised impedance. Constant-resistance circles are in the horizontal centre; constant-reactance arcs run vertically. Moving clockwise along a constant-$|\Gamma|$ circle corresponds to adding transmission line length toward the generator.

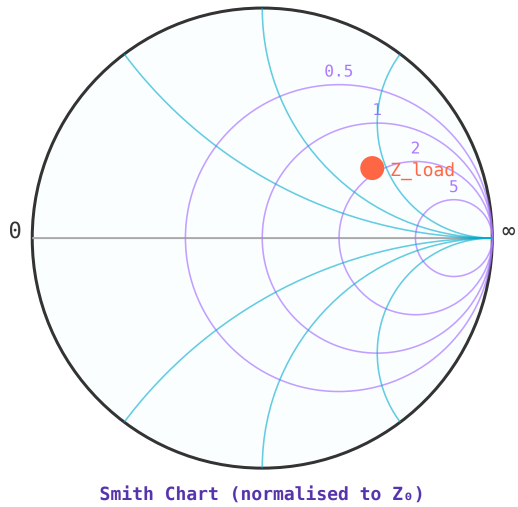

Figure 12.2: Smith Chart: constant-resistance circles (purple), constant-reactance arcs (blue). Example load shown as orange dot.

12.3 Touchstone Files

Measured S-parameters are stored in Touchstone format (.s1p, .s2p, etc.). Each line contains a frequency point followed by pairs of values encoding each Sij in MA (magnitude-angle), RI (real-imaginary), or DB (dB-angle) format.

Chapter 13

Impedance Matching

Maximum power transfer from source to load occurs when the load impedance is the complex conjugate of the source impedance. Impedance matching is the process of inserting a network between source and load to achieve this condition at the desired frequency.

13.1 Why Matching Matters

- Maximises power transfer to the load
- Minimises reflections and standing waves
- Prevents damage from reflected power in high-power systems
- Optimises noise figure in low-noise amplifiers (noise match $\neq$ power match)

13.2 L-Network

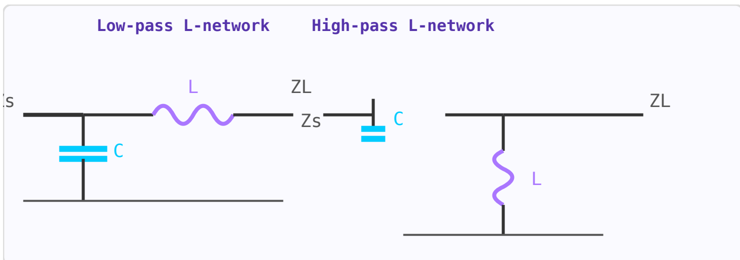

Figure 13.1: L-network topologies. Left: low-pass (shunt C + series L). Right: high-pass (series C + shunt L).

Two reactive elements (one series, one shunt) form the simplest matching network. The Q factor is fixed by the impedance ratio: $Q = \sqrt{R_{high}/R_{low} - 1}$. Only one frequency can be matched; the fractional bandwidth $\approx 1/Q$.

13.3 Pi and T Networks

Three-element networks allow Q to be chosen independently of the impedance ratio. A Pi network has two shunt elements; a T network has two series elements. Higher Q gives better harmonic suppression but narrower bandwidth.

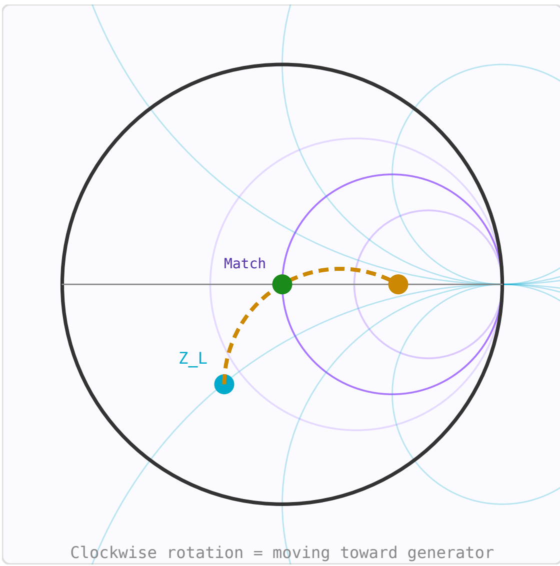

Figure 13.2: Smith Chart matching path: L_load (blue) → $\lambda/4$ rotation (orange) → matched (green).

13.4 Quarter-Wave Transformer

A $\lambda/4$ transmission line section with $Z_0 = \sqrt{Z_1 Z_2}$ matches two real impedances at the design frequency. Bandwidth is typically 10–20% for VSWR < 2.

13.5 Stub Matching

A shorted or open transmission line stub in shunt with the main line presents a pure susceptance. Single-stub matching can match any load, but requires moving the stub to the correct point on the line (found using the Smith Chart).

Part III

Guided-Wave Structures and Interconnect

Chapter 14

Microwave Transmission Lines

At microwave frequencies, transmission lines must be treated as distributed elements. Several planar and non-planar structures are used depending on frequency, power, loss, and fabrication constraints.

14.1 Microstrip

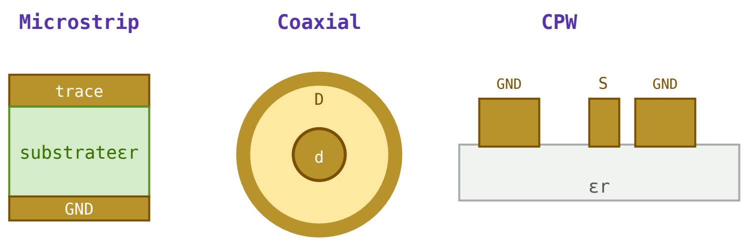

Figure 14.1: Cross-sections: microstrip (left), coaxial cable (centre), coplanar waveguide CPW (right).

The most common planar line in PCB design. A conductor trace on top of a dielectric substrate with a ground plane below. Fields are partly in the substrate and partly in air, giving an effective dielectric constant \(\varepsilon_{eff} < \varepsilon_r\). Easy to fabricate but has higher radiation loss at millimetre-wave frequencies.

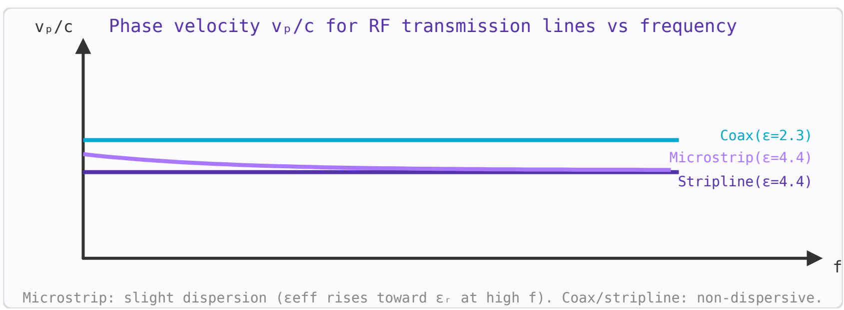

Figure 14.2: Phase velocity for common transmission lines. Microstrip is slightly dispersive; coax and stripline are not.

14.2 Stripline

Conductor buried between two ground planes in a homogeneous dielectric. All fields are in the dielectric so $\varepsilon_{eff} = \varepsilon_r$ exactly. Lower radiation loss than microstrip but harder to access for shunt connections.

14.3 Coplanar Waveguide (CPW)

Signal conductor flanked by ground conductors on the same surface. Easy shunt connections (no via needed), suitable for MMIC (monolithic microwave integrated circuit) fabrication. Widely used above 10 GHz.

14.4 Coaxial Line

TEM mode, no cutoff frequency (down to DC). The dominant transmission medium for test equipment, cables, and connectors. Standard impedances: 50 Ω (minimum attenuation/maximum power tradeoff), 75 Ω (minimum attenuation in air-filled coax).

14.5 Waveguide

Hollow metallic tube supporting TE and TM modes. Has a cutoff frequency below which propagation does not occur. Very low loss at microwave/millimetre-wave frequencies. Used in high-power radar and satellite systems.

Chapter 15

RF PCB Design

At RF frequencies, PCB traces behave as transmission lines. Their characteristic impedance must be controlled to prevent reflections, signal integrity problems, and radiation. The four standard controlled-impedance structures are microstrip, stripline, coplanar waveguide (CPW), and differential pairs. Each has different shielding characteristics, loss properties, and PCB layer requirements.

15.1 Comparison of PCB Transmission Line Types

Type	Layers needed	Shielding	Via needed for termination?	Best for
Microstrip	2 (trace + ground)	Partial (one side open)	No	General RF up to ~10 GHz, antennas
Stripline	3+ (trace between 2 grounds)	Full TEM, no radiation	Yes (for shunt elements)	High isolation, sensitive circuits, digital backplanes
CPW (coplanar waveguide)	1 (trace + ground coplanar)	Partial to full (GCPW)	No — shunt connections on same layer	MMIC, mmWave, no-via shunt connections
Differential pair	2	Partial (common-mode rejects noise)	No	USB, HDMI, LVDS, balanced RF

15.2 Coplanar Waveguide (CPW)

CPW places the signal trace between two coplanar ground planes. No ground plane is needed below the substrate, making it ideal for flip-chip and MMIC die attachment where ground connections must be made on the surface. The characteristic impedance depends on the trace width w and gap s to the ground planes:

$$Z_0 = \frac{30\pi}{\sqrt{\varepsilon_{eff}}} \frac{K(k')}{K(k)}, \quad k = \frac{w}{w + 2s}$$

where K(k) is the complete elliptic integral of the first kind. Grounded CPW (GCPW) adds a bottom ground plane, suppressing the parasitic microstrip mode and is preferred for multilayer PCBs.

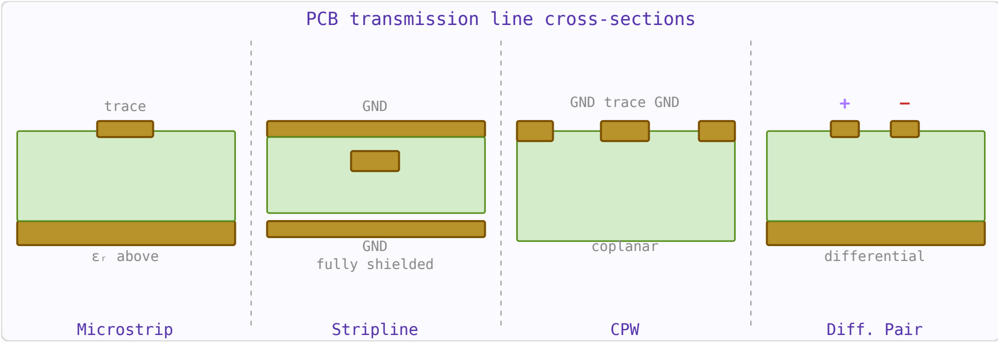

Figure 15.1: Cross-sections of the four standard PCB RF transmission line types.

15.3 Differential Pair Impedance

Edge-coupled microstrip differential pairs carry balanced signals. The key parameter is the differential impedance $Z_{diff} = 2Z_{odd}$, where Z_{odd} is the odd-mode impedance. Standard targets: 100 Ω differential for USB/HDMI/LVDS, 90 Ω for PCIe.

$$Z_{diff} = 2Z_{odd} \approx 2Z_0 \left(1 - 0.347e^{-2.655s/h}\right)$$

where s is the edge-to-edge spacing and h is the substrate height. The even-mode impedance $Z_{even} \approx 2Z_0/(1 + 0.347e^{-2.655s/h})$ determines common-mode behaviour. Tight spacing reduces Z_diff; looser spacing increases it toward 2×Z_0_single.

15.4 Stripline

Stripline is a flat conductor centred symmetrically between two ground planes. It is a TEM mode structure with no dispersion and no radiation — unlike microstrip whose fields extend into air above the trace. The impedance:

$$Z_0 = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4b}{0.67\pi w_{eff}(0.8 + t/w)}\right)$$

where b is the ground-plane separation and w_{eff} accounts for trace thickness. Because the medium is uniform, $v_p = c/\sqrt{\varepsilon_r}$ with no effective permittivity correction.

15.5 Via Parasitics

A PCB via connecting signal layers has inductance and capacitance:

$$L_{via} \approx 0.2h \ln\left(\frac{4h}{d}\right) \text{ nH}, \quad C_{via} \approx \frac{1.41\varepsilon_r h}{\ln(D/d)} \text{ pF}$$

where h is the via height (mm), d the drill diameter (mm), and D the antipad diameter (mm). At 1 GHz, a standard FR4 via (0.3 mm drill, 1.6 mm deep) has about 0.5 nH inductance and 0.15 pF capacitance — enough to cause significant reflections at microwave frequencies. Via stubs (unused portions below the signal layer) are particularly problematic and are removed by backdrilling.

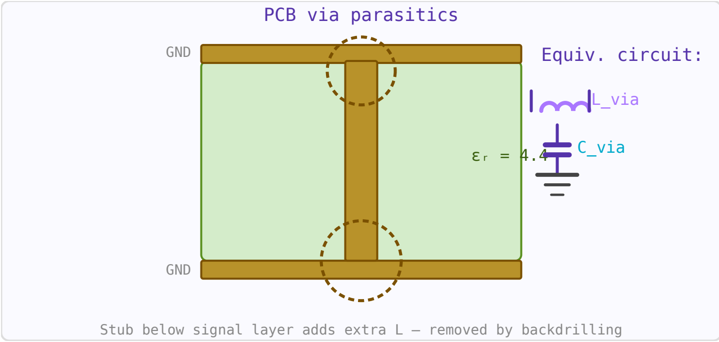

Figure 15.2: Via cross-section showing inner conductor, antipad, and the parasitic inductance + capacitance equivalent circuit.

15.6 Bond Wire Inductance

Gold or aluminium bond wires connecting a die to a package or PCB have parasitic inductance (Grover formula):

$$L_{wire} = 0.2\ell\left[\ln\left(\frac{2\ell}{d}\right) - 0.75\right] \text{ nH}$$

where ℓ and d are in mm. A typical 1.5 mm gold bond wire (d = 25 μm) has about 1.2 nH inductance, giving 7.5 Ω reactance at 1 GHz — often requiring resonating capacitors in MMIC matching networks.

15.7 Common Material Properties

Substrate	$\varepsilon_\mathbf{r}$	**tan** δ	**Typical use**
FR4	4.2–4.6	0.018–0.025	General PCB, < 3 GHz
Rogers 4003C	3.55	0.0027	RF PCB, antennas, up to 30 GHz
Rogers 4350B	3.66	0.0037	Low-cost RF alternative
PTFE (Teflon)	2.1	0.001	High-frequency, low-loss
Alumina (Al_2O_3)	9.8	0.0002	MMICs, hybrid circuits
GaAs	12.9	0.0006	MMIC substrates

Chapter 16

Coaxial Cable

Coaxial cable is the primary transmission line for RF signals above a few MHz. Choosing the wrong cable — even for a short run — can add significant insertion loss, degrade shielding, or limit power handling. This page is a quick reference for the most common types.

16.1 Cable Comparison Table

Type	OD (mm)	Z_0	Vel. factor	Atten. @ 1 GHz	Max power	Best use
RG-174	2.8	50 Ω	0.66	1.5 dB/m	25 W	Pigtails, U.FL connections, very flexible low-power
RG-316	2.6	50 Ω	0.69	1.3 dB/m	30 W	PTFE insulation; flexible, low loss vs RG-174
RG-58 (C/U)	5.0	50 Ω	0.66	0.55 dB/m	200 W	General RF cable, lab use, instrumentation up to ~500 MHz
RG-59	6.1	75 Ω	0.66	0.6 dB/m	200 W	Video, CATV, 75 Ω systems
RG-213 (U)	10.3	50 Ω	0.66	0.28 dB/m	1 kW	Base station feedlines, amateur radio HF/VHF
LMR-195	4.95	50 Ω	0.80	0.43 dB/m	250 W	Low-loss flexible for short runs, jumpers
LMR-400	10.3	50 Ω	0.85	0.135 dB/m	1.1 kW	Low-loss runs to 1 GHz; antenna feedlines

LMR-600	15.9	50 Ω	0.87	0.087 dB/m	2.5 kW	Long runs, base station, cellular infrastructure
Semi-rigid UT-085	2.2	50 Ω	0.695	0.84 dB/m	100 W	PCB-to-PCB, MMIC, excellent shielding, non-flexible
Semi-rigid UT-141	3.6	50 Ω	0.695	0.44 dB/m	300 W	Microwave benching, test fixtures

16.2 Attenuation Scaling with Frequency

Coaxial cable attenuation increases approximately as $\sqrt{}$f (due to skin effect in the conductors) plus a smaller f-dependent term from dielectric loss:

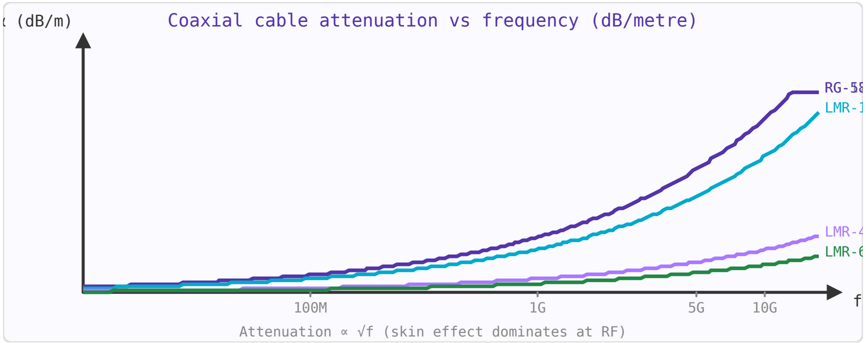

Figure 16.1: Coaxial cable attenuation increases as $\sqrt{}$f. LMR-400 has 5× less loss than RG-174 at the same frequency.

$$\alpha \approx \alpha_1\sqrt{f} + \alpha_2 f \quad [\mathrm{dB/m}]$$

At 100 MHz, LMR-400 has ~0.044 dB/m. At 2.4 GHz it's ~0.22 dB/m. A 10-metre run at 2.4 GHz loses 2.2 dB — worth using LMR-600 or a tower-mounted LNA for long runs.

16.3 Velocity Factor and Electrical Length

The velocity factor (VF) is the ratio of signal propagation speed in the cable to the speed of light. VF depends on the dielectric: solid PE ≈ 0.66, foam PE ≈ 0.78–0.88, PTFE ≈ 0.69. Electrical length = physical length / VF × 360° per wavelength. A 1-metre RG-58 cable at 300 MHz is one electrical wavelength long (0.66 × 1 m / 1 m = 0.66λ — 237°).

16.4 Power Handling Limits

Two limits apply: **thermal** (conductor heating → insulation damage, typically rated at 40°C ambient) and **voltage breakdown** (at peak voltage, the dielectric arcs over). For most cables the thermal limit is lower. Power derating at high temperatures: reduce rated power by ~1% per °C above 40°C. At altitude (lower air pressure), the breakdown voltage decreases — critical for airborne equipment.

16.5 Practical Selection Guide

- **0–30 MHz, any run length**: RG-213 or LMR-400. Loss is not critical.
- **VHF/UHF, $<$ 3 m**: RG-58 or LMR-195. Adequate loss, easy to handle.
- **VHF/UHF, $>$ 10 m**: LMR-400 or LMR-600 to keep feedline loss below 1–2 dB.
- **Microwave, $>$ 2 GHz, critical loss budget**: LMR-400 or semi-rigid.
- **Pigtails and internal connections**: RG-316 or UT-085 semi-rigid.
- **High power ($>$ 500 W)**: LMR-600 or larger hardline; ensure good connector torque.

Chapter 17

RF Connectors

Connector choice often determines system performance at microwave frequencies. Every connector junction contributes insertion loss, a small impedance discontinuity (which worsens with frequency), and a potential point of failure. Choosing the right connector requires balancing frequency range, power handling, mechanical robustness, and cost.

17.1 Connector Comparison Table

Connector	Max frequency	Impedance	Power (CW)	Torque spec	Notes
BNC	4 GHz	50 Ω / 75 Ω	80 W @ 1 GHz	Bayonet lock	Quick-connect; RF lab, oscilloscopes, video (75 Ω)
TNC	12 GHz	50 Ω / 75 Ω	100 W	1.13 N·m	BNC with threaded coupling; more rugged
N-type	18 GHz	50 Ω / 75 Ω	300 W @ 1 GHz	1.36 N·m	Workhorse for VHF/UHF/microwave; weatherproof versions
SMA	18 GHz (26 GHz semi-precision)	50 Ω	100 W @ 1 GHz	0.9 N·m	Most common RF connector; keep to 12 GHz in production
SMB	4 GHz	50 Ω	25 W	Snap-on	Compact snap-on; consumer electronics
MMCX	6 GHz	50 Ω	50 W	Snap-on	Miniature; Wi-Fi modules, GPS receivers
U.FL / IPEX	6 GHz	50 Ω	10 W	Snap-on	Very small; $<$ 30 insertion cycles; PCB to antenna pigtail
2.92 mm (K)	40 GHz	50 Ω	10 W	0.9 N·m	SMA-compatible mating; 26–40 GHz range

2.4 mm	50 GHz	50 Ω	8 W	0.9 N·m	Fragile; Ka-band test and mmWave
1.85 mm (V)	67 GHz	50 Ω	5 W	0.9 N·m	Very fragile; V-band measurements
1.0 mm	110 GHz	50 Ω	2 W	0.45 N·m	W-band; lab use only

17.2 Insertion Loss Rules of Thumb

Every connector pair (male + female) contributes roughly:

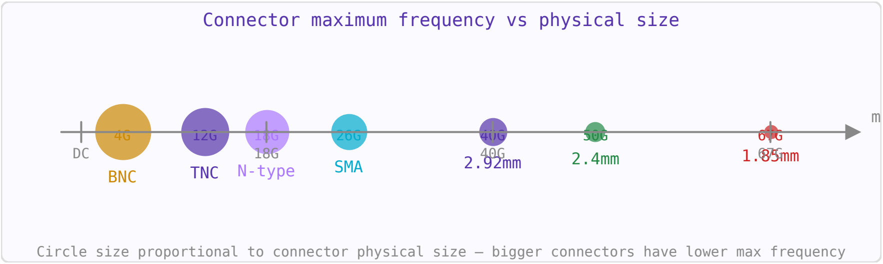

Figure 17.1: RF connector maximum frequency vs physical size. Smaller connectors enable higher frequencies but are more fragile.

- SMA pair: 0.1 dB at 10 GHz, 0.2 dB at 18 GHz
- N-type pair: 0.05 dB at 10 GHz
- 2.92 mm pair: 0.15 dB at 30 GHz, 0.3 dB at 40 GHz
- Each adapter (e.g. SMA-to-N): doubles the connector insertion loss

In a 10-connector measurement path at 18 GHz, connector losses alone can exceed 2 dB — easily dominating the DUT's own loss.

17.3 Torque and Connector Longevity

Over-torquing damages centre pin contacts. Under-torquing causes intermittent connections and unpredictable impedance discontinuities. Always use a torque wrench for precision measurements. SMA connectors are rated for ~500 mating cycles; precision SMA (0.9 N·m spec) for ~2000 cycles. 2.4 mm and 1.85 mm connectors are easily damaged by cross-threading — always align carefully before tightening.

17.4 PCB Launch Connectors

The transition from a PCB trace to a connector is a discontinuity. Edge-launch SMA connectors must be carefully matched — the launch geometry, board thickness, and connector pin dimension must all be designed together. At 20+ GHz, a poorly designed SMA launch can add 0.5–1 dB of loss and create a visible ripple in S_{11} measurements. Some vendors provide field-solver-optimised footprints for their connectors at specific frequencies.

17.5 75 Ω Systems

Cable television, broadcast video, and some telecommunications infrastructure use 75 Ω impedance (derived from the impedance of a centre-loaded dipole in free space). BNC-75, TNC-75, N-75, and F-connectors are used. Never mate 50 Ω and 75 Ω connectors — the impedance mismatch causes a permanent reflection, and physically incompatible types (N-50 vs N-75) can be damaged if forced.

Chapter 18

Waveguides

A waveguide is a hollow metallic tube that confines and guides electromagnetic waves. Unlike coaxial line (which supports TEM mode down to DC), waveguides have a cutoff frequency below which propagation does not occur. Above cutoff, waveguides offer very low loss, making them the preferred transmission medium at millimetre-wave frequencies and for high-power radar systems.

18.1 Modes

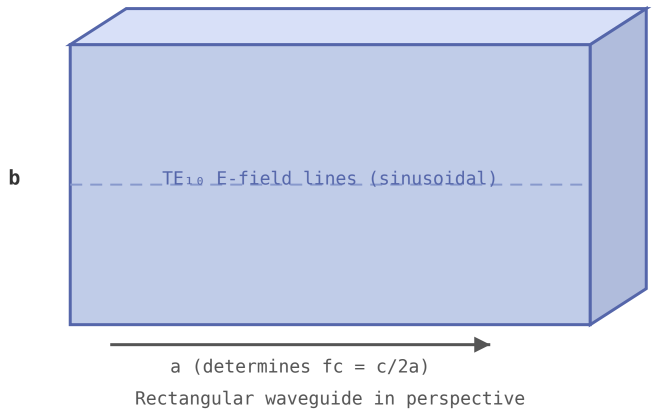

Figure 18.1: Rectangular waveguide. Cutoff frequency fc = c/(2a) for the dominant TE_{10} mode.

Rectangular waveguides support TE (transverse electric) and TM (transverse magnetic) modes. The dominant mode TE_{10} has the lowest cutoff frequency and is used in virtually all single-mode waveguide applications. The cutoff frequency of the TE_{10} mode in a rectangular waveguide of width a is:

$$f_c = \frac{c}{2a}$$

18.2 Standard Waveguide Bands

- WR-90 X-band: 8.2–12.4 GHz
- WR-42 K-band: 18–26.5 GHz
- WR-28 Ka-band: 26.5–40 GHz
- WR-10 W-band: 75–110 GHz

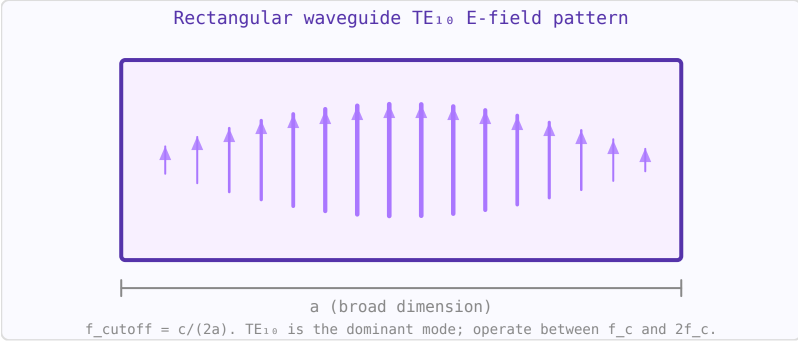

Figure 18.2: TE_{10} mode: E-field is vertical, sinusoidal in x, zero at walls. f_c = c/(2a) — only this mode propagates near cutoff.

18.3 Attenuation

Waveguide losses are due to finite conductivity of the walls. Attenuation is minimum at approximately 1.8× the cutoff frequency and rises toward both the cutoff and at very high frequencies. Rectangular waveguide attenuation is typically 0.01–0.5 dB/m, much lower than coax at the same frequency.

18.4 Applications

Waveguides are used in high-power radar systems, satellite ground station feeds, millimetre-wave communication links, and as the connecting structure between antenna feeds and LNAs in satellite receivers (where minimum loss before the LNA is critical for noise figure).

Chapter 19

VNA Measurements and Calibration

A vector network analyser (VNA) measures the complex S-parameters of a device under test (DUT) — both magnitude and phase — across a swept frequency range. Understanding what it measures, and how calibration moves the reference plane to your device, is essential for making accurate RF measurements.

19.1 What a VNA Measures

A 2-port VNA applies a stimulus at port 1 and measures the signals at both ports simultaneously. It computes:

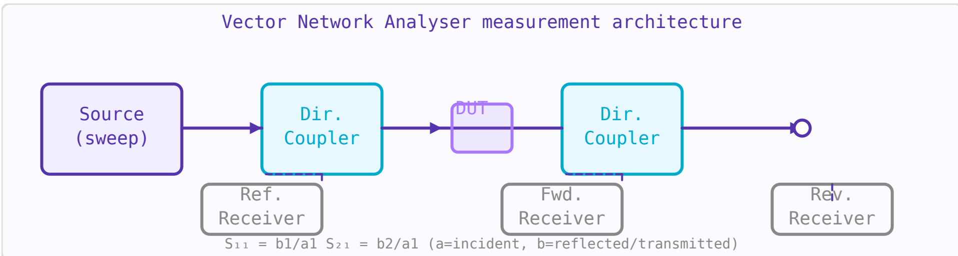

Figure 19.1: VNA architecture: swept source, directional couplers separate incident/reflected/transmitted waves, receivers measure amplitude and phase.

Parameter	Physical meaning	Typical display
S_{11}	Input reflection coefficient. How much is reflected from port 1.	Smith Chart, log mag (RL)
S_{21}	Forward transmission. Gain or insertion loss from port 1 to port 2.	Log magnitude (dB)
S_{12}	Reverse transmission. Isolation.	Log magnitude (dB)
S_{22}	Output reflection coefficient.	Smith Chart, log mag

19.2 Calibration: Moving the Reference Plane

Without calibration, a VNA measures its own cables, connectors, and switch imperfections. Calibration removes these systematic errors by measuring known standards and computing correction factors. The most common calibration is **SOL** (Short-Open-Load) or **SOLT** (Short-Open-Load-Through):

- **Short**: $\Gamma = -1$ (all reflected, 180° phase). Defines the reference plane position.

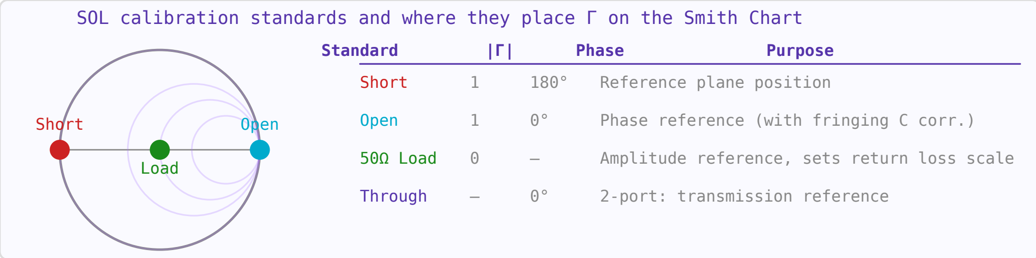

Figure 19.2: SOL calibration standards place known Γ values on the Smith Chart rim and centre, allowing systematic error correction.

- **Open**: $\Gamma = +1$ (all reflected, 0° phase, adjusted for fringing capacitance).
- **Load**: $\Gamma = 0$ (50 Ω termination, no reflection). Sets the amplitude reference.
- **Through**: Connects port 1 to port 2 directly. Characterises transmission path.

Always calibrate at the ends of the test cables, not at the VNA ports — calibration moves the reference plane to wherever you connect the standards.

19.3 Port Extension

If you cannot calibrate at the device terminals (e.g. a connector is on a PCB and you calibrate at the cable end), port extension electrically "moves" the reference plane by adding a known electrical delay. It rotates the phase of S_{11} and S_{22} to compensate for the cable length. It is an approximation — it assumes the line is lossless and dispersionless — but is adequate for many applications.

19.4 Time-Domain Gating

The VNA can apply an inverse Fourier transform to its frequency-domain data to show the time-domain response (like a TDR). This reveals reflections from connectors and discontinuities separated in time (i.e. distance). Gating in the time domain — windowing to select only a specific time region — then transforming back to frequency allows you to isolate the response of a specific section of the DUT and remove connector effects. This is particularly useful for antenna measurements and filter characterisation.

19.5 Interpreting S_{11} on a Smith Chart

A single frequency point on the Smith Chart shows the impedance at that frequency. As frequency sweeps:

- A resistor traces a point on the real axis.
- A capacitor traces clockwise along the $X = 0$ boundary toward the short-circuit point.
- A series LC traces a loop: inductive (upper half), resonant (real axis), capacitive (lower half).
- A transmission line section traces a clockwise circle centred on the origin.
- An antenna shows a small loop near the edge of the chart, touching or nearly touching the real axis at resonance.

19.6 Practical Measurement Tips

- Torque connectors to spec (SMA: 0.9 N·m / 8 in·lb). Under/over-torquing causes repeatable but incorrect results.
- Let the VNA warm up for 15–30 minutes before calibrating for best stability.
- Calibrate with the cable attached in its final position — bending it after calibration introduces errors.
- Use at least 201 frequency points across your band; use 1601+ for time-domain measurements to maximise time resolution.
- When measuring low-loss filters (< 0.5 dB), use a full 4-port SOLR or TRL calibration for best accuracy.
- Check calibration by measuring a known open or short — it should show a perfect circle on the Smith Chart rim.

Part IV

Passive Components, Resonators and Filters

Chapter 20

E-Series Standard Component Values

Resistors, capacitors, and inductors are manufactured in standardised value sets called **E-series**, defined by IEC 60063. Rather than covering every decimal value, the series places values at logarithmically equal intervals so that any calculated value falls within a specified tolerance of the nearest standard value.

20.1 Logarithmic Spacing

An E-series with N values per decade places mantissas at:

$$v_k = 10^{k/N} \quad k = 0, 1, \ldots, N-1$$

The ratio between consecutive values is always 10^(1/N). This means worst-case error is at most half that ratio, or approximately 1/(2N) × 100% of a decade — which corresponds directly to the component tolerance.

20.2 Series Summary

Series	Values/decade	Tolerance	Typical use
E6	6	±20%	General decoupling capacitors
E12	12	±10%	General-purpose resistors
E24	24	±5%	Most common resistor series
E48	48	±2%	Precision resistors
E96	96	±1%	Precision resistors, filter components
E192	192	±0.5%	High-precision networks

20.3 E6 and E12 Values (first decade)

E6 mantissas: 1.0, 1.5, 2.2, 3.3, 4.7, 6.8

E12 mantissas (includes all E6): 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2

E24 adds further interleaving: 1.0, 1.1, 1.2, 1.3, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1

20.4 Reading E-Series Values

Multiply any mantissa by a decade: a resistor marked **4.7 kΩ** uses mantissa 4.7 in the kΩ decade. A capacitor of **22 nF** uses mantissa 2.2 in the 10 nF decade. The E-series is the same for resistors, capacitors, and inductors — only the unit decade changes.

Figure 20.1: Log-scale number line for one decade showing E6 (purple) and E12 (dark) value positions.

20.5 Choosing a Series

- **E12**: adequate for bias resistors, pull-ups, and non-critical dividers
- **E24**: standard choice for RF filter component selection
- **E96 / E192**: use when the component value directly sets a frequency, gain, or impedance with <1% accuracy requirement
- Higher-N series cost more and have longer lead times — use only where the tolerance budget demands it

20.6 Finding the Nearest Value

To find the nearest E-series value to a calculated target R_target: (1) compute the decade exponent exp = floor($\log_{10}$(R_target)), (2) extract the mantissa m = R_target / 10^exp, (3) scan all N mantissas and pick the closest. The E-Series Finder calculator automates this for any R, C, or L target.

Chapter 21

Coupled Resonators

When two resonant LC circuits are brought near each other they exchange energy through mutual inductance or capacitive coupling. This splits the single resonant frequency into two modes and produces a characteristic double-peaked bandpass response. Understanding coupled resonators is essential for designing IF transformers, bandpass filters, and resonator-based sensors.

21.1 Coupling Mechanisms

- **Magnetic (inductive)**: mutual inductance M between the two coils. $k = M/L$ for identical resonators ($M < L$, so $0 < k < 1$). Dominant in IF transformers and wound coil pairs.
- **Electric (capacitive)**: a gap capacitor Cm between the two tanks. Used in high-Q cavity and ceramic resonator filters.
- **Mixed**: both mechanisms simultaneously; common in printed resonators.

21.2 Mode Splitting

For two identical resonators (same L, C, resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$), coupling splits the resonant frequency into an even mode (lower, f_-) and an odd mode (upper, f_+):

$$f_\pm = \frac{f_0}{\sqrt{1 \mp k}}$$

The mode splitting $\Delta f = f_+ - f_-$ grows with coupling coefficient k. For small k: $\Delta f \approx k \cdot f_0$.

21.3 Passband Bandwidth

For high-Q resonators the -3 dB passband bandwidth of the coupled pair equals:

$$\mathrm{BW} \approx k \cdot f_0 \qquad Q_{\text{loaded}} = \frac{f_0}{\mathrm{BW}} = \frac{1}{k}$$

A larger coupling coefficient k gives a wider, flatter passband; a smaller k gives a narrower passband with higher loaded Q.

21.4 Critical Coupling

The transfer response shape depends on the ratio of coupling to resonator loss. Define the **critical coupling coefficient**:

$$k_{\text{crit}} = \frac{1}{Q} \qquad Q = \frac{\omega_0 L}{R} \text{ (unloaded Q)}$$

- **Under-coupled** (k < k_crit): single peaked response, insertion loss at f_0
- **Critically coupled** (k = k_crit): maximally flat single peak, maximum power transfer at f_0
- **Over-coupled** (k > k_crit): double-peaked response, dip at f_0

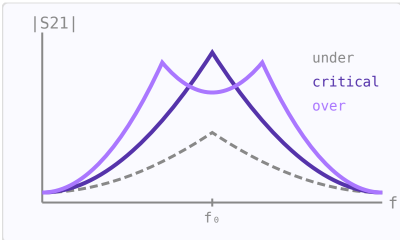

Figure 21.1: Transfer response shapes for three coupling regimes.

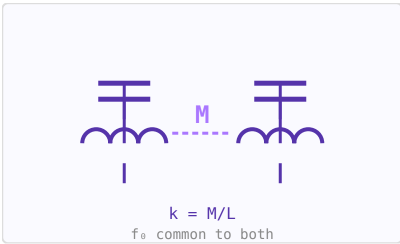

Figure 21.2: Two magnetically coupled LC tanks with mutual inductance M.

21.5 Bandpass Filter Design

A two-resonator bandpass filter is designed by choosing k to achieve the desired bandwidth and response shape. For a maximally flat (Butterworth) 2-pole bandpass filter, critical coupling gives the flattest passband. Chebyshev response requires slight over-coupling. Multi-resonator filters chain additional coupled pairs, with each coupling coefficient k_{i,i+1} set from the prototype g-values.

21.6 IF Transformers

The coupled resonator principle is the basis of the IF transformer found in superheterodyne receivers. The primary and secondary coils are wound on a shared ferrite core and are slightly over-coupled to achieve a flat-topped passband around the IF frequency (commonly 455 kHz or 10.7 MHz). Adjusting the core position changes the coupling coefficient and therefore the bandwidth.

Chapter 22

RF Attenuators

An attenuator is a passive network that reduces signal power by a precise, frequency-independent amount. Unlike a simple resistive divider, a well-designed attenuator maintains its characteristic impedance at all ports across a wide frequency range. Attenuators are used to set signal levels, improve source or load impedance, isolate stages from reflections, and calibrate measurement systems.

22.1 Key Parameters

Parameter	Symbol	Typical values
Attenuation	A (dB)	1–40 dB in 1 dB steps for switched pads; fixed values for inline pads
Characteristic impedance	Z_0	50 Ω (RF/microwave), 75 Ω (cable TV/video)
Power rating	P_max (W)	0.1–50 W for most SMD/connectorised types
Frequency range	DC–f_max	DC to 18 GHz for SMA types; 40+ GHz for 2.92 mm types
VSWR		Typically $<$ 1.2:1 across rated bandwidth

22.2 Pi (Π) Attenuator

The Pi pad uses two equal shunt resistors flanking a series resistor. It is the most common topology for RF PCBs. For attenuation A dB with system impedance Z_0:

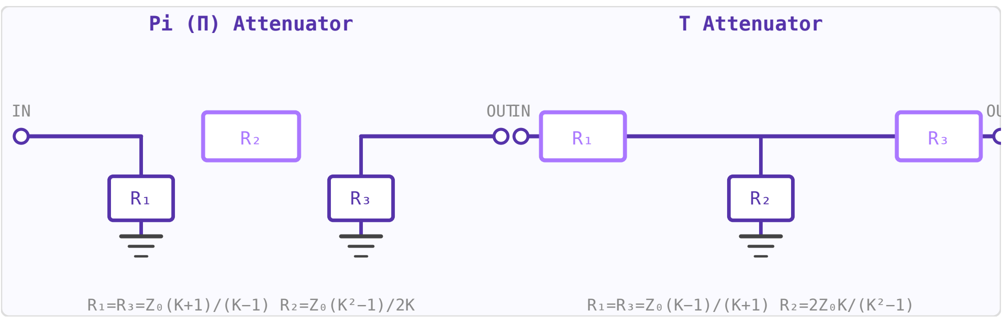

Figure 22.1: Pi attenuator (shunt-series-shunt) and T attenuator (series-shunt-series). K = 10^(A/20).

$$K = 10^{A/20}, \quad R_{shunt} = Z_0 \frac{K+1}{K-1}, \quad R_{series} = Z_0 \frac{K^2-1}{2K}$$

22.3 T Attenuator

The T pad uses two equal series resistors with a shunt resistor to ground at the midpoint. Better suited for cable assemblies where both conductors need to be attenuated symmetrically:

$$R_{series} = Z_0 \frac{K-1}{K+1}, \quad R_{shunt} = \frac{2Z_0 K}{K^2-1}$$

22.4 L-pad (Impedance-Matching Attenuator)

An L-pad matches two different source and load impedances while also providing attenuation. Unlike the Pi and T, the L-pad is inherently asymmetric. The attenuation is fixed once the impedances are set; it cannot be chosen independently. One series resistor and one shunt resistor are needed:

$$R_{series} = \sqrt{Z_S Z_L} \frac{\sqrt{Z_S/Z_L} - 1}{1}, \quad R_{shunt} = \frac{Z_S Z_L}{R_{series} + Z_S}$$

(exact formula depends on which port is the high-Z side).

22.5 Bridged-T Attenuator

The Bridged-T uses the system impedance Z_0 as one of its elements, making it efficient for large attenuation values with fewer components. With a bridging resistor $R_1 = Z_0(K-1)$ and a shunt resistor $R_2 = Z_0/(K-1)$ plus two Z_0 arms.

22.6 Temperature and Power Dissipation

Each resistor in the attenuator dissipates a fraction of the incident power. For high-power attenuators, the power in each element must be calculated from the impedance equations. The series element in a Pi pad dissipates $(1 - 1/K^2)$ of the input power.

Component values for Pi attenuator in 50 Ω system

A (dB)	K	R_shunt (Ω)	R_series (Ω)
1 dB	1.122	869.5 Ω	5.8 Ω
2 dB	1.259	436.2 Ω	11.6 Ω
3 dB	1.413	292.4 Ω	17.6 Ω
6 dB	1.995	150.5 Ω	37.4 Ω
10 dB	3.162	96.2 Ω	71.2 Ω
20 dB	10.000	61.1 Ω	247.5 Ω
30 dB	31.623	53.3 Ω	789.8 Ω

Figure 22.2: Standard Pi attenuator values. R_shunt = Z_0(K+1)/(K−1), R_series = $Z_0(K^2-1)/2K$.

22.7 Practical Notes

- Standard RF attenuator values: 1, 2, 3, 6, 10, 20, 30 dB.
- Switched attenuators in LNAs and signal generators use PIN diode or MEMS switching to change attenuation state.
- Surface-mount 0402/0603 resistors are used up to ~6 GHz; above that, thin-film chip attenuators are preferred.
- Place bypass capacitors on the shunt resistors to maintain flat response into high frequencies.

Chapter 23

Power Dividers and Combiners

Power dividers split one signal into two or more outputs. Power combiners do the reverse: sum multiple signals into one output. The Wilkinson divider is the classic design because it achieves simultaneous matching, isolation, and lossless operation using only transmission line sections and a single resistor.

23.1 Wilkinson Power Divider

The equal-split Wilkinson divider uses two $\lambda/4$ transmission line arms of impedance $Z_0\sqrt{2}$ ($\approx$ 70.7 Ω in a 50 Ω system) and an isolation resistor $R = 2Z_0$ between the two output ports:

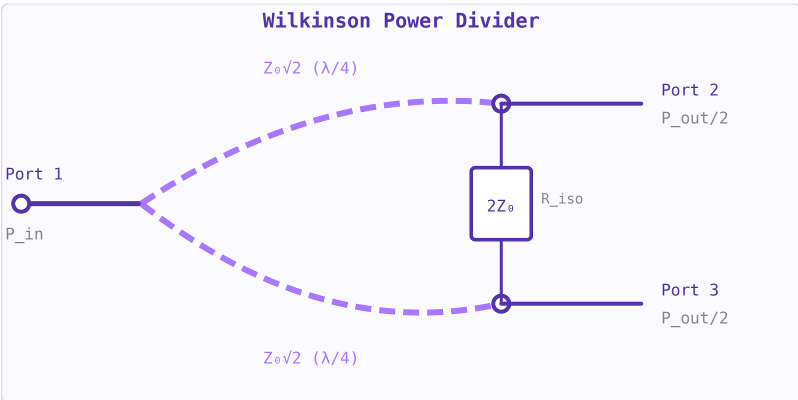

Figure 23.1: Equal-split Wilkinson divider: two $\lambda/4$ arms at $\mathrm{Z}_0\sqrt{2} \approx 70.7\ \Omega$, isolation resistor $2\mathrm{Z}_0 = 100\ \Omega$ (for 50 Ω system).

$$Z_{arm} = Z_0\sqrt{2}, \quad R_{iso} = 2Z_0$$

Key properties: all three ports are simultaneously matched; the output ports are isolated from each other (signals applied to port 2 are absorbed by R and do not appear at port 3); the divider is lossless when port 2 and port 3 signals are in phase (the isolation resistor carries no current).

23.2 Unequal-Split Wilkinson

For a power split ratio $K^2 : 1$ (port 2 gets K^2 times more power than port 3):

$$Z_1 = Z_0\sqrt{\frac{1+K^2}{K^3}}, \quad Z_2 = Z_0\sqrt{K(1+K^2)}, \quad R_{iso} = Z_0\left(K + \frac{1}{K}\right)$$

For K=1 (equal split): $Z_1 = Z_2 = Z_0\sqrt{2}$, $R = 2Z_0$. For K=2 (6 dB more to port 2): $Z_1 \approx 35.4\ \Omega$, $Z_2 \approx 112\ \Omega$, $R = 125\ \Omega$ (in 50 Ω system).

23.3 N-Way Dividers

An N-way Wilkinson divider can be built by cascading 2-way sections or using a star junction with N arms. The N-way lumped design uses N $\lambda/4$ arms of impedance $Z_0\sqrt{N}$ and isolation resistors NZ_0 between adjacent output ports.

23.4 Hybrid Couplers

A 90° hybrid (branch-line coupler) is a four-port device where the two outputs are 90° apart in phase. It is used in balanced amplifiers and I/Q mixers. A 180° hybrid (rat-race or magic T) gives outputs that are either in-phase or anti-phase, used for sum-difference networks and balanced mixers.

Hybrid coupler types and phase splits

Type	Phase split	BW	Use
Wilkinson	0°	N/A	in-phase split
90° Hybrid	0°/90°	~20%	I/Q mixers, balanced amps
180° Rat-race	0°/180°	~20%	Balanced mixers, Σ/Δ
Lange Coupler	0°/90°	octave	Wideband balanced amps

Figure 23.2: Comparison of common RF power divider and hybrid coupler types.

Device	Phase split	Bandwidth	Use
Wilkinson divider	0° (in-phase)	Moderate (~20%)	Power splitting, combining
90° hybrid coupler	0°/90°	~20%	Balanced amps, I/Q mixers
180° rat-race	0°/180°	~20%	Balanced mixers, sum/difference
Lange coupler	0°/90° (3 dB)	Wide (octave)	Wideband balanced amps

23.5 Coherent Power Combining

When N amplifiers feed a combining network, the combined output power is:

$$P_{out} = \eta N P_{each}$$

where η is the combining efficiency (0.93–0.98 for corporate networks). The gain over a single amplifier is $10\log_{10}(N) + 10\log_{10}(\eta)$ dB. If k amplifiers fail, coherent combining means power drops as $(N-k)^2/N^2$ of the full output — graceful degradation rather than total failure.

23.6 Physical Implementation

- **PCB microstrip**: The most common implementation. $\lambda/4$ at 2.45 GHz on FR4 ($\varepsilon_r \approx 4.4$) is about 18–20 mm long.

- **Lumped element**: Below ~1 GHz, LC equivalents replace transmission line sections, saving board space.
- **SMD Wilkinson**: Available as pre-packaged components from manufacturers at fixed frequencies.

Chapter 24

BALUNs and Hybrids

A BALUN (BALanced-UNbalanced) converts between single-ended (coaxial) and balanced (differential) connections. Hybrids are 4-port networks that split power with a defined phase relationship. Both are critical building blocks in RF systems.

24.1 Why BALUNs are Needed

A dipole antenna is a balanced structure (symmetric about its feed point). Connecting it directly to coaxial cable (unbalanced) allows current to flow on the cable outer conductor, distorting the radiation pattern and causing interference. A BALUN prevents this by presenting high common-mode impedance.

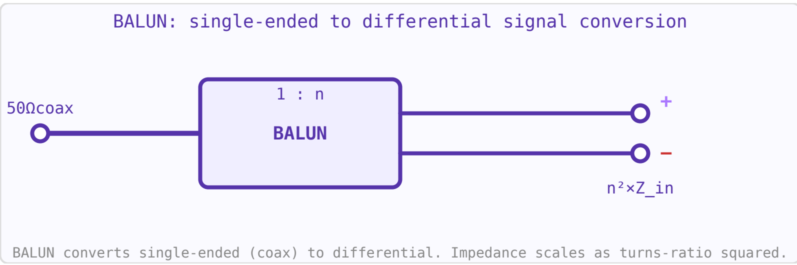

Figure 24.1: BALUN circuit symbol. Impedance ratio n^2:1 allows simultaneous conversion and impedance transformation.

24.2 LC BALUN

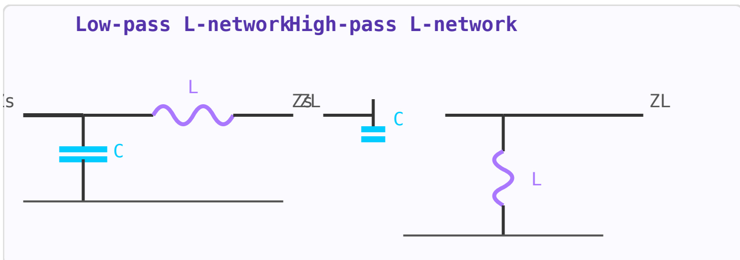

Figure 24.2: L-network as a BALUN building block. Two reactive elements provide impedance transformation and unbalanced-to-balanced conversion.

Uses inductors and capacitors to provide the balanced-to-unbalanced conversion with impedance transformation. The lattice topology uses four elements in a bridge circuit, providing $\pm 45°$ phase shift on the two balanced outputs and excellent amplitude balance over a moderate bandwidth.

24.3 Marchand BALUN

Two coupled quarter-wave resonators provide wideband balun action (often 2-3 octaves). The open-circuited stubs resonate at f_0 and present the appropriate impedance transformation. Widely used in wideband power amplifiers, mixers, and antenna feeds.

24.4 90° Hybrid (Branch-Line)

A 4-port network where power splits equally between two output ports with a 90° phase difference. Used in quadrature modulators, I/Q signal processing, and phased array feed networks. Implemented as a square of four $\lambda/4$ transmission line sections.

24.5 180° Hybrid (Rat-Race / Magic-T)

Splits power equally with 0° or 180° phase shift. Used in balanced amplifiers, mixers, and Butler matrices for phased array beamforming.

Chapter 25

Microwave Passive Components

Beyond transmission lines and filters, a range of passive components are essential in microwave systems: couplers, dividers, circulators, and attenuators.

25.1 Directional Couplers

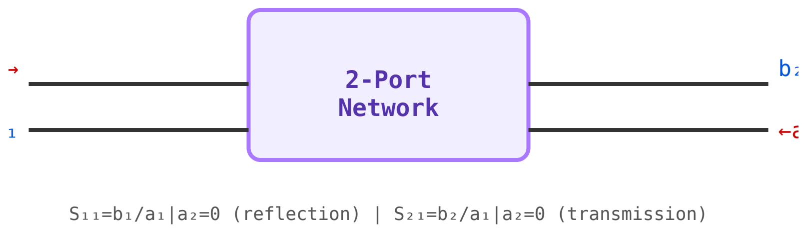

Figure 25.1: A directional coupler is a 4-port device — extending the 2-port S-matrix concept to include forward and reverse coupled ports.

A directional coupler samples a fraction of the forward or reverse traveling wave on a transmission line without significantly disturbing it. Key parameters: coupling factor (C, dB), directivity (D, dB), and insertion loss. Used for power monitoring, signal sampling, and reflectometer circuits in VNAs.

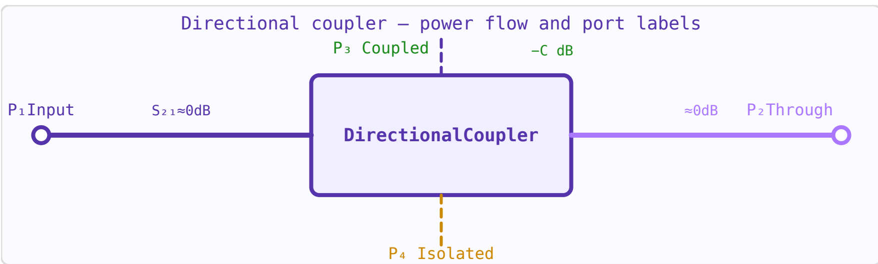

Figure 25.2: Directional coupler: power flows port 1→2 (through, ≈0 dB loss) and 1→3 (coupled, −C dB). Port 4 is isolated.

25.2 Power Dividers / Combiners

A Wilkinson power divider splits input power equally into two output ports (−3 dB each) while maintaining isolation between the two outputs. The isolation resistor absorbs any power imbalance between ports. Used in antenna arrays, balanced amplifiers, and mixer feeds.

25.3 Circulator

A three-port non-reciprocal device: power entering port 1 exits port 2; power entering port 2 exits port 3; power entering port 3 exits port 1. Uses magnetised ferrite. Critical applications: isolating a PA from antenna reflections, separating transmit and receive paths (T/R switch replacement), and protecting LNAs from high-power pulses in radar.

25.4 Attenuators

Fixed attenuators (pads) reduce signal power by a known amount while maintaining Z_0 impedance. Pi and T resistive networks are common topologies. Used for impedance bridging, reducing reflections, and setting test signal levels. Accuracy and flatness with frequency are key specifications.

25.5 PIN Diode Switches

PIN diodes operate as variable resistors at RF: forward biased → low resistance (ON); reverse biased → high resistance (OFF). Used for fast antenna switching, T/R switching in radar, and variable attenuators. Switching speed: nanoseconds to microseconds depending on carrier lifetime.

Chapter 26

RF Filters

RF filters are two-port networks that pass signals in certain frequency bands while attenuating others. They are fundamental in every RF system for channel selection, harmonic suppression, and image rejection.

26.1 Filter Types

- **Low-pass (LPF)** — passes below cutoff fc; used for harmonic suppression, anti-aliasing
- **High-pass (HPF)** — passes above fc; used to block DC or low-frequency interference
- **Band-pass (BPF)** — passes a band centred on f_0; used for channel selection
- **Band-stop / Notch** — rejects a specific band; used to suppress interferers

26.2 Butterworth Response

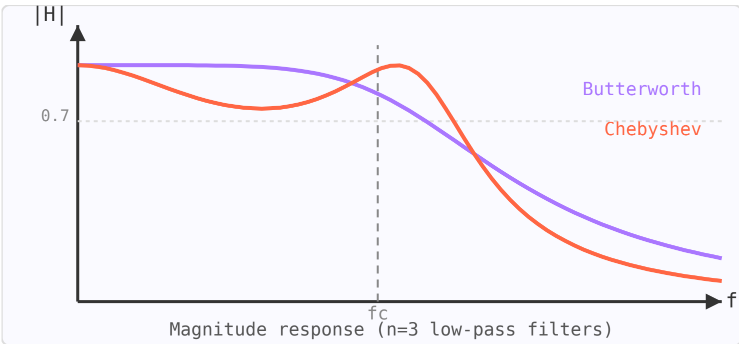

Figure 26.1: n=3 low-pass filter responses: Butterworth (flat passband, purple) vs Chebyshev (steeper roll-off, orange).

Maximally flat magnitude in the passband — no ripple. The magnitude response is:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^{2n}}}$$

Roll-off: 20n dB/decade beyond ω_c. Gentle transition band.

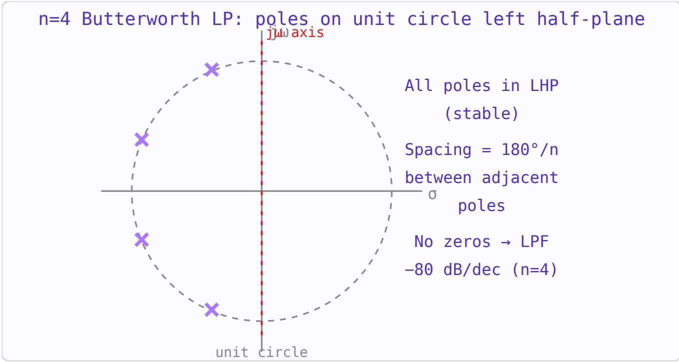

Figure 26.2: Butterworth LP pole-zero plot. n=4 poles equally spaced on the left half of the unit circle in the s-plane.

26.3 Chebyshev Response

Allows equiripple in the passband in exchange for a steeper transition band. For the same order, Chebyshev gives more attenuation in the stopband than Butterworth. The ripple (dB) is a design parameter.

26.4 LC Ladder Synthesis

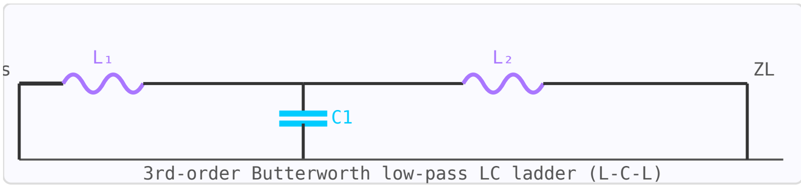

Figure 26.3: 3rd-order LC ladder low-pass filter between source and load impedances.

Prototype low-pass filter element values g_k are tabulated for Butterworth and Chebyshev responses. These normalised values are frequency- and impedance-scaled to the target cutoff frequency and port impedance using:

$$L_k = \frac{g_k R_0}{\omega_c} \quad C_k = \frac{g_k}{R_0 \omega_c}$$

Chapter 27

Stepped-Impedance Low-Pass Filters

A stepped-impedance low-pass filter (SI-LPF) replaces the lumped inductors and capacitors of a conventional LC ladder with sections of high-impedance and low-impedance transmission line. This approach is practical on PCB microstrip where lumped components at GHz frequencies behave poorly, and no vias or discrete parts are needed.

27.1 Operating Principle

A short section of **high-impedance** line (narrow trace, small physical width) behaves as a series inductor because its shunt capacitance is small relative to its series reactance. A short section of **low-impedance** line (wide trace) behaves as a shunt capacitor. Alternating these sections realises the LC ladder prototype.

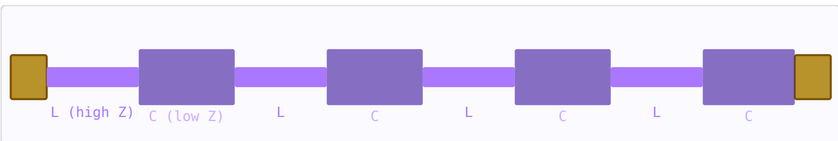

Figure 27.1: Top view of a stepped-impedance LPF: narrow (high-Z, purple) and wide (low-Z, dark) alternating sections.

27.2 Element Mapping

The prototype g-values are mapped to electrical lengths using the system impedance Z_0, high-line impedance Zh, and low-line impedance Zl:

$$\beta\ell_{\text{high}} = \frac{g_k\, Z_0}{Z_h} \qquad \text{(series inductor element)}$$

$$\beta\ell_{\text{low}} = \frac{g_k\, Z_l}{Z_0} \qquad \text{(shunt capacitor element)}$$

Here $\beta\ell$ is the electrical length in radians, evaluated at the cutoff frequency fc. The physical length is:

$$\ell = \frac{\beta\ell}{2\pi} \cdot \frac{c}{f_c\,\sqrt{\varepsilon_{\text{eff}}}}$$

27.3 Prototype g-Values

Use standard Butterworth or Chebyshev low-pass prototype tables. For a 3rd-order Butterworth: $g_1 = 1.0$, $g_2 = 2.0$, $g_3 = 1.0$. For Chebyshev 0.5 dB ripple, 5th-order: $g_1 = 1.7058$, $g_2 = 1.2296$, $g_3 = 2.5408$, $g_4 = 1.2296$, $g_5 = 1.7058$. Higher order gives steeper roll-off but longer board area.

27.4 Design Rules

- Choose Zh/Z_0 as high as fabrication allows (typically Zh = 80–120 Ω for 50 Ω system)
- Choose Zl as low as practical (typically Zl = 15–30 Ω); limited by board width constraints
- Higher Zh/Zl ratio → shorter sections → better approximation to lumped elements
- All sections must have electrical lengths $\beta\ell < 45°$ at the cutoff frequency for the lumped-element approximation to be valid
- Spurious passband appears at 2fc (or 4fc for symmetric designs); stepped-Z is not suitable above 2–3× cutoff

27.5 Microstrip Trace Width

The required trace width for a target impedance Z on a substrate of thickness h and permittivity εr is found by the inverse Hammerstad-Jensen formulas (implemented in the Microstrip Calculator). As a rough guide: for 50 Ω on FR4 (εr ≈ 4.4, h = 1.6 mm), w ≈ 3 mm; for 100 Ω, w ≈ 0.5 mm; for 25 Ω, w ≈ 10 mm.

Part V

Noise, Amplifiers and Linearity

Chapter 28

Noise in RF Systems

Noise ultimately limits the sensitivity of any RF receiver. Understanding noise sources, how they combine in a cascaded system, and how to minimise their contribution is fundamental to receiver design.

28.1 Thermal (Johnson-Nyquist) Noise

Any resistor at temperature T generates noise power due to thermal agitation of electrons. The available noise power in bandwidth B is:

$$P_n = kTB$$

At room temperature (290 K): $kT = -174$ dBm/Hz. This is the noise floor of any passive system at room temperature.

28.2 Noise Figure

Noise Figure (NF) is the degradation of SNR through a component, referenced to a 290 K source:

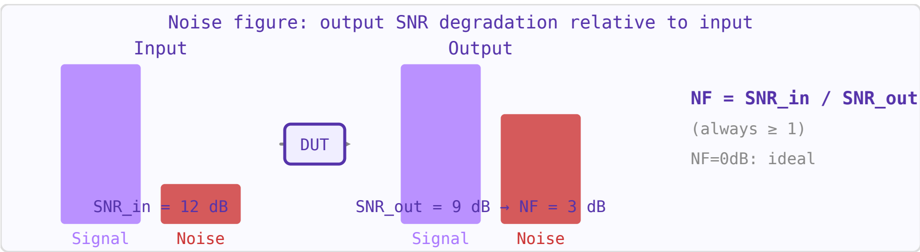

Figure 28.1: Noise figure quantifies how much a device degrades SNR. A 3 dB NF device adds as much noise as a matched resistor at 290 K.

$NF = SNR_{in} - SNR_{out}$ *(dB)*

A perfect noiseless amplifier has NF = 0 dB. A 50 Ω resistive attenuator of X dB has NF = X dB.

28.3 Friis Formula for Cascaded Systems

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

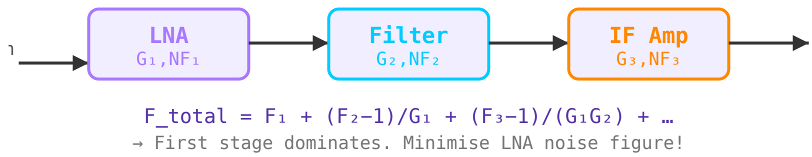

Figure 28.2: Cascaded receiver chain. LNA noise figure dominates because later stages are divided by preceding gains.

The first stage dominates. This is why an LNA with low NF and high gain at the front of a receive chain is critical. Lossy elements (cables, switches) before the LNA directly add to system NF.

28.4 System Noise Temperature

Equivalent noise temperature $T_e = (F - 1) \times 290$ K. Useful for low-noise systems such as radio telescopes and satellite receivers where temperatures of a few kelvin are achieved.

Chapter 29

RF Amplifiers

RF amplifiers increase signal power. Key specifications include gain, noise figure, output power, linearity, and stability. Different applications demand very different tradeoffs: a receiver LNA optimises noise figure; a transmit PA optimises efficiency and output power.

29.1 Low-Noise Amplifier (LNA)

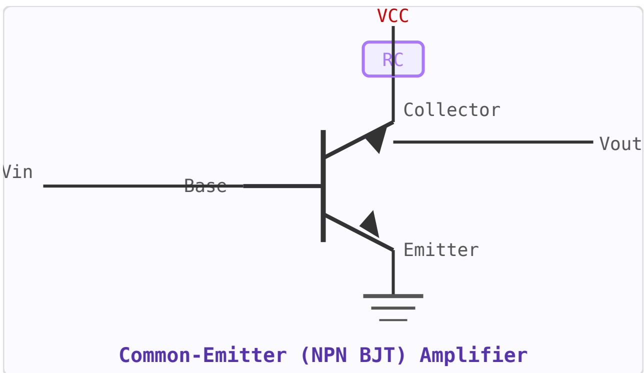

Figure 29.1: Common-emitter NPN amplifier. RC = collector load; voltage divider sets quiescent base current.

The first active stage in a receiver chain. Its noise figure dominates the overall system NF (Friis formula), so minimising LNA NF is critical. The input matching network is designed for noise match rather than power match. Typical specifications: NF = 0.3–1.5 dB, gain = 15–25 dB.

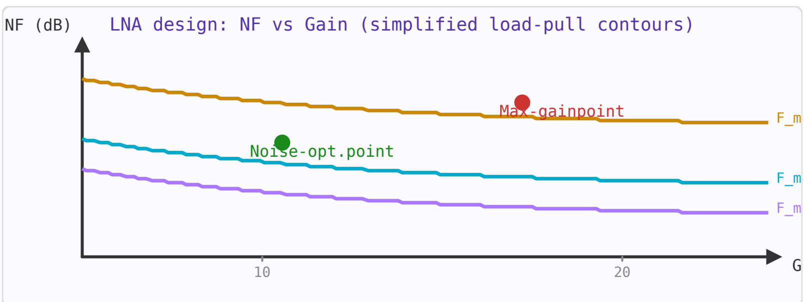

Figure 29.2: LNA tradeoff: minimum NF and maximum gain are different operating points. For first-stage LNA, prioritise NF.

29.2 Power Amplifier (PA)

The final stage of a transmitter. PA efficiency (PAE — power added efficiency) is critical for battery life in mobile devices. Class A amplifiers have low distortion but poor efficiency (~25–30%). Class B/AB amplifiers improve efficiency (~50–60%). Class E/F switching amplifiers can achieve theoretical efficiencies above 90%.

29.3 Gain and Stability

Transistor amplifiers are potentially unstable at certain frequencies due to internal feedback. The Rollett stability factor $K > 1$ (and \(|\Delta| < 1\)) is the condition for unconditional stability. Amplifiers are often stabilised using resistive loading or feedback networks.

29.4 Linearity: P1dB and IP3

The 1 dB compression point (P1dB) is the input power at which the gain has dropped by 1 dB from its small-signal value. The third-order intercept point (IP3) characterises intermodulation distortion — the spurious mixing products generated when two closely spaced signals both enter the amplifier. As a rule of thumb, $IIP3 \approx P1dB + 10$ dBm.

Chapter 30

Amplifier Stability

An RF amplifier can oscillate if the reflection coefficient seen looking into either port exceeds unity magnitude for some source or load impedance. Stability analysis identifies whether this can happen using the device's S-parameters, without having to check every possible termination.

30.1 S-Parameter Determinant (Δ)

The first quantity to compute is the two-port determinant:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

This combines all four S-parameters into a single complex number whose magnitude appears in both the K-factor and μ-factor criteria.

30.2 Rollett Stability Factor K

The Rollett K-factor was the first widely adopted stability criterion (1962):

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2\,|S_{12}|\,|S_{21}|}$$

The device is **unconditionally stable** (stable for any passive source and load) when both:

$$K > 1 \quad \text{AND} \quad |\Delta| < 1$$

Both conditions must hold simultaneously. K > 1 alone is necessary but not sufficient. When K < 1 the device is only conditionally stable — it may oscillate with certain terminations.

30.3 μ-Factor (Edwards-Sinsky)

The μ-factor (1992) provides a single-parameter unconditional stability test that is both necessary and sufficient:

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta\,S_{11}^{*}| + |S_{12}\,S_{21}|}$$

The device is unconditionally stable if and only if $\mu > 1$. A larger μ indicates a greater stability margin. The output-port version μ' uses S22 in the numerator and S11 in the denominator symmetrically.

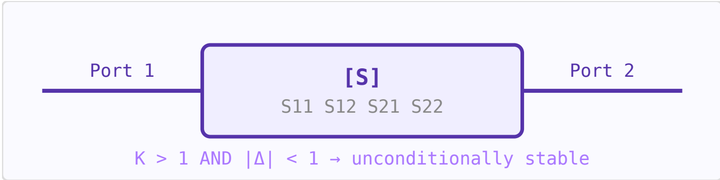

Figure 30.1: Two-port device described by S-parameters. Stability is assessed from S11, S12, S21, S22 alone.

30.4 Maximum Available Gain (MAG)

When the device is unconditionally stable (K > 1, $|\Delta| < 1$), there exists an optimal simultaneous conjugate match at both ports. The resulting maximum available gain is:

$$G_{\text{MAG}} = \left|\frac{S_{21}}{S_{12}}\right| \left(K - \sqrt{K^2 - 1}\right)$$

30.5 Maximum Stable Gain (MSG)

When K < 1 (conditionally stable), MAG is undefined. The maximum stable gain is the theoretical upper bound achievable by adding loss to bring K $= 1$:

$$G_{\text{MSG}} = \left|\frac{S_{21}}{S_{12}}\right|$$

MSG is always higher than the achievable gain in practice — it is an optimistic upper bound used to compare devices before choosing stabilisation resistors.

30.6 Practical Notes

- S-parameters are frequency-dependent — check stability across the entire frequency range, not just the operating band
- Low-noise transistors often have K < 1 at low frequencies; add series or shunt resistors to stabilise out-of-band
- Stability circles on the Smith Chart show the boundary of terminations that cause $|\Gamma\text{in}| = 1$ or $|\Gamma\text{out}| = 1$
- A device with $\mu = 1.5$ at a given frequency has 50% more stability margin than one with $\mu = 1.01$

Chapter 31

Power Amplifier Design

Power amplifiers (PAs) are among the most challenging RF circuits to design because efficiency and linearity pull in opposite directions. A perfectly linear PA wastes most of its supply power as heat. A highly efficient PA distorts the signal. Modern PA design — especially for cellular and wireless standards — is the art of managing this tradeoff.

31.1 PA Classes

PA classes are defined by the transistor's conduction angle — the fraction of the RF cycle during which the device conducts current:

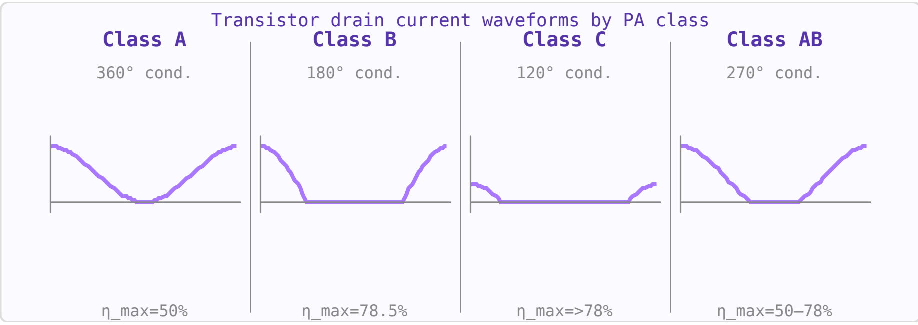

Figure 31.1: Drain current waveforms by amplifier class. Less conduction → higher efficiency but more distortion.

Class	**Conduction angle**	**Max efficiency**	**Linearity**	**Use**
A	360° (always on)	50%	Excellent	Small-signal, wideband, instrumentation
AB	180°–360°	50–78%	Good	Most practical linear RF PAs (LTE, Wi-Fi)
B	180°	78.5%	Moderate	Push-pull audio and some RF
C	$< 180°$	$> 78.5\%$	Poor	Narrowband constant-envelope (FM, CW radar)
D	Switch	100% (ideal)	Needs filtering	Audio switching amplifiers, some RF

E	Switch + resonant	100% (ideal)	Needs filtering	High efficiency narrowband RF
F	Resonant harmonics	100% (ideal)	Narrowband	High efficiency RF (mobile base stations)

31.2 Drain Efficiency vs Power-Added Efficiency (PAE)

$$\eta_{drain} = \frac{P_{out}}{P_{DC}} \qquad \text{PAE} = \frac{P_{out} - P_{in}}{P_{DC}}$$

PAE is the more meaningful figure for RF PAs because it accounts for the drive power required. For a PA with 15 dB gain, the difference between the two is small. For a PA with only 6 dB gain (common at mmWave), PAE is significantly lower than drain efficiency.

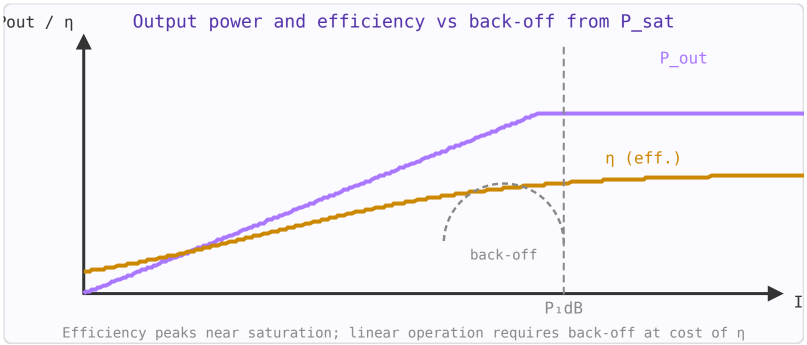

Figure 31.2: PA output power (blue) and efficiency (orange) vs input drive. P_1dB marks onset of compression.

31.3 1-dB Compression and Saturation

As input power increases, the output eventually saturates. The **P_1dB** (1-dB compression point) is where the gain has dropped by 1 dB from its small-signal value. Above P_1dB the device is nonlinear and generates harmonics and intermodulation products. For linear operation, PAs are backed off from P_1dB by 6–10 dB, accepting lower efficiency.

31.4 Linearity Requirements by Modulation

Modulation	PAPR	Typical back-off	Notes
CW / FM	0 dB	0 dB	Constant envelope — no linearity needed
BPSK/QPSK	3–4 dB	3–6 dB	Low PAPR, tolerates moderate nonlinearity
16-QAM	6–7 dB	6–8 dB	Moderate requirements
64-QAM	8–9 dB	8–10 dB	LTE typical; high back-off needed
OFDM (Wi-Fi/5G)	10–12 dB	10–12 dB	High PAPR; DPD or Doherty needed for efficiency

PAPR = Peak-to-Average Power Ratio.

31.5 Doherty Architecture

The Doherty PA is the dominant architecture for modern base station and handset PAs because it maintains high efficiency at back-off. It uses two amplifiers — a "main" (carrier) PA that operates in class AB/B, and a "peaking" PA that only turns on near peak power. At back-off, only the main PA operates (high efficiency). At peak power, both combine, restoring efficiency. A Doherty PA can achieve $> 50\%$ efficiency at 6 dB back-off, compared to 12–15% for a class A PA.

31.6 Digital Predistortion (DPD)

DPD applies an inverse of the PA's nonlinearity digitally, before the signal reaches the PA, so that the output is linear. It requires a feedback path to measure and adapt to the PA's actual behaviour. DPD allows a PA to operate closer to compression (higher efficiency) while meeting EVM and spectral mask requirements. It is standard in modern cellular base stations.

31.7 Output Matching

PA transistors are typically low-impedance devices that need an output matching network to present the optimal load impedance for maximum power and efficiency. The load line resistance $R_{opt} = V_{dd}^2/(2P_{out})$ sets the ideal load for a class A stage. For a 28 V, 10 W PA: $R_{opt} = 784/20 = 39.2\ \Omega$ — conveniently near 50 Ω, which is why GaN on 28 V supply is so popular for base station PAs.

Chapter 32

Intermodulation Distortion and IP3

When two or more signals pass through a nonlinear device (an amplifier, mixer, or any component with a nonlinear transfer function), new frequency components are created. These intermodulation (IM) products can fall within the passband of the system and cannot be removed by filtering. Third-order intermodulation (IM3) is the dominant concern in most RF receivers and transmitters.

32.1 Why Nonlinearity Matters

A perfectly linear device produces outputs only at its input frequencies. A nonlinear device with input $x(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t)$ produces output terms at frequencies $n\omega_1 \pm m\omega_2$ for all integers n, m. The third-order products at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ are close to the original signals and fall in-band.

32.2 Third-Order Intercept Point (IP3)

In a two-tone test, both the desired output and the IM3 products are plotted as a function of input power. On a log-log scale, the desired output has slope 1 (1 dB output per 1 dB input); the IM3 products have slope 3 (3 dB output per 1 dB input). Extrapolating these two lines, they intersect at the **IP3 point**.

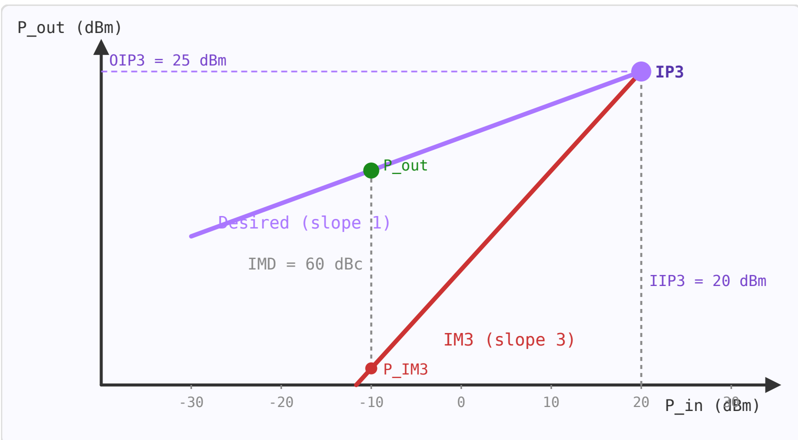

Figure 32.1: Output power vs input power. Signal: slope 1. IM3 products: slope 3. They extrapolate to the IP3 intercept point.

$$P_{IM3,out} = 3P_{in} - 2 \cdot IIP3 + G \qquad OIP3 = IIP3 + G$$

where all powers are in dBm and G is the gain in dB. The OIP3 is typically 10–15 dB above the 1 dB compression point.

32.3 Input and Output IP3

Parameter	Symbol	Typical values	Notes
Input IP3	IIP3	-10 to $+40$ dBm	Referred to device input
Output IP3	OIP3	0 to $+50$ dBm	OIP3 = IIP3 + Gain
1 dB compression	P_1dB	$\approx$ IIP3 $-$ 9.6 dB	Empirical approximation
Low-noise amplifier	IIP3	0 to $+15$ dBm	Trades with noise figure
Power amplifier	OIP3	P_1dB + 10 dB	Backed off from saturation
Passive mixer	IIP3	$+15$ to $+25$ dBm	Higher than active

32.4 IMD Ratio & Dynamic Range

The intermodulation distortion ratio (IMD) is the power difference between the desired signal and the IM3 product at the output:

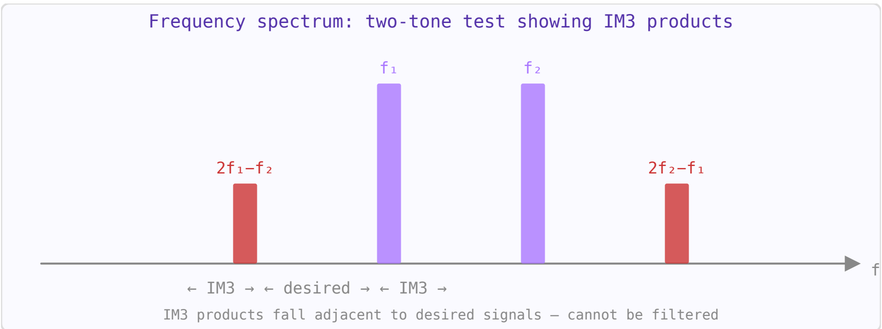

Figure 32.2: Two-tone test: input tones f_1 and f_2 (purple) generate IM3 products at $2f_1-f_2$ and $2f_2-f_1$ (red) close to the passband.

$$\mathrm{IMD} = 2(IIP3 - P_{in})\ \mathrm{dBc}$$

The spurious-free dynamic range (SFDR) is the input power range over which the system can operate without IM3 products exceeding the noise floor:

$$\mathrm{SFDR} = \tfrac{2}{3}(IIP3 - P_{noise})$$

32.5 Cascaded IP3

For a chain of stages, the cascaded IIP3 is dominated by later stages (because the signal is amplified). The reciprocal rule gives:

$$\frac{1}{IIP3_{total}} = \frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1 G_2}{IIP3_3} + \cdots$$

where all quantities are in linear (not dB) units. This is why the last high-gain stage in a receiver chain typically limits linearity.

32.6 Practical Implications

- **LNA IIP3**: typically +5 to +15 dBm. Must handle the full received signal range without generating IM3 products that mask weak desired signals.
- **PA linearity**: power amplifiers operate backed off from their P_1dB to maintain acceptable IM3. ACPR (adjacent channel power ratio) is the PA equivalent of IMD.
- **Mixer IIP3**: passive mixers (diode or FET ring) have higher IP3 than active mixers but higher conversion loss.

Part VI

Signal Generation and Frequency Conversion

Chapter 33

Oscillators

An oscillator generates a periodic signal without an external input. In RF systems, oscillators provide the local oscillator (LO) signal for frequency conversion and the carrier signal for transmitters. Phase noise (short-term frequency instability) is the key quality metric.

33.1 Oscillation Conditions

The Barkhausen criterion: a feedback amplifier oscillates when loop gain $|A\beta| = 1$ and loop phase shift is 0° (or 360°). Common topologies: Colpitts (capacitive voltage divider), Hartley (inductive divider), Clapp, Pierce (crystal oscillators).

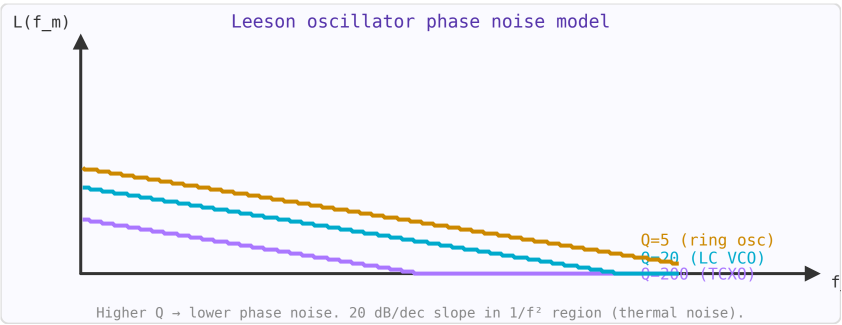

Figure 33.1: Leeson's model: phase noise L(f_m) for oscillators with different Q factors. Higher Q = lower noise at all offsets.

33.2 Phase Noise

Phase noise $\mathcal{L}(f_m)$ (dBc/Hz) describes the spectral purity of an oscillator — the noise power in a 1 Hz bandwidth at offset f_m from the carrier, relative to the carrier power. The Leeson model:

$$\mathcal{L}(f_m) = 10 \log \left[\frac{2FkT}{P_s} \left(1 + \frac{f_0^2}{4Q_L^2 f_m^2} \right) \right]$$

Phase noise decreases with increasing loaded Q (Q_L) and oscillator power (P_s).

33.3 Voltage-Controlled Oscillator (VCO)

A VCO is an oscillator whose frequency is tuned by a control voltage, via a varactor diode (voltage-variable capacitor). Tuning sensitivity K_v (MHz/V) characterises the VCO. Higher

K_v increases frequency range but also increases phase noise sensitivity to supply noise.

33.4 Phase-Locked Loop (PLL)

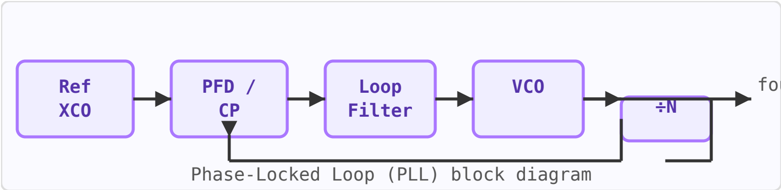

Figure 33.2: PLL block diagram: VCO locked to reference crystal via phase-frequency detector and loop filter.

A PLL phase-locks a VCO to a stable reference oscillator (usually a crystal), allowing the VCO's tunable frequency range with the reference's spectral purity. Components: phase-frequency detector (PFD), charge pump, loop filter, VCO, frequency divider (÷N). Within the loop bandwidth, VCO phase noise is suppressed to near the reference noise floor.

Chapter 34

Phase Noise and Jitter

Phase noise and jitter are two equivalent descriptions of timing uncertainty in an oscillator or clock signal. Phase noise describes instability in the frequency domain — the spectral purity of an oscillator. Jitter describes it in the time domain — the variation of edge transitions from their ideal positions. They are related by an integral transformation.

34.1 Phase Noise Definition

The single-sideband (SSB) phase noise L(f) is defined as the ratio of power in a 1 Hz bandwidth at offset f from the carrier to the total carrier power:

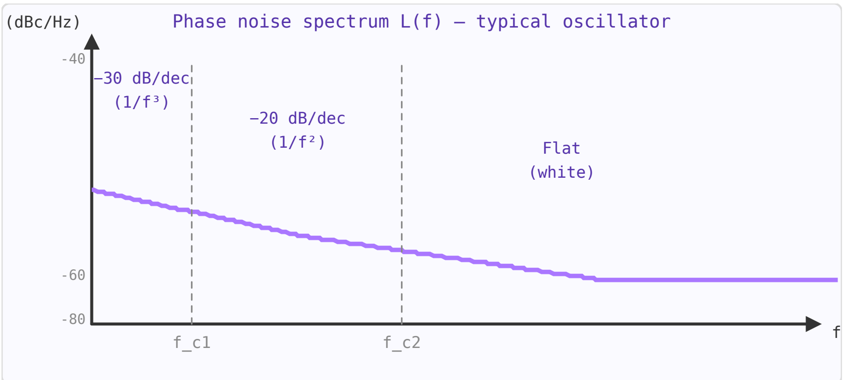

Figure 34.1: Phase noise L(f) showing three regions: $1/f^3$ (flicker FM), $1/f^2$ (white FM / Leeson), and flat (white phase noise floor).

$$\mathcal{L}(f) = \frac{P_{noise,SSB}(f_0 + f,\, 1\text{ Hz BW})}{P_{carrier}} \quad [\text{dBc/Hz}]$$

Phase noise is plotted on a log-log scale. Typical regions:

Offset region	Slope	Dominant mechanism
Very close-in (< 10 Hz)	$1/f^3$ (−30 dB/decade)	Flicker FM noise
Close-in (10 Hz – 10 kHz)	$1/f^2$ (−20 dB/decade)	White FM, Leeson's model
Far-from-carrier	Flat (0 dB/decade)	White phase noise (thermal)

34.2 Leeson's Model

For a feedback oscillator with resonator Q factor and noise figure F:

$$\mathcal{L}(f_m) = 10\log_{10}\left[\frac{2FkT}{P_s}\left(\frac{f_0}{2Q_L f_m}\right)^2\right]$$

where f_m is the offset frequency, f_0 is the carrier frequency, Q_L is the loaded Q, P_s is the signal power, F is the amplifier noise figure (linear), k = 1.38×10^{-23} J/K (Boltzmann), and T is the temperature in Kelvin. Higher Q and higher signal power both reduce phase noise.

34.3 Phase Noise to Jitter Conversion

RMS jitter integrates the phase noise over a bandwidth from f_1 to f_2:

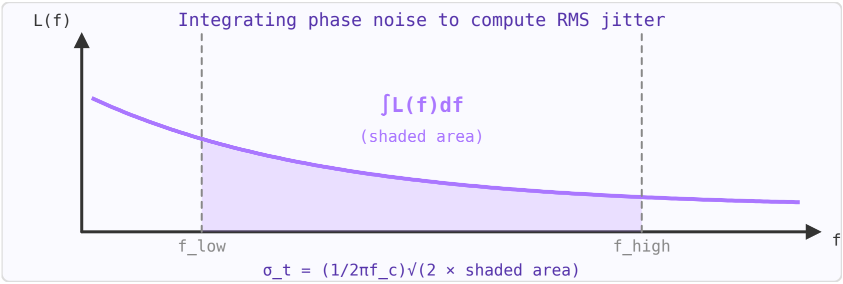

Figure 34.2: RMS jitter equals the square root of twice the integrated phase noise (linear) over the specified bandwidth.

$$\sigma_t = \frac{1}{2\pi f_0}\sqrt{2\int_{f_1}^{f_2}\mathcal{L}(f)\,df} \quad \text{[s RMS]}$$

where L(f) is in linear units (not dBc/Hz). The factor 2 accounts for double-sideband power. The integration limits define what jitter you care about — typically the Nyquist band for a data converter application, or the loop bandwidth for a PLL.

34.4 Typical Values

Oscillator type	**Phase noise @ 1 kHz**	**Phase noise @ 1 MHz**	**RMS jitter**
MEMS oscillator	−120 dBc/Hz	−160 dBc/Hz	200–500 fs
TCXO (100 MHz)	−130 dBc/Hz	−165 dBc/Hz	50–200 fs
OCXO (100 MHz)	−150 dBc/Hz	−170 dBc/Hz	10–50 fs
LC VCO (2.4 GHz)	−85 dBc/Hz	−135 dBc/Hz	1–5 ps
Ring oscillator (1 GHz)	−50 dBc/Hz	−100 dBc/Hz	10–50 ps

34.5 Jitter Types

- **Period jitter**: Variation of consecutive clock periods. Relevant for digital setup/hold timing.
- **Cycle-to-cycle jitter**: Difference between adjacent periods. Related to close-in phase noise.
- **Long-term jitter (accumulated)**: Deviation over many cycles. Related to far-from-carrier noise.
- **RMS jitter**: Standard deviation σ. Peak-to-peak jitter $\approx 6\sigma$ to 7σ at 10^{-12} BER.

34.6 Practical Impact

For an ADC sampling at frequency f_s, phase noise causes an SNR limit called the *aperture jitter limit*:

$$\mathrm{SNR}_{jitter} = -20\log_{10}(2\pi f_{in}\,\sigma_t)$$

At f_in = 1 GHz with σ_t = 100 fs, the jitter-limited SNR is only 24 dB — equivalent to a 4-bit ADC. This is why high-frequency ADC clock sources must have sub-100 fs jitter.

Chapter 35

Phase-Locked Loops and Frequency Synthesis

A phase-locked loop (PLL) is a feedback system that synchronises an oscillator to a reference signal in both frequency and phase. PLLs are ubiquitous in RF systems: they generate the local oscillator frequency in every radio receiver and transmitter, provide clock recovery in serial data links, and clean up noisy frequency sources.

35.1 PLL Building Blocks

Block	Function	Key parameter
Phase detector (PD) / PFD	Compares phase of reference and divided VCO output. Generates error signal.	Gain K_φ (A/rad or V/rad)
Charge pump (CP)	Converts PFD pulses to current. Charges loop filter.	Current I_CP (μA–mA)
Loop filter (LF)	Integrates CP output. Sets loop bandwidth and stability.	Bandwidth f_n, phase margin
VCO	Voltage-controlled oscillator. Output frequency $\propto$ control voltage.	Gain K_VCO (MHz/V)
Divider ÷N	Divides VCO frequency by N to match reference. N sets output frequency.	Division ratio N

35.2 How It Works

The PFD compares the phase of the reference (f_ref) to the divided VCO output (f_VCO/N). When they match, the loop is locked and f_VCO = N × f_ref. If the VCO drifts high, the PFD generates a correction signal that reduces the control voltage, pulling the frequency back. The loop bandwidth determines how fast this correction happens.

35.3 Open-Loop Transfer Function

$$G(s) = \frac{K_\phi I_{CP}}{2\pi} \cdot F(s) \cdot \frac{K_{VCO}}{s} \cdot \frac{1}{N}$$

where F(s) is the loop filter transfer function. The loop bandwidth f_n is approximately where |G(jω)| = 1. Phase margin should be 45–60° for stable, well-damped response.

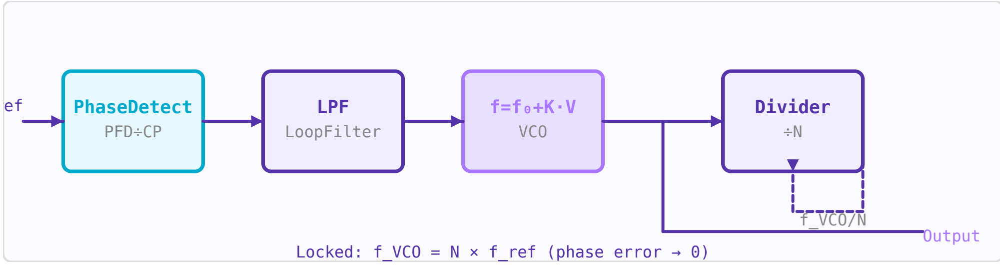

Figure 35.1: PLL block diagram: PFD compares f_ref to f_VCO/N; loop filter drives VCO to lock frequency and phase.

35.4 Lock Time and Bandwidth Tradeoff

The loop bandwidth controls a fundamental tradeoff:

- **Wide bandwidth**: Fast lock time, good VCO noise suppression, poor reference spur rejection, more reference noise passes through.
- **Narrow bandwidth**: Slow lock time, poor VCO noise suppression at close-in offsets, excellent reference spur rejection.

Typical loop bandwidths: 1–100 kHz for frequency synthesisers, 1–10 MHz for fast-hopping systems.

35.5 Phase Noise in a PLL

Inside the loop bandwidth, the PLL output tracks the reference and the output phase noise follows the reference noise amplified by $20{\cdot}\log_{10}(\mathrm{N})$. Outside the loop bandwidth, the output phase noise equals the free-running VCO phase noise. The total output phase noise is the worst of the two at each offset:

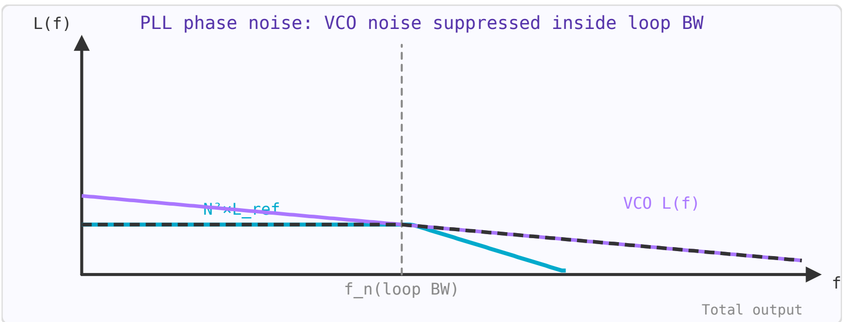

Figure 35.2: PLL output phase noise: inside loop BW = reference noise × N^2. Outside = free-running VCO noise. Total = worst of the two.

$$\mathcal{L}_{out}(f_m) \approx \begin{cases} \mathcal{L}_{ref}(f_m) + 20\log_{10} N & f_m \ll f_n \\ \mathcal{L}_{VCO}(f_m) & f_m \gg f_n \end{cases}$$

35.6 Integer-N vs Fractional-N

Architecture	Channel spacing	Phase noise	Reference spurs
Integer-N	= f_ref	20·log(N) above reference noise	Low
Fractional-N (Σ-Δ)	< f_ref (any)	Higher N → worse noise; Σ-Δ adds quantisation noise	Fractional spurs

Integer-N synthesis requires f_ref = channel spacing. For a 200 kHz GSM channel with a 200 kHz reference, the division ratio N can be 4000–9000 for 800–1800 MHz output — the $20 \cdot \log_{10}(N)$ penalty is 72–79 dB above the reference noise. Fractional-N uses a Σ-Δ modulator to rapidly switch between integer values of N, achieving fractional average division ratios and allowing wider f_ref for better phase noise.

35.7 Reference Spurs

Reference spurs are discrete spectral lines in the VCO output at offsets of ±f_ref (and harmonics) from the carrier. They arise from imperfect charge pump matching — leakage current causes small periodic control voltage ripples at the reference rate. Typical specification: < -70 dBc. Reference spurs cause interference to adjacent channels.

35.8 Practical PLL Design Steps

1. Choose f_ref = GCD of all required output frequencies (integer-N) or use fractional-N for fine frequency resolution.
2. Select VCO with adequate tuning range, low K_VCO (for low sensitivity to supply noise), and target phase noise.
3. Choose loop bandwidth: 1/10 of f_ref as a starting point to reject reference spurs.
4. Design a 3rd or 4th order loop filter (2 poles, 1–2 zeros) for 50° phase margin.
5. Simulate phase noise, lock time, and spur levels before fabricating.

Chapter 36

Mixers and Frequency Conversion

A mixer multiplies two signals in the time domain, producing sum and difference frequency components. This allows frequency conversion — moving signals from one frequency band to another. Mixers are at the heart of every superheterodyne radio, radar, and wireless communication system.

36.1 Basic Operation

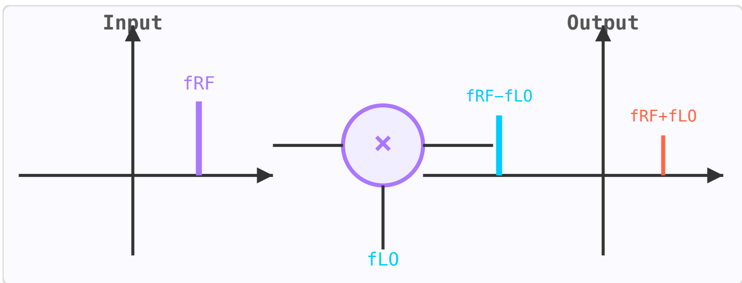

Figure 36.1: Mixer produces sum (fRF+fLO, orange) and difference (fRF−fLO, blue) frequency components.

An ideal mixer with RF input at f_{RF} and local oscillator (LO) at f_{LO} produces output at $f_{IF} = f_{RF} \pm f_{LO}$. The desired sideband is selected by the IF filter; the unwanted image frequency falls at $f_{LO} - f_{IF}$ (for a low-side LO) and must be rejected.

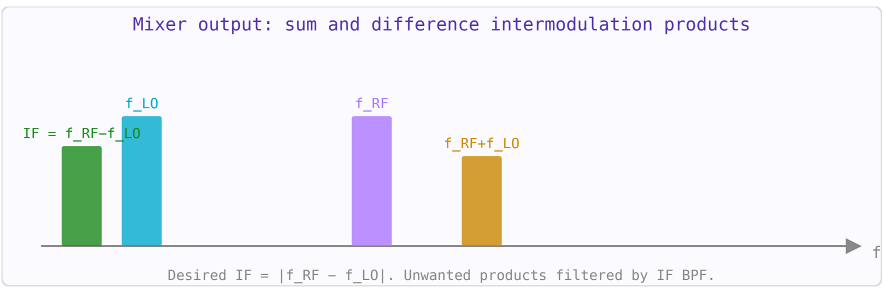

Figure 36.2: Mixing products: desired IF (green), input tones (blue/purple), and upper sideband (orange).

36.2 Conversion Loss and Noise Figure

Passive mixers (using diodes or FETs) have conversion loss — the IF output is weaker than the RF input. Typical conversion loss: 5–8 dB for a diode ring mixer. The NF of a passive mixer equals approximately its conversion loss. Active mixers provide conversion gain (0–15 dB) at the cost of higher DC power and potentially more distortion.

36.3 Image Rejection Mixer (IQ Mixer)

An IQ (in-phase/quadrature) mixer uses two mixers with LO signals 90° apart, producing in-phase (I) and quadrature (Q) baseband outputs. When the outputs are combined correctly, the image frequency is cancelled. Image rejection of 30–40 dB is achievable in practice; better rejection requires careful amplitude and phase balance.

36.4 Spurs and Isolation

Real mixers generate mixing products at $mf_{LO} \pm nf_{RF}$ (spurious responses or "spurs"). Mixer datasheets specify LO-to-RF, LO-to-IF, and RF-to-IF isolation to characterise feedthrough. Double-balanced mixers provide better isolation and spur rejection than single-ended designs.

Chapter 37

Modulation Schemes

Modulation encodes information onto a carrier wave by varying its amplitude, frequency, or phase. The choice of modulation scheme involves tradeoffs between spectral efficiency (bits/Hz), power efficiency (Eb/N0 required), and robustness to noise and interference.

37.1 Amplitude Modulation (AM)

The carrier amplitude varies with the message signal. Simple to demodulate but spectrally inefficient and susceptible to amplitude noise. Double-sideband suppressed carrier (DSB-SC) removes the carrier for power efficiency; single-sideband (SSB) halves the bandwidth.

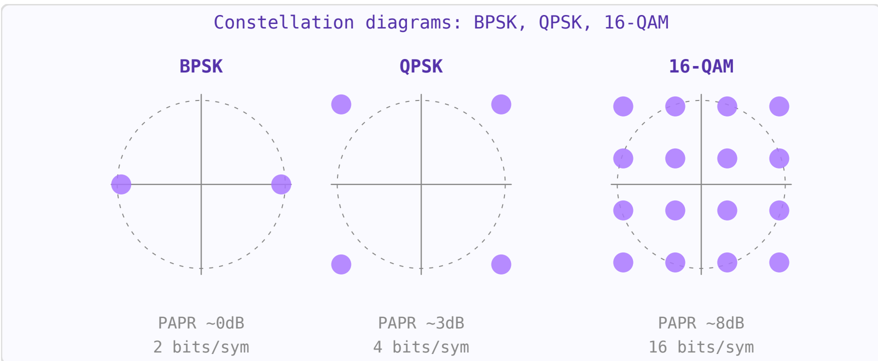

Figure 37.1: Constellation diagrams. More dots per unit area = higher spectral efficiency but requires better SNR.

37.2 Frequency Modulation (FM)

The carrier frequency varies with the message. Provides noise immunity above the FM threshold but requires wider bandwidth than AM. Used in broadcast radio ($\pm$75 kHz deviation) and VHF voice communications.

37.3 Phase-Shift Keying (PSK)

Digital data is encoded as discrete phase shifts. BPSK (2 phases, 1 bit/symbol) is robust; QPSK (4 phases, 2 bits/symbol) doubles spectral efficiency. Higher orders (8-PSK, 16-PSK) pack more bits per symbol but require higher SNR.

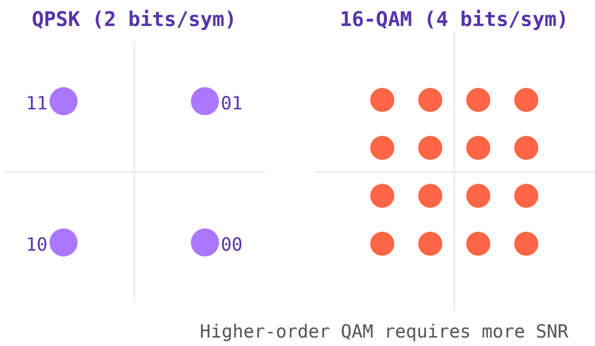

Figure 37.2: QPSK (4 symbols, 2 bits/sym) and 16-QAM (16 symbols, 4 bits/sym) constellation diagrams.

37.4 Quadrature Amplitude Modulation (QAM)

Combines amplitude and phase modulation. 16-QAM carries 4 bits/symbol; 64-QAM carries 6 bits/symbol; 256-QAM carries 8 bits/symbol. Used extensively in LTE, 5G, WiFi (802.11ax uses 1024-QAM), and cable TV. Higher QAM orders require much higher SNR and are sensitive to phase noise and nonlinearity.

37.5 OFDM

Orthogonal Frequency Division Multiplexing splits the channel into many narrow subcarriers, each modulated with QAM. The cyclic prefix guards against multipath inter-symbol interference. OFDM is the basis of WiFi (802.11a/g/n/ac/ax), LTE, 5G NR, and digital broadcast (DVB-T, DAB).

Part VII

Antennas and Propagation

Chapter 38

Antenna Fundamentals

An antenna is a transducer that converts between guided electromagnetic waves (in a transmission line) and free-space radiation. Every antenna has a reciprocal receive and transmit behaviour described by the same parameters.

38.1 Key Parameters

- **Gain (G)** — ratio of radiation intensity in the peak direction to that of an isotropic radiator, measured in dBi.
- **Directivity (D)** — same as gain but ignoring losses. $G = \eta \cdot D$ where η is radiation efficiency.
- **Beamwidth (HPBW)** — angular width of the main beam between half-power (-3 dB) points.
- **Bandwidth** — frequency range over which VSWR < 2 (or another specification).
- **Polarisation** — orientation of the E-field: linear, circular, or elliptical.
- **Radiation resistance** — equivalent resistance that would dissipate the same power as is radiated.

38.2 Half-Wave Dipole

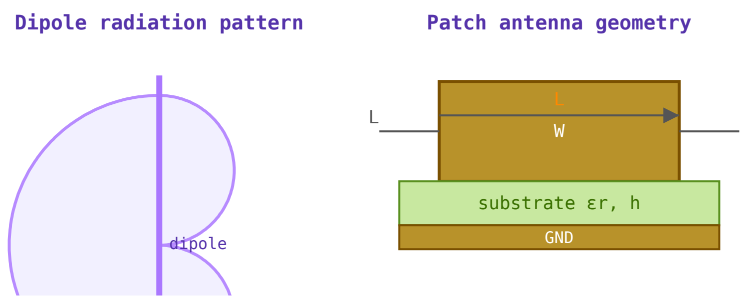

Figure 38.1: Left: dipole E-plane figure-8 radiation pattern (gain $\approx$ 2.15 dBi). Right: rectangular patch antenna geometry.

The reference antenna. Total length $\approx \lambda/2$ (slightly less due to end effects). Gain = 2.15 dBi. Radiation resistance $\approx 73\ \Omega$. Omnidirectional in the plane perpendicular to its axis.

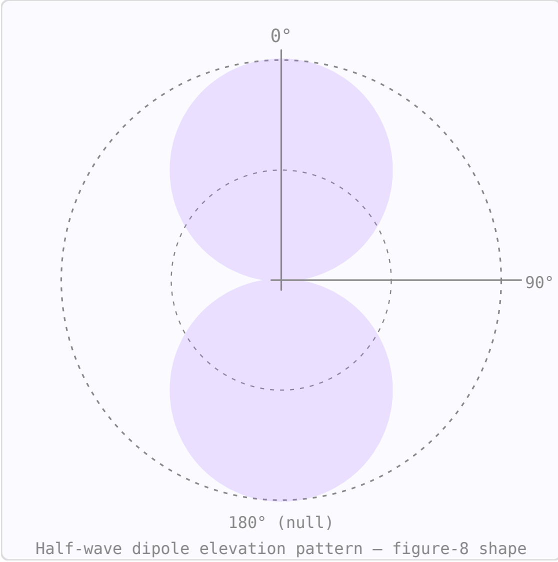

Figure 38.2: Dipole radiation pattern: maximum at broadside (90°), nulls at 0° and 180°.

38.3 Patch (Microstrip) Antenna

A rectangular conductor on a PCB substrate, resonant when the patch length L $\approx \lambda/(2\sqrt{\varepsilon}\text{reff})$. Gain $\approx$ 5–8 dBi, bandwidth 2–5% (can be increased with thicker substrates). Widely used in GPS, WiFi, and cellular handsets due to its flat, PCB-compatible form factor.

38.4 Array Factor

When N identical elements are arranged in a line with spacing d and progressive phase shift δ, the combined pattern is the element pattern multiplied by the array factor AF(θ). The main beam can be steered electronically by changing δ without physically moving the array — the principle of phased array radar and 5G beamforming.

Chapter 39

Antenna Polarisation

Polarisation describes the orientation of the electric field vector of the radiated electromagnetic wave. Mismatched polarisation between transmit and receive antennas causes signal loss, regardless of alignment. In MRI, the transmit/receive coil polarisation determines imaging efficiency.

39.1 Linear Polarisation

A half-wave dipole radiates a wave whose E-field oscillates in a fixed plane. A vertical dipole is vertically polarised. A horizontal dipole is horizontally polarised. Two dipoles at right angles with the same signal but 0° phase difference are co-polarised along their common axis; with 90° phase difference they produce circular polarisation.

39.2 Circular Polarisation (CP)

In circular polarisation, the E-field vector rotates at the carrier frequency. The rotation can be right-hand (RHCP — E rotates clockwise looking in the direction of propagation) or left-hand (LHCP). CP antennas are used for satellite links (independent of receiver orientation), GPS (RHCP), and MRI body coils (CP mode gives $\sqrt{2}$ more B_1 efficiency than linear).

Figure 39.1: RHCP: E-field rotates clockwise (CW) when viewed looking in direction of propagation. LHCP: counter-clockwise.

$$\text{Axial ratio (AR)} = \frac{|E_{major}|}{|E_{minor}|}$$

Perfect circular polarisation: AR = 1 (0 dB). Linear polarisation: AR = ∞. Typical CP antenna spec: AR $<$ 3 dB. A RHCP antenna rejects LHCP signals by the cross-polarisation isolation — ideally infinite, practically 20–30 dB.

39.3 Polarisation Mismatch Loss

When the polarisation of the incident wave does not match the antenna, received power is reduced. The polarisation mismatch loss (or polarisation efficiency η_p):

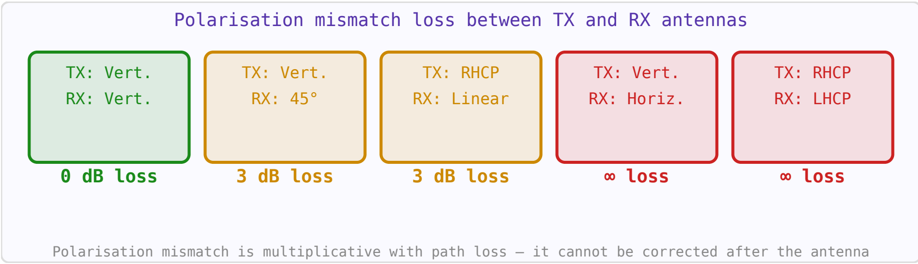

Figure 39.2: Polarisation mismatch loss for common antenna orientation combinations.

$$\eta_p = |\hat{\rho}_{\text{ant}} \cdot \hat{\rho}_{\text{wave}}|^2$$

For linear/circular mismatch: 3 dB loss (worst case). For RHCP/LHCP: theoretically infinite loss. Vertical/horizontal linear: infinite loss at 90° cross-polarisation.

TX polarisation	RX polarisation	Mismatch loss
Vertical linear	Vertical linear	0 dB
Vertical linear	Horizontal linear	∞ dB (no coupling)
RHCP	RHCP	0 dB
RHCP	LHCP	∞ dB (ideal)
RHCP	Vertical linear	3 dB
45° slant linear	Vertical linear	3 dB

39.4 Reading Radiation Pattern Plots

A radiation pattern shows antenna gain or field intensity as a function of angle, typically in a polar plot. Key features:

- **Main lobe**: Direction of maximum radiation. Peak value is the antenna gain.
- **−3 dB beamwidth (HPBW)**: Angular width where gain is within 3 dB of the peak. Narrower = more directive.
- **Sidelobe level (SLL)**: Gain of the strongest sidelobe relative to main lobe. Specified in dBc.
- **Front-to-back ratio (F/B)**: Gain of main lobe vs gain directly behind. Important for interference rejection.
- **Null**: Direction of minimum radiation. Can be steered for interference cancellation.

39.5 Gain, Directivity and Efficiency

$$G = \eta_r \cdot D \qquad D = \frac{4\pi}{\Omega_A}$$

where η_r is radiation efficiency (0–1) and Ω_A is the beam solid angle in steradians. Gain differs from directivity by the radiation efficiency — a highly directive antenna with 50% efficiency has 3 dB less gain than its directivity. An isotropic radiator (theoretical, impossible to build) has D = 1 = 0 dBi.

39.6 Polarisation in MRI

MRI uses circularly polarised B_1 fields. A linearly polarised coil couples to only one sense of rotation — half the available RF power is wasted driving the "wrong" CP mode. A quadrature (CP) coil uses two orthogonal loops fed 90° apart in phase, coupling efficiently to both the transmit field and the NMR signal. This gives $\sqrt{2}$ improvement in SNR and requires half the transmit power for the same flip angle.

Chapter 40

Yagi-Uda Antennas

The Yagi-Uda antenna (commonly called a Yagi) is a directional antenna composed of a fed dipole, a reflector behind it, and one or more directors in front. Only the driven element is connected to the feed line — the others are parasitic and couple electromagnetically. It is the dominant design for VHF/UHF terrestrial and amateur radio links and satellite ground stations.

40.1 Element Roles

- **Reflector**: one element, positioned ~0.25λ behind the driven element. Slightly longer than $\lambda/2$ ($\approx 0.505\lambda$). Reflects energy forward, increasing F/B ratio.
- **Driven element**: the only fed element, ≈ 0.47–0.48λ long. Input impedance in the array is ~25 Ω (straight dipole) or ~200 Ω (folded dipole).
- **Directors**: shorter than $\lambda/2$ (≈ 0.43–0.46λ), spaced 0.25–0.42λ forward of the driven element. Each director adds roughly 1–2 dBd of gain. Gain saturates above ~9–10 elements.

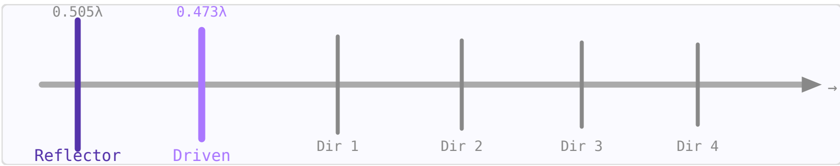

Figure 40.1: 5-element Yagi: reflector (dark), driven element (purple, fed), and three directors (grey).

40.2 Gain vs Element Count (NBS/Viezbicke Tables)

The NBS Technical Note 688 (Viezbicke, 1976) tabulates optimised element lengths and spacings for maximum gain, derived from moment-method (NEC) calculations for element diameter $d/\lambda = 0.0085$.

Elements	Gain (dBd)	Gain (dBi)	F/B (approx)
2	3.8	5.9	5 dB
3	5.9	8.1	10 dB
5	8.2	10.4	14 dB
7	10.0	12.2	18 dB
9	11.3	13.5	22 dB

dBi = dBd + 2.15. Adding directors past ~9–10 elements yields diminishing returns (<0.5 dB per element) for significant boom length cost.

40.3 Feed Impedance and Matching

The driven-element feed impedance is reduced from the free-space dipole value (~73 Ω) by the mutual coupling of adjacent elements:

- **Straight dipole**: Zin ≈ 25–35 Ω in the array — requires a matching network (gamma match, beta match) to 50 Ω coax
- **Folded dipole**: naturally 4× higher impedance → Zin ≈ 200–300 Ω → use a 4:1 balun or $\lambda/4$ transformer to 50 Ω

40.4 Bandwidth and Scaling

A well-optimised Yagi has an operating bandwidth of approximately 2–5% of the centre frequency — narrow enough that element length must be scaled accurately. To move a design to a new frequency, scale all dimensions by the ratio of the original wavelength to the new wavelength (f_old/f_new).

Chapter 41

Helical Antennas

The helical antenna is a wire wound in a helix and fed at one end against a ground plane. Depending on the circumference-to-wavelength ratio it can operate in two distinct modes. The **axial mode** (end-fire) is the most useful for gain antennas — it produces a circularly polarised beam along the helix axis and is widely used in satellite ground stations, CubeSats, and amateur radio.

41.1 Normal Mode vs Axial Mode

- **Normal mode**: circumference C $\ll \lambda$; radiation broadside (perpendicular to axis); linearly polarised; low gain. Used as a compact antenna when space is limited.

- **Axial mode**: circumference C $\approx \lambda$; radiation along the axis; circularly polarised; high gain. This is the practical gain antenna.

41.2 Axial-Mode Condition

The helix operates in axial mode when:

$$0.75 < \frac{C}{\lambda} < 1.33 \quad \text{where } C = \pi D$$

The optimal pitch angle (angle of the helix wire relative to the horizontal plane) is approximately $\alpha \approx 14°$, which corresponds to a turn spacing S $\approx \lambda/4$ when C $\approx \lambda$.

41.3 Gain and Beamwidth (Kraus Formulas)

$$G \approx 15\,N\left(\frac{C}{\lambda}\right)^2\frac{S}{\lambda} \quad \text{(linear, not dB)}$$

$$\text{HPBW} \approx \frac{52°}{\frac{C}{\lambda}\sqrt{N\frac{S}{\lambda}}}$$

where N is the number of turns. Gain increases with N — a 10-turn helix at C = λ, S = $\lambda/4$ gives G $\approx 15{\times}10{\times}1{\times}0.25 = 37.5$ ($\approx$ 15.7 dBi).

41.4 Input Impedance

The input impedance of the axial-mode helix is nearly resistive and approximated by:

$$Z_{\text{in}} \approx 140\,\frac{C}{\lambda}\ \ [\Omega]$$

At the optimum C/λ = 1, Zin ≈ 140 Ω. A quarter-wave stripline transformer between the helix feed and the 50 Ω coax connector matches the impedance: the transformer impedance is Z_t = √(140 × 50) ≈ 84 Ω.

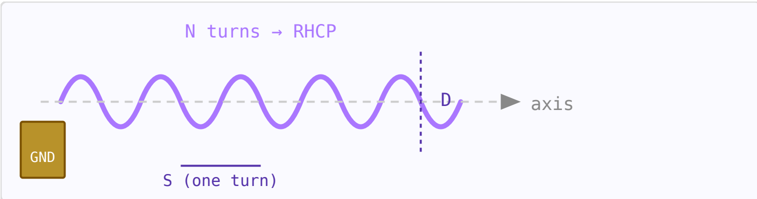

Figure 41.1: Axial-mode helix: N turns of diameter D, turn spacing S, fed against a ground plane.

41.5 Circular Polarisation

The axial-mode helix produces circular polarisation naturally. The sense (RHCP or LHCP) is determined by the winding direction:

- **Right-hand winding** (thumb along axis, fingers curl in direction of travel) → RHCP
- **Left-hand winding** → LHCP

For satellite reception (LEO downlinks, GPS), RHCP is standard. The axial ratio of a well-designed axial-mode helix is typically better than 1 dB (very close to perfect circular polarisation) over the 3 dB beamwidth.

41.6 Design Procedure

1. Choose target frequency f → compute λ = c/f
2. Set D = λ/π (so C = λ) and S = $\lambda/4$ for optimal pitch angle
3. Choose N for desired gain: G ≈ 15N×1×0.25 = 3.75N → e.g. N = 10 gives G ≈ 37.5 (15.7 dBi)
4. Compute Zin ≈ 140 Ω; design $\lambda/4$ transformer to 50 Ω (Z_t ≈ 84 Ω stripline)
5. Ground plane diameter should be at least 0.75λ to 1λ

Part VIII

RF Systems: Spectrum, Receivers, Links and Radar

Chapter 42

RF Frequency Bands

The electromagnetic spectrum from 3 Hz to 300 GHz is divided into named bands by the ITU. Each band has characteristic propagation behaviour, regulatory allocations, and typical applications. This page is a compact reference for the most important RF and microwave bands.

42.1 ITU Frequency Band Designations

Band	Name	Frequency	Wavelength	Propagation
ELF	Extremely Low Frequency	3–30 Hz	10,000–100,000 km	Penetrates seawater; submarine communications
SLF	Super Low Frequency	30–300 Hz	1,000–10,000 km	Power grid frequency; submarine comms
LF	Low Frequency	30–300 kHz	1–10 km	Ground wave; LORAN, DECCA navigation, AM longwave
MF	Medium Frequency	300 kHz–3 MHz	100 m–1 km	Ground wave + night-time skywave; AM broadcast
HF	High Frequency	3–30 MHz	10–100 m	Ionospheric skywave; intercontinental amateur, shortwave broadcast, OTHR
VHF	Very High Frequency	30–300 MHz	1–10 m	Line of sight + troposcatter; FM radio, TV, aviation, land mobile
UHF	Ultra High Frequency	300 MHz–3 GHz	10 cm–1 m	Line of sight; cellular, Wi-Fi, GPS, DVB-T, radar
SHF	Super High Frequency	3–30 GHz	1–10 cm	Line of sight (rain scatter); satellite, microwave links, 5G, radar

EHF	Extremely High Frequency	30–300 GHz	1–10 mm	Line of sight (rain/O_2 absorption); mmWave 5G, automotive radar, imaging

42.2 IEEE Radar / Microwave Band Letters

Band	Frequency	Applications
L	1–2 GHz	GPS (1.575 GHz), cellular 4G/5G, ATC radar, satellite
S	2–4 GHz	Wi-Fi 2.4 GHz, Bluetooth, weather radar, cellular
C	4–8 GHz	Satellite communications, C-band radar, Wi-Fi 5 GHz
X	8–12 GHz	Airborne/marine radar, SAR, satellite TV, PCB measurement
Ku	12–18 GHz	Satellite broadcast (DBS), VSAT, Ku-band radar
K	18–27 GHz	Automotive radar (24 GHz)
Ka	27–40 GHz	Satellite broadband, 5G mmWave, imaging radar
V	40–75 GHz	60 GHz WiGig, automotive radar (77 GHz)
W	75–110 GHz	Automotive radar (79 GHz), imaging, atmospheric
mm	110–300 GHz	THz imaging, spectroscopy, 5G future bands

42.3 Key Frequency Allocations

Frequency	Allocation / Service
433.92 MHz	ISM band (EU): wireless sensors, key fobs, LoRa
868 MHz	ISM band (EU): LoRa, Zigbee, smart meters
915 MHz	ISM band (US/AU): LoRa, RFID
1.575 GHz	GPS L1 (BPSK, C/A code)
1.227 GHz	GPS L2 (military P-code)
2.400–2.4835 GHz	ISM: Wi-Fi (802.11b/g/n), Bluetooth, ZigBee, microwave ovens
5.150–5.850 GHz	U-NII: Wi-Fi (802.11a/n/ac/ax)
5.8 GHz	ISM band: cordless phones, some Wi-Fi
24 GHz	ISM band: short-range radar, motion sensors
60 GHz	ISM band: 802.11ad/ay WiGig, high-speed short-range links
77 GHz	Automotive radar (long-range)

42.4 Antenna Size Reference

Frequency	$\lambda/2$ dipole	$\lambda/4$ monopole	Patch (in FR4)

433 MHz	346 mm	173 mm	~82 mm
868 MHz	173 mm	86 mm	~41 mm
915 MHz	164 mm	82 mm	~39 mm
2.45 GHz	61 mm	31 mm	~14.5 mm
5.8 GHz	26 mm	13 mm	~6.1 mm
24 GHz	6.2 mm	3.1 mm	~1.5 mm
77 GHz	2.0 mm	1.0 mm	~0.47 mm

Chapter 43

The Superheterodyne Receiver

The superheterodyne receiver, invented by Edwin Armstrong in 1918, translates the received RF signal to an intermediate frequency (IF) where filtering and amplification are more practical. It remains the dominant architecture in virtually every radio, cellular phone, and test instrument made today.

43.1 Block Diagram Signal Flow

The signal path from antenna to baseband:

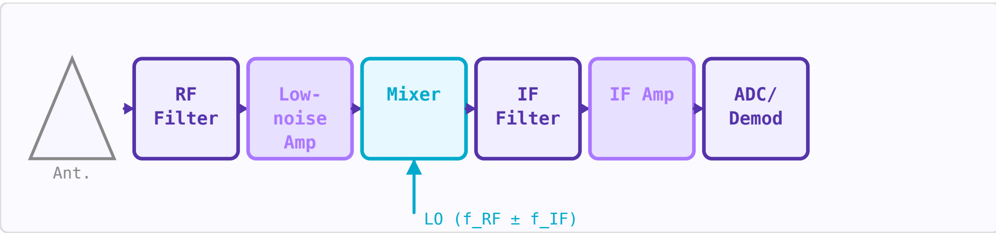

Figure 43.1: Superheterodyne receiver signal flow. The RF filter sets image rejection; the IF filter provides channel selectivity.

1. **Antenna** → receives signal + noise + interference across all frequencies.
2. **RF bandpass filter (BPF)** → passes the desired RF band, rejects out-of-band signals. Critical for preventing image interference and protecting the LNA.
3. **Low-noise amplifier (LNA)** → first amplification stage. Dominates system noise figure (Friis: first stage matters most). Typically 15–25 dB gain, 1–3 dB NF.
4. **Image rejection filter** → rejects the image frequency f_image = f_LO − f_IF (for low-side injection). Must be placed before the mixer.
5. **Mixer** → multiplies RF by local oscillator (LO), producing sum and difference frequencies. Desired output is |f_RF − f_LO| = f_IF.
6. **IF bandpass filter** → selects the IF and provides adjacent-channel rejection. This is where most selectivity comes from — fixed-frequency filters are practical at IF.
7. **IF amplifier** → provides most of the system gain at IF (easier to build stable high-gain amplifiers at lower frequencies).
8. **Demodulator / ADC** → extracts information from the IF signal.

43.2 The Image Frequency Problem

The mixer responds to two input frequencies — the desired signal at f_RF and an "image" at f_image = f_LO ± f_IF. Noise or interference at the image frequency is also downconverted to the IF and corrupts the desired signal. Image rejection is the ratio (in dB) of how much the receiver attenuates the image relative to the desired frequency:

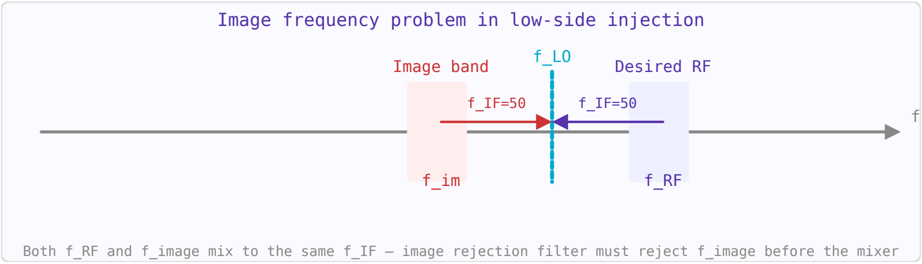

Figure 43.2: The image frequency f_image = f_LO − f_IF (low-side injection) produces the same IF as the desired signal.

$$f_{image} = f_{RF} \pm 2f_{IF}$$

Higher IF → image is further from f_RF → easier to filter. This motivates the choice of IF. A high first IF (e.g. 70 MHz or 1.4 GHz) gives good image rejection; a second lower IF can then improve channel selectivity.

43.3 Noise Figure Through the Chain

Using the Friis formula, the cascaded noise figure:

$$F_{sys} = F_{filter} + \frac{F_{LNA} - 1}{G_{filter}} + \frac{F_{mixer} - 1}{G_{filter}G_{LNA}} + \cdots$$

A passive RF filter before the LNA (e.g. with 2 dB insertion loss) adds directly to the noise figure: F_filter = 1 + loss(linear) = 1 + (L−1)/L. This is why a low-noise LNA must come as early in the chain as possible, and why surface-mount filters with high insertion loss degrade sensitivity.

43.4 Dynamic Range

The receiver must handle signals from the noise floor (−100 to −120 dBm) to strong in-band signals that could be 80 dB stronger. Two limits define the dynamic range:

- **Minimum detectable signal (MDS)**: noise floor + minimum required SNR. Determined by noise figure and bandwidth.
- **Blocking / desensitisation**: strong off-channel signals that compress the LNA or generate intermodulation products that mask the desired signal. Determined by IIP3 and P_1dB.

$$\text{Dynamic Range} = P_{1dB,input} - \text{MDS} \quad [\text{dB}]$$

43.5 Superheterodyne vs Direct Conversion (Zero-IF)

Feature	Superheterodyne	Direct conversion (zero-IF)
IF frequency	Fixed non-zero value	0 Hz (baseband I/Q output)
Image rejection	Achieved by RF filter	No image problem (I/Q cancellation)
Channel filtering	Fixed IF filter (ceramic, SAW)	Baseband low-pass filter (active)
Integration	Harder — off-chip IF filters needed	Easier — fully on-chip in CMOS
Key problems	Image, LO harmonics, IF filter	DC offset, IQ imbalance, 1/f noise
Used in	Test instruments, broadcast receivers	Smartphones, SDR, Wi-Fi, Bluetooth

Chapter 44

Receiver Cascade Analysis

A receiver chain is a sequence of amplifiers, filters, and frequency converters from antenna to digital output. The system noise figure determines sensitivity, the gain distribution determines dynamic range, and the IP3 cascade determines linearity. These three analyses — Friis NF, gain chain, and cascaded IP3 — must be done together to optimise the receiver architecture.

44.1 The Three-Parameter Cascade

For each stage, you need three numbers: gain G (dB), noise figure NF (dB), and input-referred IP3 (IIP3, dBm). From these, the cascaded quantities are computed.

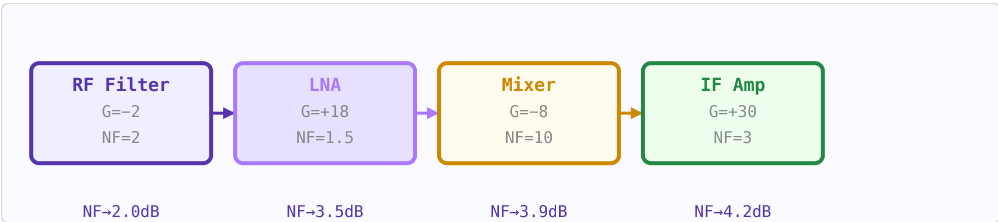

Figure 44.1: Cumulative noise figure stabilises after the LNA. First stage and its gain dominate the system NF.

44.2 Friis Formula for Cascaded Noise Figure

$$F_{sys} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

where F = 10^(NF/10) (linear). The first stage dominates if G_1 is large. A low-NF, high-gain LNA first is critical.

44.3 Cascaded IIP3

$$\frac{1}{IIP3_{sys}} = \frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1 G_2}{IIP3_3} + \cdots$$

(all in linear watts, not dBm). The last high-gain stage dominates linearity — opposite to noise figure.

44.4 Worked Example: 2.4 GHz Receiver

Stage	G (dB)	NF (dB)	IIP3 (dBm)	Cumul. NF	Cumul. IIP3
RF filter (insertion loss)	−2	2	∞	2.0 dB	—
LNA	+18	1.5	+10	3.5 dB	+8.5 dBm
Image rejection filter	−2	2	∞	3.5 dB	+8.5 dBm
Mixer	−8	10	+20	4.3 dB	+4.5 dBm
IF amplifier	+30	3	+25	4.3 dB	−6.3 dBm

The cumulative noise figure stabilises at 4.3 dB after the LNA — the LNA gain means all downstream stages add negligible noise. The IIP3 degrades to −6.3 dBm due to the high IF amplifier gain — if this is insufficient, the IF gain must be spread across more stages or reduced.

44.5 Minimum Detectable Signal

$$MDS = -174 + NF_{sys} + 10\log_{10}(BW) + SNR_{min} \quad \text{[dBm]}$$

For the example above with 5 MHz bandwidth and 10 dB minimum SNR: MDS = −174 + 4.3 + 67 + 10 = −92.7 dBm.

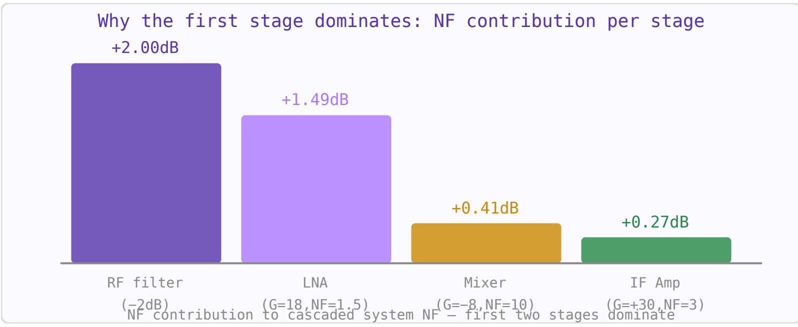

Figure 44.2: Each stage's marginal contribution to cascaded NF (linear units). Later stages contribute nearly nothing after a high-gain LNA.

44.6 Spurious-Free Dynamic Range

$$SFDR = \frac{2}{3}(IIP3_{sys} - MDS)$$

With IIP3 = −6.3 dBm and MDS = −92.7 dBm: SFDR = 2/3 × 86.4 = 57.6 dB. This means the receiver can handle a 57.6 dB range of input signal levels without the IM3 products exceeding the noise floor.

44.7 Gain Distribution Rules of Thumb

- Place the LNA as early as possible (after only necessary filtering) to establish noise figure.
- Total front-end gain before the ADC should be enough that ADC noise is irrelevant: typically 60–90 dB total gain for a −100 dBm sensitivity target into an ADC with −70 dBm full scale.
- Distribute gain across multiple stages to keep no single stage near compression. Each high-gain stage needs correspondingly high IIP3.

- Use variable gain amplifiers (VGAs) to handle the signal range: the AGC loop keeps the ADC input level constant regardless of received signal power.
- Don't forget the ADC itself: its ENOB and input full-scale power set the final SNR.

44.8 AGC and Automatic Gain Control

An AGC loop monitors the signal level (usually after the IF filter) and adjusts attenuators or VGAs to keep the ADC input at a constant level. Good AGC design requires: fast attack time (handles sudden strong signals before saturation), slow decay time (prevents noise pumping), and sufficient control range (> dynamic range of expected input signals).

Chapter 45

Link Budgets

A link budget is a systematic accounting of all gains and losses in a communication system from transmitter to receiver. It determines whether a link will achieve adequate signal-to-noise ratio under specified conditions, and by how much margin.

45.1 Link Budget Equation

$$P_r = P_t + G_t - L_{tx} - FSPL - L_{misc} + G_r - L_{rx}\ (dBm)$$

Then compare received power to the minimum detectable signal (MDS):

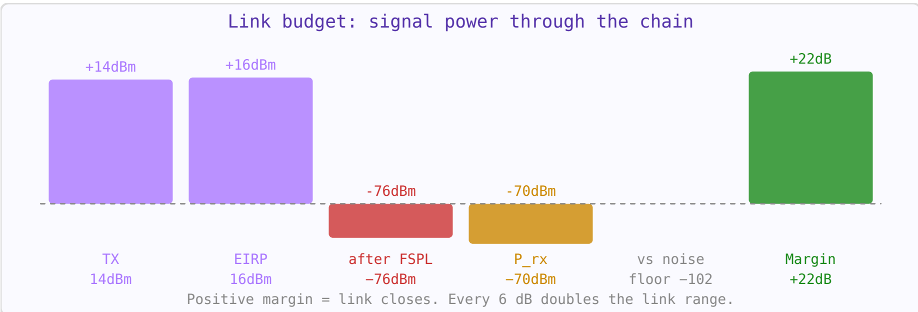

Figure 45.1: Link budget bar chart: TX power flows through gains and losses. Green margin bar = working link.

$$MDS = kTB + NF + SNR_{min}\ (dBm)$$

Link margin $= P_r - MDS$. A positive margin means the link works; typical design margins are 3–20 dB depending on the fade margin required.

45.2 Free-Space Path Loss

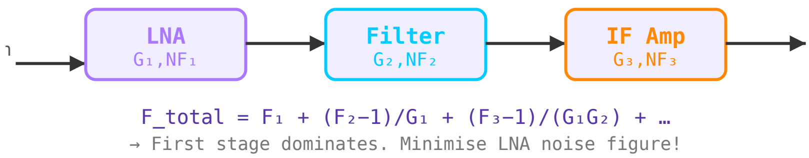

Figure 45.2: Link budget chain: each block contributes gains (antennas) or losses (cables, path loss). The sum determines received power vs noise floor.

$$FSPL = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right)$$

At 2.4 GHz, FSPL over 100 m $\approx$ 80 dB. FSPL grows with distance^2 and frequency^2 — doubling distance adds 6 dB; doubling frequency adds 6 dB.

45.3 Additional Loss Mechanisms

- **Multipath fading** — reflections cause constructive/destructive interference. Fast fading margin: 20–30 dB for 99.9% link reliability.
- **Rain attenuation** — significant above 10 GHz; up to several dB/km in heavy rain.
- **Atmospheric absorption** — oxygen absorption peak at 60 GHz (15 dB/km) limits this band to short-range applications.
- **Polarisation mismatch** — misaligned transmit/receive polarisations cause 0–∞ dB loss.

Chapter 46

Radar Fundamentals

Radar (Radio Detection And Ranging) uses reflected electromagnetic waves to detect objects and measure their range, velocity, and angular position. The radar equation relates transmitted power to received echo power.

46.1 Radar Equation

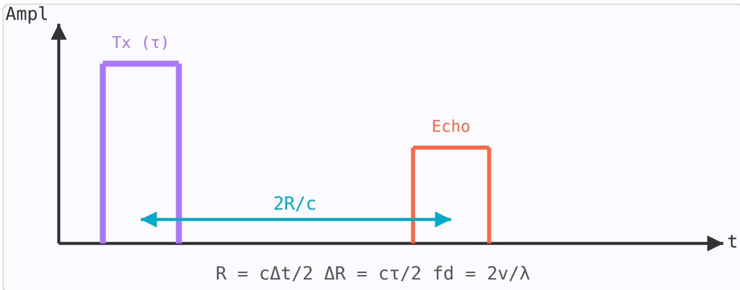

Figure 46.1: Radar timing diagram: transmit pulse (purple, width τ) and echo (orange). Range $\mathrm{R} = \mathrm{c}\Delta\mathrm{t}/2$; resolution $\Delta\mathrm{R} = \mathrm{c}\tau/2$.

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4}$$

where σ is the target radar cross section (RCS) in m^2, and R is range. Note the R^4 dependence — radar range halves when range doubles require 16× more transmit power.

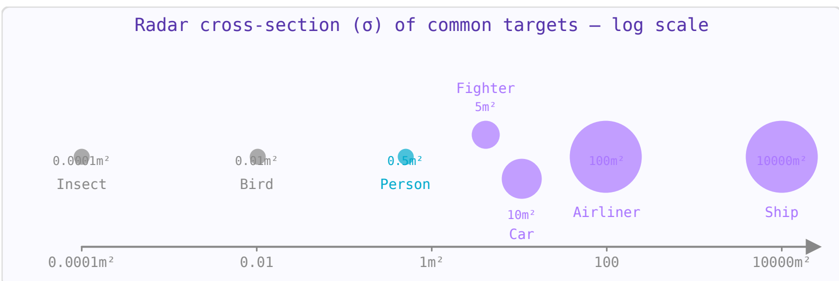

Figure 46.2: Radar cross-section of typical targets. RCS varies enormously — from 10^{-4} m^2 for insects to 10^4 m^2 for ships.

46.2 Range Resolution

The ability to distinguish two closely spaced targets in range depends on pulse width τ: $\Delta R = c\tau/2$. Shorter pulses $\rightarrow$ better resolution. Pulse compression (chirp or phase coding) allows high range resolution while maintaining high average power.

46.3 Doppler Effect

A target moving with radial velocity v_r shifts the echo frequency by the Doppler shift: $f_d = 2v_r/\lambda$. At 10 GHz, a 100 km/h target produces $f_d \approx 1.85$ kHz. Moving target indicator (MTI) and pulse-Doppler processing use this to separate moving targets from stationary clutter.

46.4 Radar Cross Section (RCS)

RCS (σ) characterises how effectively a target reflects radar energy back toward the receiver. It depends on target size, shape, material, and radar frequency. Stealth aircraft are designed to minimise RCS through shape (angled surfaces scatter energy away from radar) and radar-absorbing materials.

Chapter 47

FMCW and Pulsed Radar

The radar equation gives the received signal power, but the waveform determines how accurately range and velocity can be measured. The three main radar waveform families — pulsed, CW, and FMCW — each make a different tradeoff between range resolution, velocity measurement capability, and hardware complexity.

47.1 Pulsed Radar

A pulsed radar transmits short bursts of RF energy and measures the round-trip travel time to determine range. The transmitter is silent while the receiver listens for echoes:

$$R = \frac{c \cdot \tau}{2} \qquad \Delta R = \frac{c}{2B}$$

where τ is the pulse round-trip time, and ΔR is the range resolution set by the signal bandwidth B (not the pulse duration, for pulse-compressed systems). A 1 μs pulse → 150 m range resolution. A 10 MHz bandwidth (pulse-compressed or stepped frequency) → 15 m resolution.

Parameter	Formula	Notes
Max unambiguous range	R_max = c/(2·PRF)	PRF = pulse repetition frequency
Range resolution	ΔR = c/(2B)	B = bandwidth; 150 m/μs for CW pulse
Duty cycle	DC = τ·PRF	Typically 0.001–0.1 for pulsed systems
Average power	P_avg = P_peak · DC	Sets detection range via SNR

47.2 CW Radar

Continuous wave radar transmits a single frequency continuously. It cannot measure range (no time reference) but measures velocity extremely well via the Doppler shift:

$$f_D = \frac{2v_r f_0}{c} = \frac{2v_r}{\lambda}$$

where v_r is the radial velocity (positive = approaching). A 10 GHz radar and 100 km/h target: f_D = 2 × 27.8 / 0.03 = 1.85 kHz. Police speed guns and simple motion detectors use CW radar.

47.3 FMCW Radar

Frequency-modulated CW (FMCW) combines range and velocity measurement by sweeping the transmit frequency linearly. The echo from a target at range R arrives with a time delay τ = 2R/c, creating a constant "beat frequency" f_b between the current transmit frequency and the received echo:

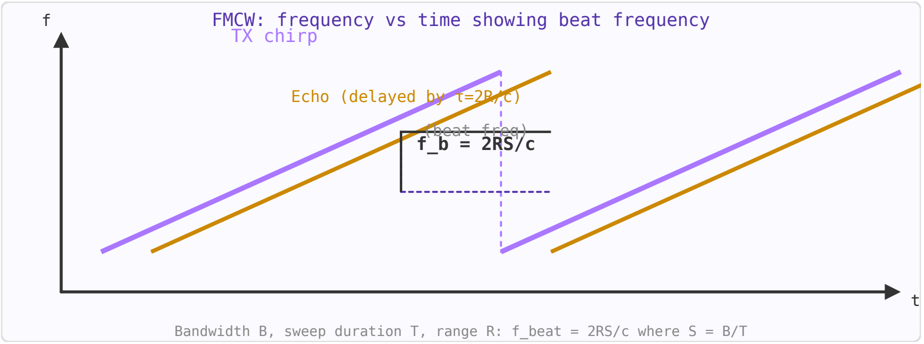

Figure 47.1: FMCW: TX sweeps from f_low to f_high; echo arrives delayed by τ=2R/c; beat frequency f_b $\propto$ range.

$$f_b = \frac{2R \cdot S}{c} \qquad S = \frac{B}{T_{sweep}}$$

where S is the frequency sweep rate (Hz/s), B the sweep bandwidth, and T_sweep the sweep duration. By measuring f_b with an FFT, range is determined directly. Modern automotive radar (77 GHz, 4 GHz bandwidth) achieves ΔR = c/(2×4GHz) = 3.75 cm range resolution.

47.4 Range-Doppler Matrix

A moving target creates both a range displacement and a Doppler frequency. In FMCW, velocity causes a phase rotation of the beat signal between sweeps. By taking a 2D FFT (range FFT per sweep, then Doppler FFT across sweeps), the range-Doppler matrix simultaneously shows all targets' range and velocity:

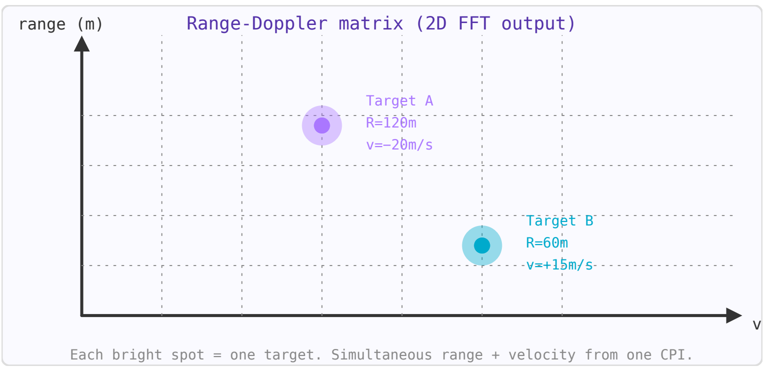

Figure 47.2: Range-Doppler matrix from 2D FFT. Range (vertical axis) and velocity (horizontal) resolved simultaneously.

$$\Delta v = \frac{\lambda}{2NT_{sweep}} \qquad v_{max} = \frac{\lambda}{4T_{sweep}}$$

where N is the number of sweeps in the coherent processing interval (CPI).

47.5 Ambiguity Function

The radar ambiguity function $\chi(\tau,$ f_D) describes the range-Doppler response of a waveform — how much a target at (range τ_0, Doppler f_D$_0$) leaks into adjacent range-Doppler cells. A perfect ambiguity function has a single narrow spike at the origin and zero everywhere else (impossible). The uncertainty principle of radar states: ΔR $\times$ Δv $\geq$ λc/4. Short pulses give good range resolution but poor velocity resolution; long CPI gives good velocity but requires coherence across many sweeps.

47.6 FMCW Applications

- **Automotive radar (77 GHz)**: 4 GHz sweep, 3.75 cm range resolution, ±200 km/h velocity, 200 m range. Used for adaptive cruise control, AEB, blind-spot detection.
- **Level measurement**: Industrial FMCW sensors measure tank fill level to mm accuracy over tens of metres.
- **Ground-penetrating radar (GPR)**: 100 MHz–3 GHz sweep through soil to detect buried utilities, unexploded ordnance, or archaeological features.
- **Indoor positioning**: 60 GHz UWB FMCW can locate people to $<$ 10 cm accuracy.

Part IX

EMC and RF Safety

Chapter 48

EMC Shielding Effectiveness

Shielding effectiveness (SE) quantifies how much an enclosure or barrier reduces the electromagnetic field at a point. It is expressed in dB as the ratio of the field without the shield to the field with it. Good shielding is essential for meeting regulatory limits (FCC Part 15, CE/ETSI), preventing interference between circuits, and protecting sensitive receivers from external interference.

48.1 Shielding Mechanisms

A metallic shield attenuates fields by three mechanisms:

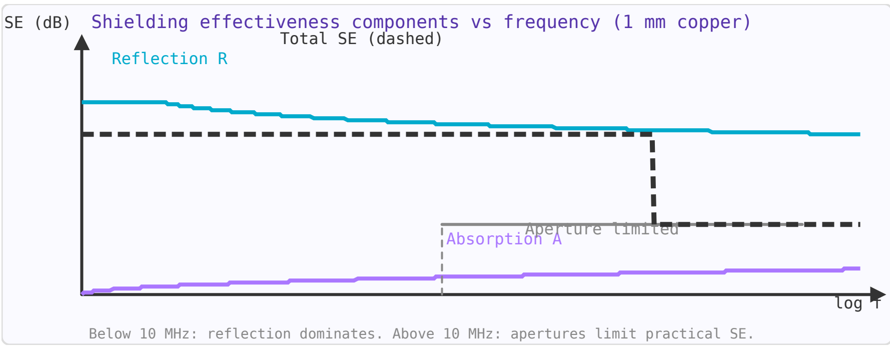

Figure 48.1: Shielding effectiveness components for 1 mm copper: reflection (blue), absorption (purple), and aperture limit (grey).

$$SE = R + A + B \quad \text{[dB]}$$

- **Reflection (R)**: At the air-metal interface, impedance mismatch reflects waves. Significant even for thin shields. Dominant at low frequencies for electric fields.
- **Absorption (A)**: Energy is dissipated as the wave propagates through the metal. A = $20 \cdot \log_{10}(e) \times t/\delta = 8.686 \times t/\delta$ (dB), where t is shield thickness and δ is skin depth. Increases with frequency.
- **Multiple reflections (B)**: For thin shields ($t < \delta$), waves bounce between the two surfaces, reducing the net absorption. B is negative and usually negligible when $A > 15$ dB.

48.2 Shielding Effectiveness vs Frequency

Frequency	δ (Cu)	A (1mm Cu)	R (plane wave)	Total SE (1 mm Cu)
1 kHz	2.1 mm	4 dB	132 dB	~136 dB
100 kHz	0.21 mm	40 dB	112 dB	~152 dB
1 MHz	66 μm	131 dB	102 dB	>200 dB (limited by seams)
1 GHz	2.1 μm	>400 dB	62 dB	Limited by apertures

Above ~10 MHz, even a very thin copper sheet is electromagnetically opaque — the limitation is *apertures* (holes, slots, seams), not the metal itself.

48.3 Aperture Leakage

Holes and slots in shielding enclosures are the dominant leakage path at high frequencies. A rectangular aperture of length L acts as a half-wave resonant slot antenna at f = c/(2L). Below resonance, SE decreases by 20 dB/decade as frequency increases:

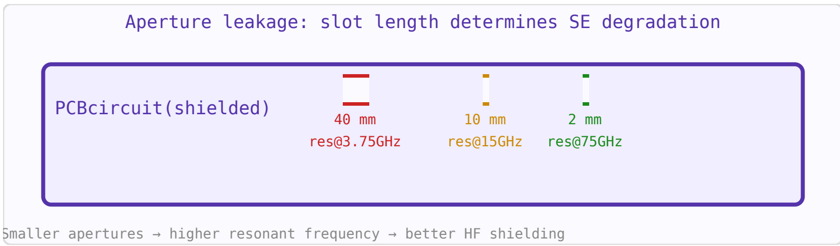

Figure 48.2: Aperture resonant frequency = c/(2L). Smaller holes give better HF shielding at the cost of airflow.

$$SE_{aperture} \approx 20 \log_{10}\left(\frac{\lambda}{2L}\right) \text{ dB} \quad (f \ll c/2L)$$

Practical consequences:

- A 30 cm slot (ventilation opening) resonates at 500 MHz — completely transparent above that.
- A 3 mm slot resonates at 50 GHz — fine for cellular but useless at mmWave.
- Replace single large holes with many small holes of the same total area: N holes of diameter d instead of one hole of diameter D=d$\sqrt{N}$ gives ~20·$\log_{10}(\sqrt{N})$ more SE.
- Honeycomb waveguide below cutoff arrays provide ventilation with >100 dB SE.

48.4 Seams and Joints

The weakest point of most shields is where panels join. A seam with periodic fasteners (screws every L mm) acts like an array of slots of length L. For good HF shielding, seams must be:

- Soldered or welded for maximum conductivity

- EMI gaskets (beryllium copper finger stock, wire mesh, conductive foam) for removable panels
- Screws spaced to $\lambda/10$ at the highest frequency of concern
- Conductive paint or plating on plastic enclosures (provides 30–50 dB SE)

48.5 Cable Shield Transfer Impedance

Cable shields leak energy through their transfer impedance Z_T (Ω/m) — the ratio of voltage induced on the inner conductor to the current flowing on the outer shield:

$$Z_T = \frac{V_{induced}/\text{unit length}}{I_{shield}} \quad [\Omega/\text{m}]$$

Cable type	**Z_T @ 10 MHz**	**SE @ 10 MHz**
RG-58 (single braid)	10–50 mΩ/m	50–60 dB
RG-214 (double braid)	1–5 mΩ/m	70–80 dB
Semi-rigid coax	< 0.1 mΩ/m	> 100 dB
Foil + braid	2–10 mΩ/m	60–75 dB

Always connect cable shields at both ends for RF shielding. Single-end grounding (for audio hum prevention) leaves the shield as an antenna above a few kHz.

48.6 PCB Shielding Cans

Stamped metal shielding cans solder directly to a PCB ground pour to isolate sensitive circuits (RF front-ends, oscillators). Key design rules: no gaps $> \lambda/20$ at the highest frequency; soldered continuously or with fences of vias spaced $< \lambda/20$; include a test access point (small hole or removable lid) for production testing.

Chapter 49

RF Safety and Human Exposure

RF electromagnetic fields are non-ionising: their photon energies (eV scale at microwave frequencies) are far too low to break chemical bonds or directly damage DNA. However, they can heat tissue through resistive losses, and at high power densities near high-power transmitters, this heating is the primary safety concern.

49.1 Exposure Limits: ICNIRP and FCC

Two sets of exposure guidelines dominate internationally:

- **ICNIRP (International Commission on Non-Ionizing Radiation Protection)**: Used in Europe and most of the world. Sets reference levels in terms of power density (W/m^2) and electric field strength (V/m).
- **FCC OET-65 (US)**: Used in the United States. Defines Maximum Permissible Exposure (MPE) for both occupational/controlled and general public/uncontrolled environments.

Frequency	**ICNIRP general public (W/m^2)**	**FCC uncontrolled (W/m^2)**
100 MHz	2	0.2
400 MHz	2	0.57
1 GHz	5	1.0
2.45 GHz	10	1.0
10 GHz	10 (up to 300 GHz)	5

49.2 Specific Absorption Rate (SAR)

SAR measures the rate at which RF energy is absorbed per unit mass of tissue:

$$SAR = \frac{\sigma |E|^2}{\rho} \quad \text{[W/kg]}$$

Whole-body average SAR limits:

- ICNIRP / IEEE: 0.08 W/kg whole-body (general public); 0.4 W/kg (occupational)
- Localised head/torso SAR: 2 W/kg averaged over any 10 g of tissue
- Localised limbs: 4 W/kg averaged over any 10 g of tissue

Mobile phones are tested against a 1.6 W/kg limit (FCC) or 2.0 W/kg limit (ICNIRP) averaged over 10 g of tissue. These limits include a safety factor of ~50× relative to the threshold for measurable biological effects from heating.

49.3 Minimum Safe Distance from Transmitters

In the far field, power density decreases as $1/R^2$. The minimum safe distance for an EIRP of P_EIRP watts to achieve power density S (W/m^2) at the limit:

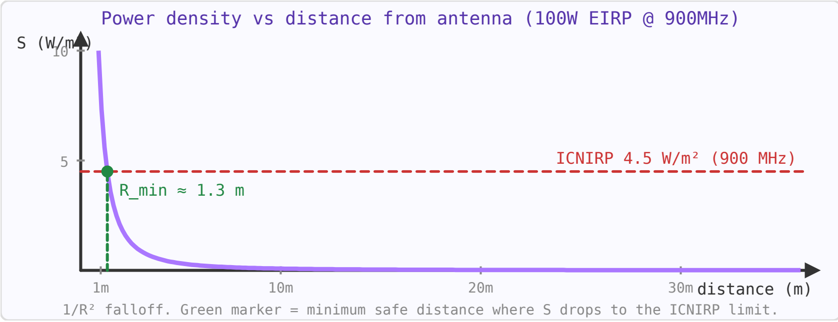

Figure 49.1: Power density falls as $1/R^2$ in the far field. For 100 W EIRP at 900 MHz, the ICNIRP 2 W/m^2 limit is reached at ~2 m.

$$R_{min} = \sqrt{\frac{P_{EIRP}}{4\pi S}}$$

For a 100 W EIRP omnidirectional transmitter at 900 MHz (ICNIRP 2 W/m^2 limit): R_min = $\sqrt{}(100/(4\pi\times2))$ = 2.0 m. For a high-gain antenna system with 1 kW EIRP: R_min = 6.3 m.

49.4 Near-Field vs Far-Field Exposure

The transition from near field to far field occurs at the Fraunhofer distance: R_ff = $2D^2/\lambda$ (D = antenna aperture). In the near field, power density does not follow the $1/R^2$ law — it can be higher or lower depending on the antenna. Near-field SAR calculations require full EM simulation (FDTD or FEM), not the simple far-field formula.

Practical near-field exposure situations:

- Holding a mobile phone against the head (3–10 mm from antenna)
- Wearing a smartwatch or fitness tracker
- Working near a base station antenna at maintenance distance
- Industrial RF heating equipment (RF sealers, dielectric heaters)

49.5 Practical RF Safety Guidelines

- Never stand in the main beam of a high-gain antenna within the minimum safe distance — be aware that gain increases the EIRP dramatically.
- RF safety limits apply to time-averaged power over 6 minutes (ICNIRP) or 30 minutes (FCC). Pulsed or intermittent transmitters use their duty cycle to calculate the average.
- Reflective surfaces can increase local power density above the direct beam level — consider reflected paths in safety assessments.
- No safety concern at Wi-Fi, Bluetooth, or mobile phone power levels at normal use distances (metres from the source).

- Implanted medical devices (pacemakers, cochlear implants) may be sensitive to strong RF fields at specific frequencies — maintain 20–30 cm distance from transmitters and check device specifications.

Note: This page covers general RF exposure. For MRI-specific SAR limits and the body coil SAR calculation, see the MRI SAR article.

Part X

MRI Physics and Coil Engineering

Chapter 50

NMR Basics

Nuclear Magnetic Resonance (NMR) is the physical phenomenon underlying MRI. It exploits the quantum mechanical property of nuclear spin to create detailed maps of atomic nuclei in a sample — in MRI, predominantly the ^{1}H (proton) nuclei in water and fat.

50.1 Nuclear Spin

Nuclei with an odd number of protons or neutrons possess a net quantum mechanical spin and an associated magnetic dipole moment $\boldsymbol{\mu}$. In a static magnetic field B_0, these moments precess around the field axis at the Larmor frequency:

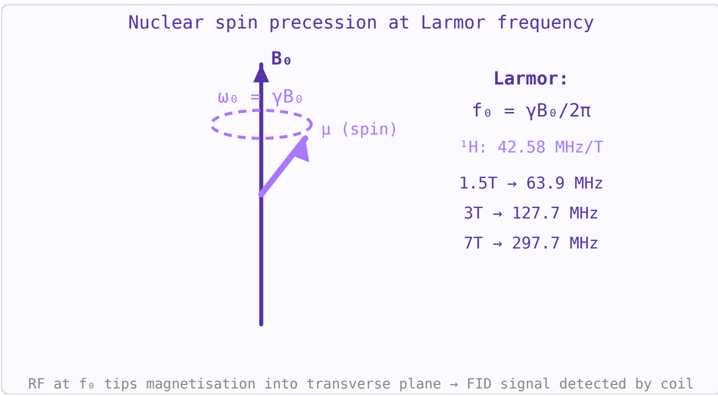

Figure 50.1: Larmor precession: spin μ precesses around B_0 at $\omega_0 = \gamma B_0$. RF excitation at this resonance frequency creates NMR signal.

$$f_0 = \frac{\gamma B_0}{2\pi}$$

where γ is the gyromagnetic ratio (for ^{1}H: $\gamma/2\pi = 42.577$ MHz/T). At 3 T, ^{1}H precesses at ~128 MHz — in the RF range.

50.2 Equilibrium Magnetisation

In a static field B_0, spins align preferentially along the field, creating a bulk magnetisation M_0 parallel to B_0 (the z-axis). This magnetisation is the signal source in MRI.

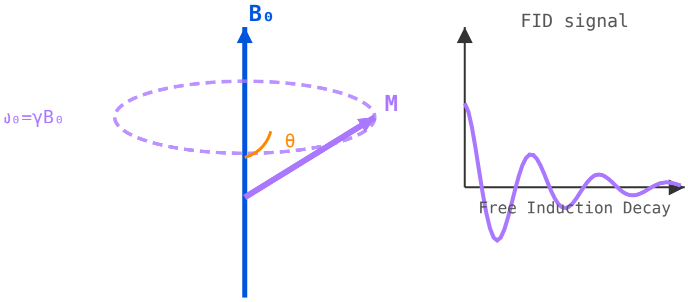

Figure 50.2: Left: M precessing around B_0 at $\omega_0 = \gamma B_0$. Right: free induction decay (FID) signal after a 90° pulse.

50.3 RF Excitation

A radiofrequency pulse at the Larmor frequency (the B1 field) tips M_0 away from z. A 90° pulse rotates M_0 entirely into the transverse (xy) plane where it precesses and induces a signal in the receive coil. A 180° pulse inverts M_0.

50.4 Bloch Equations and Relaxation

After excitation, the transverse magnetisation decays (T2 relaxation — spin-spin interactions) while the longitudinal magnetisation recovers (T1 relaxation — spin-lattice interactions). Different tissues have different T1/T2 values, creating the soft-tissue contrast that makes MRI so powerful. Typical values in brain tissue: T1 $\approx$ 900 ms, T2 $\approx$ 80 ms at 3 T.

Chapter 51

MRI Hardware

An MRI scanner is a complex system combining superconducting magnet technology, precision gradient electronics, and RF engineering. Each subsystem must work in concert to produce a diagnostic image.

51.1 Main Magnet

Modern clinical MRI scanners use superconducting solenoid magnets operating at liquid helium temperature (4 K) to achieve field strengths of 1.5–3 T (clinical) or 7–14 T (research). The magnet must have exceptional homogeneity — typically <1 ppm over the imaging volume — achieved by shimming with additional coils or ferromagnetic pieces.

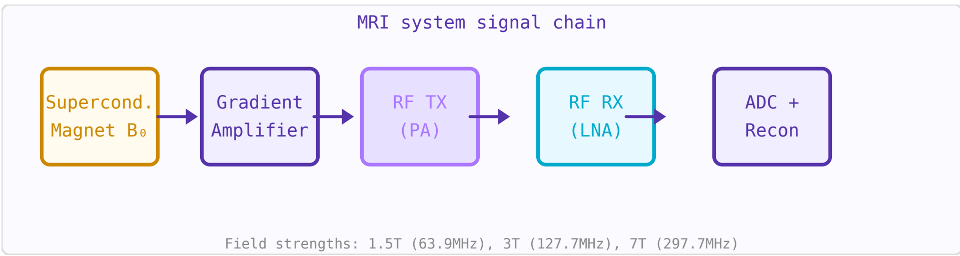

Figure 51.1: MRI system: superconducting magnet, gradient amplifiers, RF transmit power amp, LNA receiver, and digital reconstruction.

51.2 Gradient Coils

Three sets of gradient coils (Gx, Gy, Gz) superimpose linear magnetic field gradients on B_0. This makes the Larmor frequency spatially dependent, encoding spatial information. Gradient switching generates the characteristic loud knocking sound during an MRI scan. Gradient amplitude: 20–80 mT/m; slew rate: up to 200 T/m/s.

51.3 RF Transmit System

A high-power RF amplifier (typically 1–35 kW) drives the body or volume coil to produce the B1 excitation field at the Larmor frequency. The transmitter must deliver precise pulse shapes and calibrate the flip angle across the imaging volume. Frequency: 64 MHz (1.5 T), 128 MHz (3 T), 300 MHz (7 T).

51.4 RF Receive System

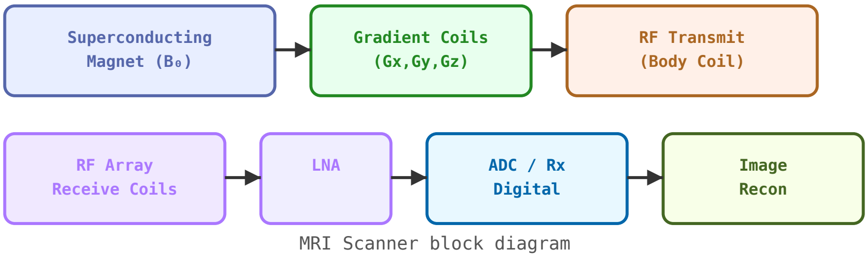

Figure 51.2: MRI scanner block diagram showing magnet, gradient coils, RF transmit/receive chains, and image reconstruction.

Receive coils (surface coils or phased arrays) detect the tiny precessing magnetisation signal. The signal is typically on the order of microvolts. It passes through a low-noise preamplifier (LNA), is downconverted to baseband, digitised, and reconstructed via Fourier transform into an image.

51.5 Shielding

The RF system operates inside an RF-shielded room (Faraday cage) to prevent external interference from contaminating the signal. The magnet's fringe field is contained by passive or active shielding to protect nearby equipment and comply with safety zones.

Chapter 52

MRI Gradient Coils

Gradient coils are the mechanism by which MRI encodes spatial information into the NMR signal. They produce linear spatial variations in the B_0 field (dB_z/dx, dB_z/dy, dB_z/dz) that allow slice selection, frequency encoding, and phase encoding. Their performance — gradient strength and switching speed — directly determines image resolution, scan time, and what pulse sequences are possible.

52.1 Why Gradients Are Needed

In a uniform B_0 field, all protons resonate at the same Larmor frequency $f_0 = \gamma B_0$. To determine where a signal comes from, we need to make f_0 vary with position. A gradient G_z adds a field increment G_z·z, so protons at position z resonate at $f = \gamma(B_0 + G\_z \cdot z)$. By sweeping the gradient and detecting the frequency spectrum, we reconstruct the 1D spin distribution — repeated in three directions gives 3D imaging.

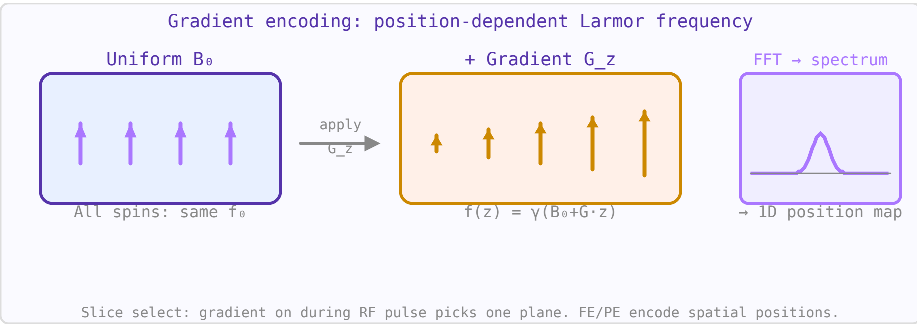

Figure 52.1: Gradient fields make Larmor frequency position-dependent. FFT of the NMR signal then maps frequencies to spatial positions.

52.2 Key Performance Metrics

Parameter	Symbol	Typical values	Impact
Gradient strength	G_max	20–80 mT/m	Minimum slice thickness, resolution, diffusion b-value
Slew rate	SR = dG/dt	100–200 T/m/s	Echo time (TE), echo spacing, sequence speed
Rise time	t_rise = G_max/SR	0.3–1 ms	Gradient echo timing

Linearity	% deviation from ideal	< 5% over DSV	Geometric distortion in images
Efficiency	η = G_max/I_max	5–30 mT/m/A	Power amplifier requirements
DSV (diameter spherical volume)		40–50 cm	Field of view with acceptable linearity

52.3 Coil Geometries

- **Maxwell pair**: Two circular loops with opposing currents, separated by $\sqrt{3}$·r. Produces a uniform dB_z/dz gradient along the bore axis (z-gradient). Used in early resistive MRI systems.

- **Golay coil (saddle coil)**: Four curved conductor arcs producing transverse gradients (x or y). The Golay geometry minimises higher-order field terms by careful choice of arc angle (120°) and axial spacing.

- **Fingerprint / distributed winding**: Modern gradients are designed numerically (target field method or streamline method) to optimise uniformity, efficiency, and acoustic noise simultaneously.

52.4 Eddy Currents

Rapidly switching gradient fields induce eddy currents in the cryostat bore, former, and any nearby conducting structures. These eddy currents create opposing fields that distort the desired gradient waveform — causing B_0 field errors (shift of resonance frequency), gradient pre-emphasis distortion, and image artefacts (ghosting, distortion in EPI). Solutions:

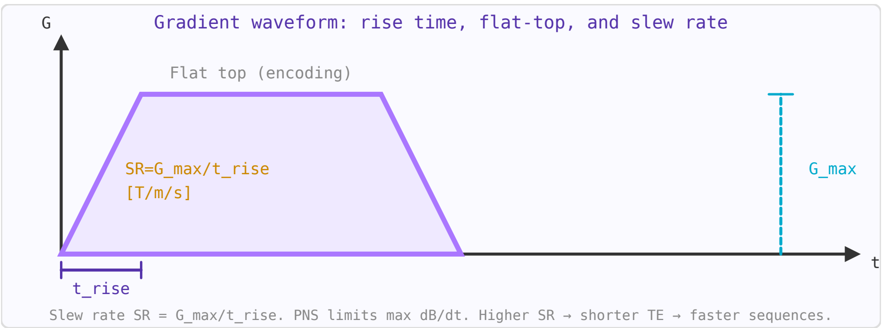

Figure 52.2: Trapezoidal gradient waveform. Rise time t_rise = G_max/SR limits shortest achievable echo time.

- **Actively shielded gradients**: A second "shielding" winding around the primary gradient cancels the field outside the gradient coil set, dramatically reducing eddy currents in the cryostat.

- **Gradient pre-emphasis**: The gradient amplifier applies a pre-distorted drive waveform that compensates for the expected eddy current decay.

- **Slotted/segmented formers**: Break conducting paths to reduce eddy current loops.

52.5 Peripheral Nerve Stimulation (PNS)

Very fast gradient switching induces electric fields in the patient's body (by Faraday's law: $E \propto dB/dt$). At high enough dB/dt, these induced fields can directly stimulate peripheral nerves, causing uncomfortable or painful muscular twitching. IEC 60601-2-33 defines PNS limits based on the rheobase (minimum current for stimulation with infinite pulse duration) and the chronaxie (pulse duration for 2× rheobase threshold). Modern scanners monitor dB/dt in real time and limit gradient slew rates to stay within PNS limits, which constrains fast pulse sequences like EPI and diffusion.

52.6 Acoustic Noise

Gradient coils experience Lorentz forces ($F = IL \times B_0$) when current flows through them in the main field. Rapidly switching these forces at audio frequencies (typically 1–4 kHz for EPI) causes the coil former to vibrate, generating the characteristic loud banging noise of MRI. Modern scanners use acoustic dampening foam, vacuum-bonded coil formers, and specific pulse sequence designs to reduce peak sound pressure levels below the IEC limit of 140 dB(A) at the patient's ears.

Chapter 53

MRI Pulse Sequences

A pulse sequence is the pattern of RF pulses and gradient waveforms used to acquire MRI data. Different sequences exploit T1, T2, and proton density contrast to highlight different tissue properties.

53.1 Spin Echo (SE)

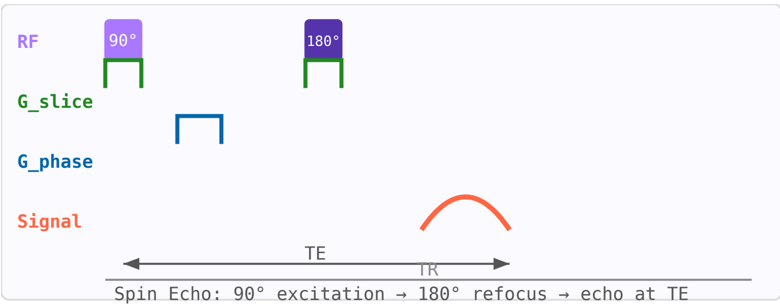

Figure 53.1: Spin echo pulse sequence: 90° excitation, 180° refocus at TE/2, echo at TE. TR sets T1 weighting.

A 90° excitation pulse followed by a 180° refocusing pulse after time TE/2 creates an echo at time TE. The 180° pulse reverses static field inhomogeneity dephasing but not true T2 decay, making SE sensitive to tissue T2. Repetition time TR controls T1 weighting. SE is robust to B0 inhomogeneity.

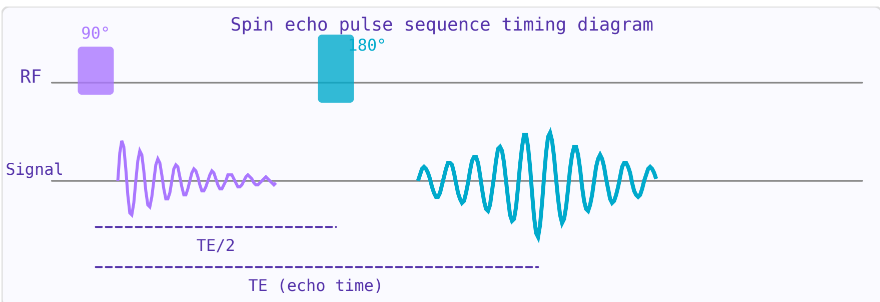

Figure 53.2: Spin echo: 90° excitation creates FID (purple); 180° refocusing pulse at TE/2 forms a spin echo at TE (cyan).

53.2 Gradient Echo (GRE)

Uses a flip angle $< 90°$ and a gradient reversal (instead of 180° RF pulse) to form the echo. Faster than SE but sensitive to B0 inhomogeneity (T2* rather than T2). Used for fast imaging, functional MRI (BOLD), and angiography.

53.3 Inversion Recovery (IR)

A 180° inversion pulse followed by a delay TI before a spin-echo acquisition. By choosing TI $=$ T1$\cdot$ln(2) for a tissue, its signal is nulled. FLAIR (Fluid Attenuated Inversion Recovery) nulls CSF to improve lesion conspicuity. STIR nulls fat.

53.4 k-Space

MRI data is acquired in the spatial frequency domain (k-space). Each data point corresponds to a different spatial frequency component of the image. The image is reconstructed by a 2D Fourier transform. Low spatial frequencies (near k-space centre) determine image contrast; high frequencies (periphery) determine edge sharpness. Trajectories through k-space define the acquisition strategy: Cartesian, radial, spiral, EPI, etc.

Chapter 54

MRI Contrast and Relaxation

MRI contrast arises from differences in tissue relaxation times T1 and T2, proton density (PD), and flow. By choosing appropriate pulse sequence parameters, specific tissue properties are emphasized, allowing discrimination of normal from pathological tissue.

54.1 T1 Relaxation (Spin-Lattice)

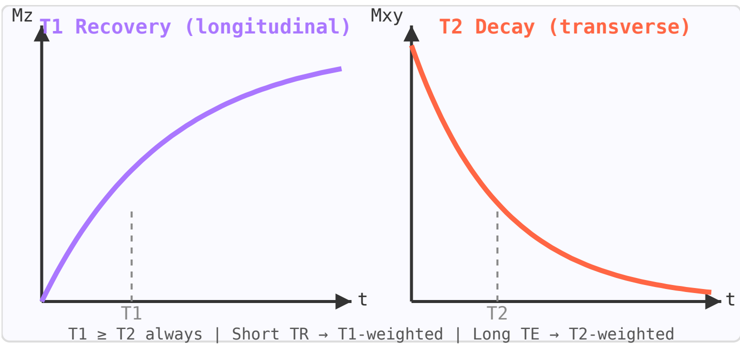

Figure 54.1: T1 recovery (purple) and T2 decay (orange) curves. T1 $\geq$ T2 always. Different tissues have different values.

T1 is the time constant for recovery of longitudinal magnetisation after excitation. Energy is transferred from the spin system to the surrounding lattice (molecular framework). T1 reflects molecular mobility: water in CSF has long T1 (~4 s at 3 T); fat has short T1 (~260 ms) due to efficient energy transfer at the Larmor frequency. Short TR $\rightarrow$ T1-weighted image.

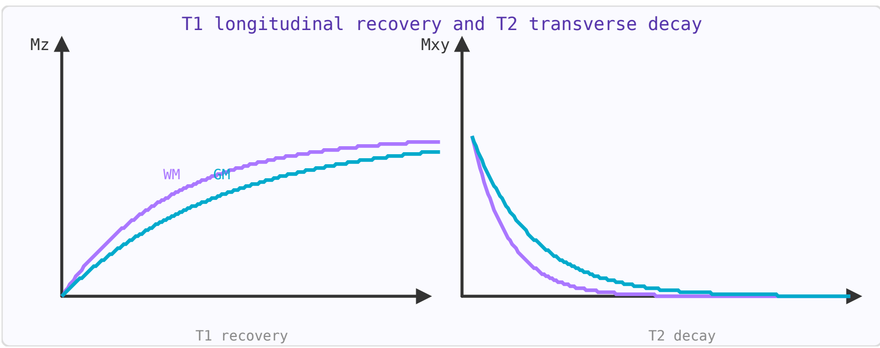

Figure 54.2: T1: longitudinal magnetisation recovers exponentially toward M_0. T2: transverse magnetisation decays. Tissue contrast comes from T1/T2 differences.

54.2 T2 Relaxation (Spin-Spin)

T2 is the time constant for decay of transverse magnetisation. Local field inhomogeneities from neighbouring spins cause dephasing. T2 is always $\leq$ T1. Fluids have long T2 (~2 s for CSF) appearing bright on T2-weighted images; solid tissues have short T2. Long TE $\rightarrow$ T2-weighted image.

54.3 T2* and Susceptibility

T2* includes additional dephasing from macroscopic B0 field inhomogeneities (susceptibility differences, main magnet imperfections). T2* $<$ T2. Gradient echo sequences are T2*-weighted. Susceptibility-weighted imaging (SWI) exploits T2* to visualise iron deposits, venous blood, and calcifications.

54.4 Contrast Agents

Gadolinium-based contrast agents (GBCAs) shorten T1 in their vicinity, making enhancing tissue appear bright on T1-weighted images. Used to detect blood-brain barrier breakdown, tumour vascularity, and inflammation. Iron oxide nanoparticles shorten T2/T2*, creating signal voids.

Chapter 55

MRI Artefacts

MRI artifacts are features in the image that do not correspond to real anatomy. Understanding artifacts is essential for correct image interpretation and for optimising acquisition protocols.

55.1 Gibbs Ringing (Truncation Artifact)

Sharp edges in the image (e.g. brain-CSF interface) produce ringing stripes parallel to the edge, caused by truncation of k-space at finite matrix size. Mitigated by acquiring more k-space lines (larger matrix) or applying k-space apodisation filters (at cost of resolution).

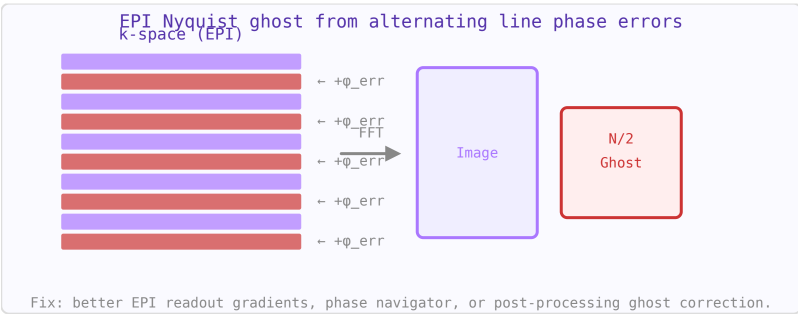

Figure 55.1: EPI Nyquist ghost: alternating k-space lines have opposite phase errors, creating a N/2-shifted ghost image.

55.2 Chemical Shift Artifact

Water and fat protons resonate at slightly different frequencies (3.5 ppm $\approx$ 450 Hz at 3 T). In the frequency-encoding direction, fat and water signals are spatially misregistered by this frequency offset, creating a bright/dark band at fat-water interfaces (e.g. kidney-perinephric fat). Mitigated by wider receiver bandwidth or fat suppression.

55.3 Motion Artifacts

Patient movement during acquisition causes ghosting (copies of moving structures) in the phase-encoding direction. Respiratory motion causes abdominal blurring. Mitigated by breath-holding, respiratory triggering/gating, navigator echoes, and radial k-space trajectories (more motion-robust).

55.4 Susceptibility Artifacts

Interfaces between materials of different magnetic susceptibility (metal implants, air-tissue, bone-tissue) distort the local B0 field, causing signal loss and geometric distortion. Worst near metallic implants. Mitigated by reducing TE, using spin echo (refocuses static inhomogeneity), or MARS (Metal Artifact Reduction Sequences).

55.5 RF Inhomogeneity (B1+ Artifact)

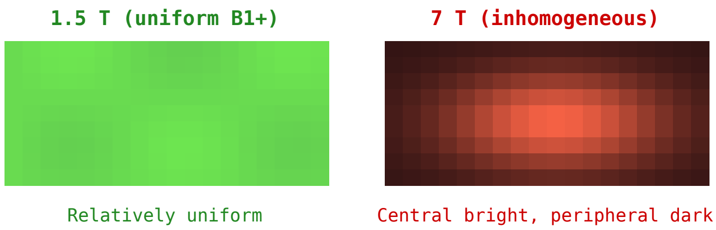

Figure 55.2: B1+ inhomogeneity: uniform at 1.5 T (left) but strong central brightening at 7 T (right) creating signal non-uniformity.

At 3 T and above, wavelength effects cause spatially non-uniform $B1^+$ fields, resulting in flip angle variations across the image. Manifests as signal brightening in the centre and darkening at the periphery of large body parts at 3 T, and complex interference patterns at 7 T. Mitigated by adiabatic pulses, parallel transmit, and B1 mapping-based pulse calibration.

Chapter 56

RF Coil Design for MRI

Designing an MRI RF coil requires careful attention to resonance, matching, decoupling, and coil Q. The goal is to maximise SNR while maintaining safety and compatibility with the scanner's RF system.

56.1 Tuning to Resonance

The coil must resonate at the Larmor frequency of the target nucleus. For a single-loop coil of inductance L, a tuning capacitor C_tune = $1/(\omega^2 L)$ brings the coil to resonance. Fixed or variable capacitors (trimmers) are used. The tuning is sensitive to nearby conductive or dielectric objects (loading), so provision for fine adjustment near the sample is necessary.

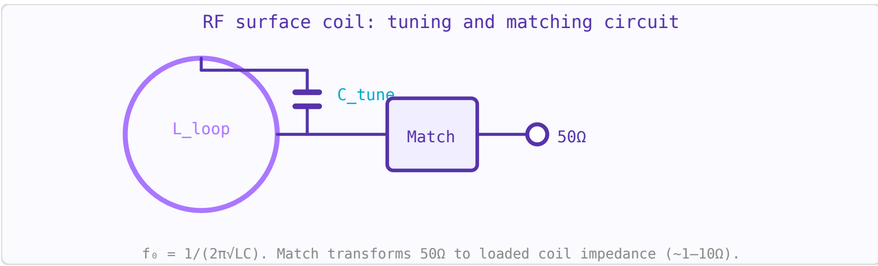

Figure 56.1: RF coil tuning: parallel capacitor C_tune resonates loop inductance L. Matching network presents 50 Ω to the coaxial cable.

56.2 Matching to 50 Ω

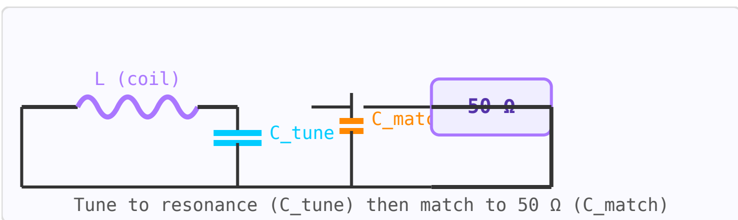

Figure 56.2: Coil tuning and matching circuit: C_tune resonates the coil at f_Larmor; C_match transforms impedance to 50 Ω.

The coil's radiation resistance is much lower than 50 Ω (typically 0.1–10 Ω for small surface coils). A matching network transforms this to 50 Ω to maximise power transfer between the coil and the preamplifier. Common approaches: capacitive tap on the tuning capacitor, additional shunt matching capacitor, or a transformer.

56.3 Quality Factor Q

The unloaded Q (Q_u) is determined by the coil's own resistance. The loaded Q (Q_L) includes sample losses. For receive coils, SNR is maximised when Q_L is dominated by sample noise (Q_L ≈ Q_u/2), not coil resistance. High-conductivity conductor (copper, silver), thick wire, and minimising resistive losses are key. At 3 T, typical unloaded Q for a 10 cm surface coil: 200–500; loaded Q: 30–80.

56.4 Decoupling

In phased array coils, adjacent elements must be decoupled to prevent inductive coupling from degrading noise performance. Methods include: overlap decoupling (geometrically overlapping adjacent loops to cancel mutual inductance), capacitive decoupling networks, and preamplifier decoupling (presenting high impedance to the coil current using a low-input-impedance preamp with an additional inductance).

56.5 Active Detuning

During RF transmission, receive coils must be detuned (switched off) to prevent them from disturbing the transmit field and from experiencing potentially damaging induced voltages. Active detuning circuits use PIN diodes to insert a large inductance or short circuit that shifts the resonant frequency away from f_0 during the transmit phase.

Chapter 57

MRI Coil Types

RF coils are the antennas of the MRI scanner. The receive coil must be placed as close as possible to the anatomy of interest to maximise the filling factor (the fraction of coil inductance filled by the sample) and thus the SNR.

57.1 Volume Coils

Surround the anatomy completely, providing a uniform B1 field. The birdcage coil is the most common volume coil — a cylindrical structure with rungs and end rings, driven in quadrature to produce circular polarisation (improving SNR by $\sqrt{2}$). Used for head, knee, and wrist imaging.

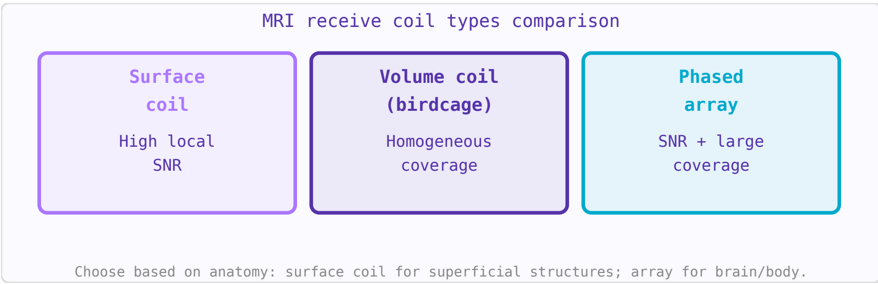

Figure 57.1: MRI coil type comparison. Each represents a different tradeoff between SNR, coverage, and homogeneity.

57.2 Surface Coils

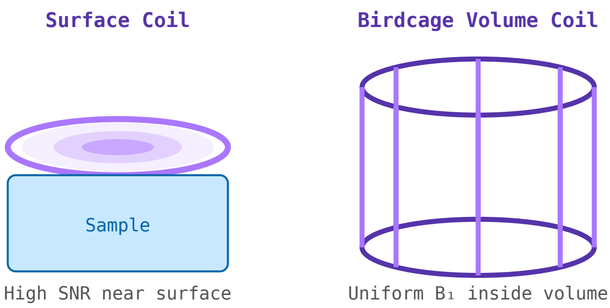

Figure 57.2: Left: surface coil — high local SNR, non-uniform. Right: birdcage volume coil — uniform B_1 throughout volume.

A single loop placed directly on the skin surface. Excellent SNR near the coil but very non-uniform sensitivity — SNR drops rapidly with depth. The depth of sensitivity is roughly equal to the coil radius. Ideal for superficial structures: spine, shoulder, temporomandibular joint.

57.3 Phased Array Coils

Multiple receive elements arranged around the anatomy, each with its own preamplifier and receive channel. Combines the high local SNR of surface coils with wider coverage. Also enables parallel imaging (SENSE, GRAPPA) where k-space undersampling is corrected using the spatial sensitivity profiles of the individual elements, reducing scan time.

57.4 Birdcage Coil

A cylindrical resonator with multiple rungs connecting two end rings. Driven in two quadrature ports (90° phase offset), it produces a highly uniform rotating $B1^+$ field ideal for transmit. Available in low-pass, high-pass, and band-pass configurations. The high-pass birdcage is preferred at high field (>3 T) as its capacitors are on the end rings rather than the rungs.

57.5 Transceiver vs Separate Tx/Rx

Some coils both transmit and receive (transceiver or T/R coil). Others use a separate body coil for transmit and a surface/array coil for receive. The receive-only configuration allows the receive coil to be optimised purely for sensitivity while the transmit coil provides uniform excitation.

Chapter 58

The Birdcage Coil

The birdcage coil is a cylindrical resonator consisting of a set of parallel conducting rungs connecting two circular end rings. Driven in quadrature (two ports 90° apart in space and in phase), it produces a highly uniform, circularly polarised B1 field ideal for both transmit and receive in MRI. It was proposed by Hayes et al. in 1985 and remains the dominant head and extremity coil design.

58.1 Resonant Modes

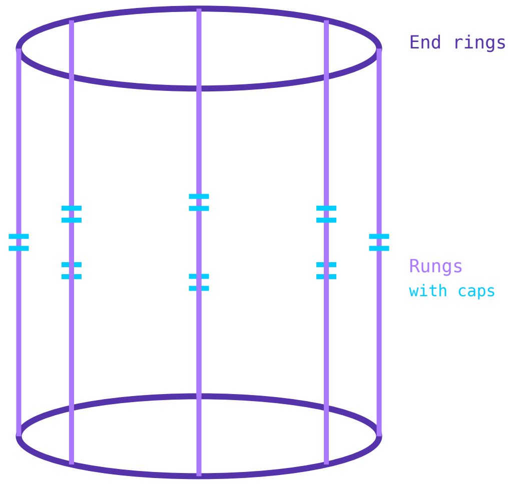

Figure 58.1: High-pass birdcage coil (N=8 rungs). Capacitors (cyan) on end rings. Rungs (purple) carry sinusoidal current distribution.

A birdcage with N rungs supports N resonant modes. The lowest frequency mode (mode 1) produces the sinusoidal current distribution required for field homogeneity. For an N-rung birdcage, the useful mode resonates at:

$$f_0 \approx \frac{1}{2\pi\sqrt{L_{rung} \cdot C_{eff}}}$$

where L_{rung} is the rung inductance and C_{eff} is determined by the capacitor placement.

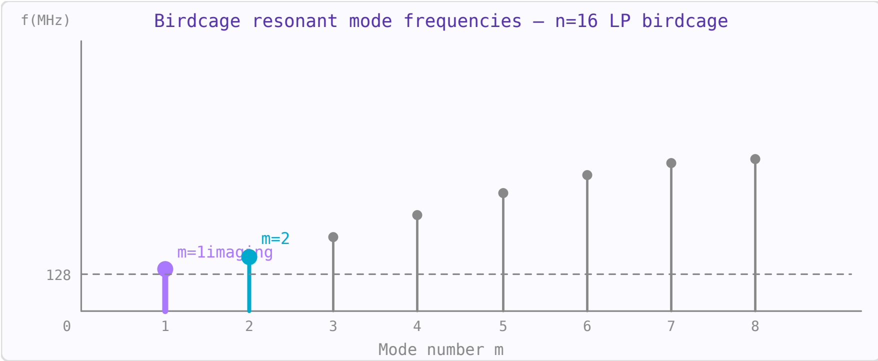

Figure 58.2: Birdcage resonant mode frequencies vs mode number m. Mode m=1 is the imaging mode (uniform field). Higher modes are spurious.

58.2 Low-Pass vs High-Pass Configurations

- **Low-pass**: capacitors on the rungs, end rings are continuous. Rungs look like capacitors at low frequency → resonance at low end of mode spectrum.
- **High-pass**: capacitors on the end rings, rungs are continuous inductors. Preferred at high field (3 T and above) because the desired mode is the highest frequency mode, further from spurious modes.
- **Band-pass**: capacitors on both rungs and end rings. Wider bandwidth and used in volume coils for body imaging.

58.3 Quadrature Drive

Driving two ports 90° apart in space and phase creates circular polarisation, producing $B1^+$ field in the rotating frame that efficiently tips magnetisation. This gives $\sqrt{2}$ (3 dB) SNR improvement over linear excitation for the same deposited power, or equivalently halves the required transmit power for the same flip angle.

Chapter 59

SNR in MRI Receive Coils

Signal-to-noise ratio (SNR) is the fundamental figure of merit of any MRI coil. Understanding the sources of noise and the factors that affect SNR guides coil design decisions.

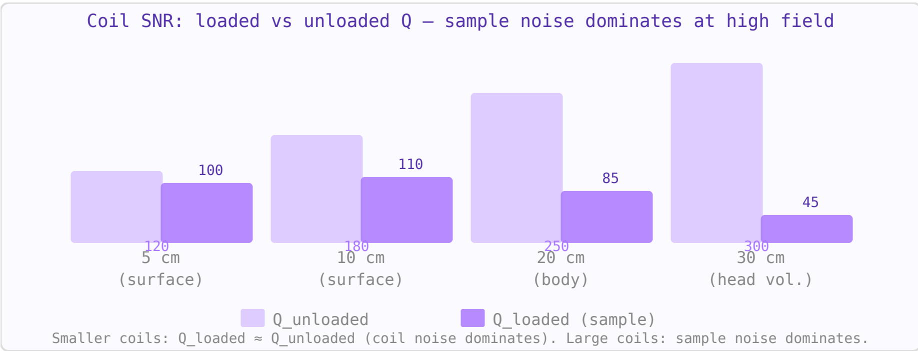

Figure 59.1: Loaded vs unloaded Q for MRI receive coils. When Q_loaded $\ll$ Q_unloaded, sample noise dominates — the coil is in the "sample-noise dominated" regime where SNR $\propto \omega_0$ B_1/coil.

59.1 Signal Source

The MRI signal is the EMF induced in the receive coil by the precessing transverse magnetisation in the sample. It scales with: field strength B0 (approximately B0^7/4 for a given experiment time), coil filling factor, and sample magnetisation density.

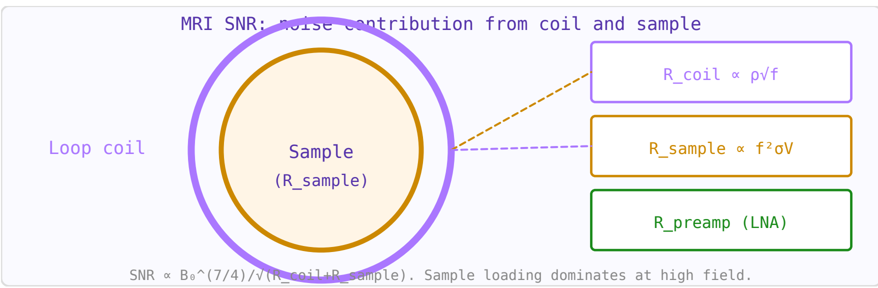

Figure 59.2: MRI coil SNR model: noise sources are coil resistance R_c, sample loading R_s, and preamplifier R_p.

59.2 Noise Sources

At MRI frequencies, two main noise sources compete:

- **Coil resistance noise** — Johnson noise from the ohmic resistance of the coil conductor. Dominates for small coils (surface coils <~5 cm at 3 T) or at low field.
- **Sample noise** — noise from induced currents (eddy currents) in the conductive biological tissue. Dominates for large coils, at high field, and near large conductive samples. Unavoidable — it sets the SNR ceiling.

59.3 Filling Factor

The filling factor η is the fraction of the coil's inductance that is "filled" by the sample volume of interest. $\eta = \int_{sample} |B_1|^2 dV / \int_{all} |B_1|^2 dV$. Maximising filling factor (placing the coil as close as possible to the anatomy) directly improves SNR.

59.4 Optimal Coil Size

A smaller coil placed close to the anatomy intercepts less noise (smaller sensitive volume) but provides high B1 per unit current near the coil. The optimal coil size roughly matches the depth of the structure of interest — a good rule of thumb is: for a structure at depth d below the surface, the optimal surface coil radius is approximately d.

59.5 Preamplifier Noise

The preamplifier (preamp) also contributes noise. For a sample-noise-dominated coil, preamp noise figure adds directly to system NF. Modern MRI preamps achieve NF of 0.3–0.7 dB. The preamp should be physically close to the coil (short coax) to minimise cable losses that degrade SNR.

Chapter 60

Preamplifier Decoupling

In phased array receive coils, adjacent elements must be isolated from each other to prevent mutual inductive coupling from increasing correlated noise and degrading combined SNR. Preamplifier decoupling is the most effective and widely used technique for achieving this isolation.

60.1 The Problem: Mutual Inductance

Two adjacent coil loops have mutual inductance M. If one coil carries current (as part of a resonant circuit), it induces a voltage in its neighbour. This reduces the SNR of both coils and is equivalent to adding correlated noise between array channels.

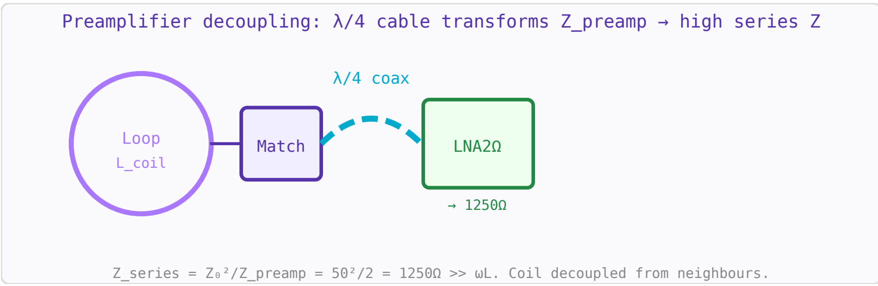

Figure 60.1: Preamplifier decoupling circuit. $\lambda/4$ cable transforms 2 Ω LNA input to 1250 Ω high series impedance in the coil loop.

60.2 The Technique

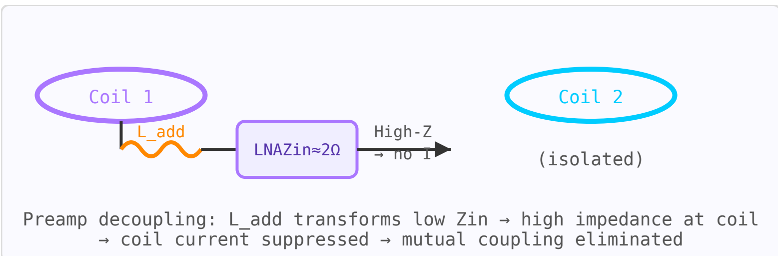

Figure 60.2: Preamplifier decoupling: L_add resonates with C_match, transforming the low preamp Zin to high impedance at the coil loop, suppressing inter-coil coupling.

A low-input-impedance preamplifier, together with a series inductor L_{add} at the coil port, presents a very high impedance to current flowing in the coil loop at the resonant frequency.

This "preamplifier decoupling" suppresses the induced current in each coil, effectively eliminating mutual coupling without requiring geometric overlap or additional decoupling networks.

$$Z_{preamplifier\ decoupling} = \frac{(\omega L_{add})^2}{Z_{in,preamp}}$$

If $Z_{in,preamp}$ is very low (~2–5 Ω) and L_{add} is chosen such that $\omega L_{add} = 1/(\omega C_m)$ (resonates with the matching capacitor), the impedance presented to coil current becomes very high (>1000 Ω), suppressing currents and hence coupling.

60.3 Geometric Overlap Decoupling

An alternative for nearest-neighbour elements is to physically overlap adjacent loops until their mutual inductance is zero (the induced flux from one loop through the other sums to zero due to partial cancellation). About 15–20% overlap is typical. Used in combination with preamplifier decoupling for arrays with many elements.

Chapter 61

Phased-Array Coils

A phased array is an antenna system consisting of multiple radiating elements whose relative phases (and sometimes amplitudes) are controlled electronically. By varying the phase distribution, the combined beam can be steered in any direction without mechanical movement.

61.1 Beam Steering

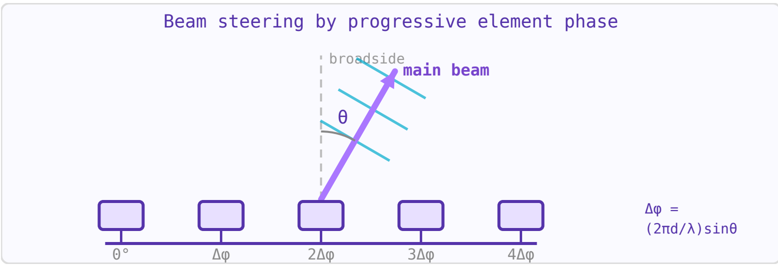

Figure 61.1: Phased array uses progressive phase shifts (implemented by phase shifters) to steer the main beam electronically.

To steer the main beam to angle θ_0, a progressive phase shift $\psi = kd \sin \theta_0$ is applied across the array, where $k = 2\pi/\lambda$ and d is element spacing. Phase shifters (analog or digital) implement this delay electronically. Beam switching speed: microseconds, compared to seconds for mechanical steering.

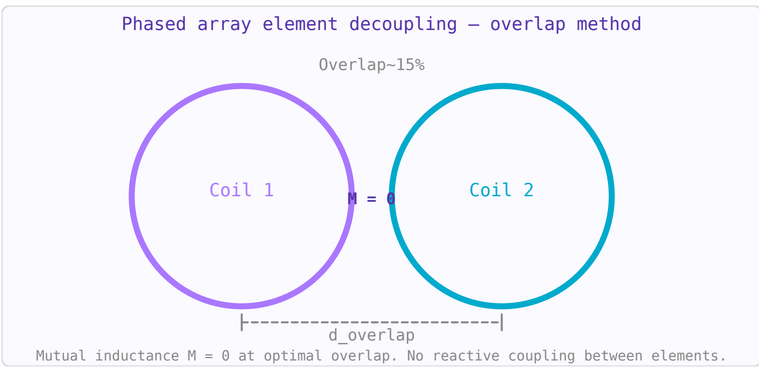

Figure 61.2: Overlapping coil decoupling: optimal overlap distance (≈15% of diameter) sets mutual inductance M = 0.

61.2 Grating Lobes

When element spacing $d > \lambda/2$, additional main beams (grating lobes) appear at angles other than θ_0. Standard design rule: $d \leq \lambda/2$ for grating-lobe-free scanning to $\pm 90°$.

61.3 Beamforming

Digital beamforming (DBF) processes the signals from individual array elements digitally, allowing simultaneous multiple beams, adaptive null steering (pointing nulls toward interference sources), and MIMO operation. 5G base stations use digital beamforming with massive MIMO (64–256 elements) for spatial multiplexing of many users.

61.4 Applications

- Military: AESA (Active Electronically Scanned Array) radar — simultaneous multi-function operation
- Satellite communications: electronically steered flat-panel antennas (e.g. Starlink terminals)
- 5G NR mmWave: compact phased array modules at 28/39 GHz for beamforming
- MRI: RF phased array transmitters for parallel transmission (pTx) at 7 T

Chapter 62

Parallel Imaging

Parallel imaging uses the spatial sensitivity profiles of multiple receive array elements to recover images from undersampled k-space data, allowing faster acquisition without image quality penalty (at the cost of some SNR).

62.1 The Basic Principle

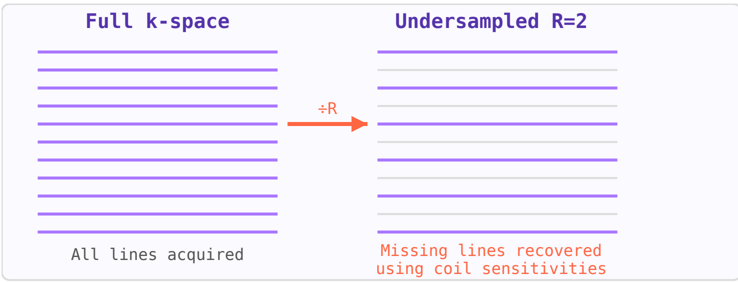

Figure 62.1: k-space: full acquisition (left) vs R=2 undersampled (right). Parallel imaging recovers missing lines from coil sensitivities.

If k-space is undersampled by an acceleration factor R (only every Rth phase-encode line is acquired), the Nyquist criterion is violated and aliasing (fold-over artifacts) occurs in the image. However, if multiple receive channels are available, each with a different spatial sensitivity, the aliased signals from different locations can be separated using the known sensitivity patterns.

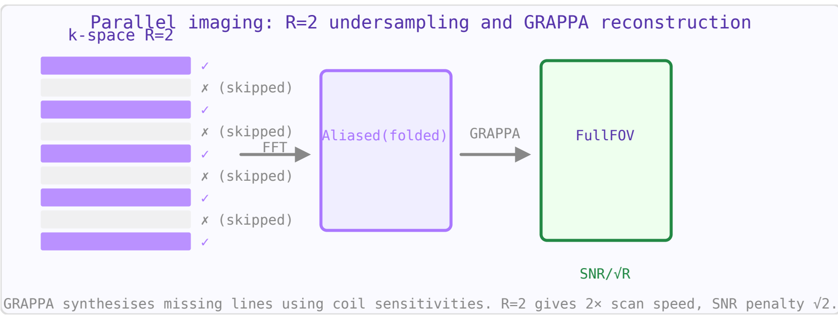

Figure 62.2: Parallel imaging: every 2nd k-space line acquired (R=2), GRAPPA reconstructs missing lines using multi-coil data.

62.2 SENSE

Sensitivity Encoding (SENSE) operates in image space. The aliased image from each channel is combined using a matrix inversion based on the measured coil sensitivity maps. Requires explicit sensitivity calibration and produces a noise amplification characterised by the geometry factor (g-factor).

62.3 GRAPPA

GeneRalised Autocalibrating Partial Parallel Acquisition (GRAPPA) operates in k-space. Missing k-space lines are synthesised from acquired lines using calibration coefficients estimated from a central fully-sampled auto-calibration signal (ACS) region. More robust to motion between calibration and imaging than SENSE.

62.4 SNR and g-Factor

Parallel imaging reduces scan time by factor R but always reduces SNR by at least $\sqrt{R}$ (less data = more noise). The g-factor (geometry factor) ≥ 1 further penalises SNR due to the conditioning of the unaliasing matrix. $SNR_{PI} = SNR_{full}/(g\sqrt{R})$. High-field MRI (7 T) benefits most from parallel imaging because the SNR advantage at high field partially compensates for the $\sqrt{R}$ SNR loss.

Chapter 63

B1 Field Mapping

B1 mapping refers to measuring the spatial distribution of the RF transmit field ($B1^+$) inside the subject. At 3 T and above, wavelength effects cause significant $B1^+$ inhomogeneity, which manifests as flip angle variations across the image. B1 maps are used for quantitative MRI, SAR estimation, and parallel transmit calibration.

63.1 Why B1 Mapping Matters at High Field

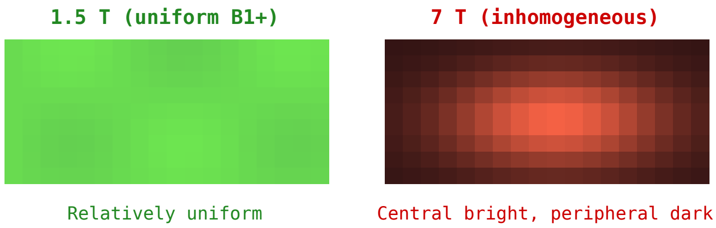

Figure 63.1: Simulated B1+ maps: 1.5 T (left, uniform) vs 7 T (right, central brightening due to constructive EM interference).

At 3 T (128 MHz), the wavelength in tissue (~25 cm) becomes comparable to body dimensions, causing constructive and destructive interference of the RF field. A nominal 90° pulse may deliver 70° in some regions and 110° in others. Quantitative MRI techniques (T1 mapping, T2 mapping, MR spectroscopy) require accurate flip angles for valid results.

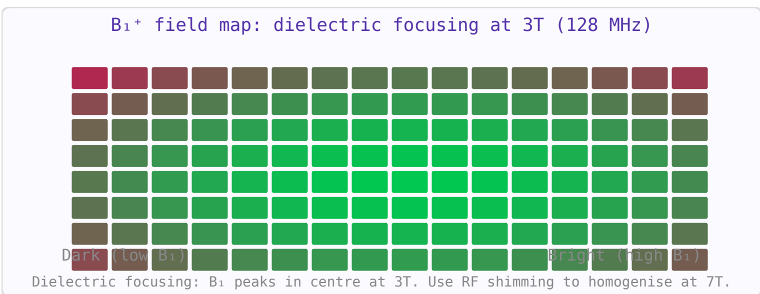

Figure 63.2: B_1^+ field map simulation at 3T. Constructive interference of RF waves creates a bright central spot in the phantom.

63.2 Double Angle Method (DAM)

Two spin-echo images are acquired with the same TR but with nominal flip angles α and 2α. The signal ratio gives: $|S(2\alpha)/S(\alpha)| = 2\cos(\alpha_{actual})$, from which the actual flip angle can be calculated pixel-by-pixel. Simple but slow (requires long TR for T1 recovery).

63.3 Actual Flip Angle Imaging (AFI)

Two interleaved acquisitions with different TRs (TR1 $\ll$ TR2) but the same flip angle. The signal ratio depends only on flip angle and the TR ratio (not T1 for reasonable conditions), allowing fast B1 mapping.

63.4 Parallel Transmit (pTx)

At 7 T, multiple transmit channels (typically 8) each drive a separate coil element. By controlling the amplitude and phase of each channel, the $B1^+$ distribution can be shaped (RF shimming) to improve uniformity or focus power. B1 maps for each channel are a prerequisite for pTx calibration.

Chapter 64

SAR and RF Safety in MRI

The interaction of RF electromagnetic fields with biological tissue at MRI frequencies deposits energy as heat. Safety standards limit this deposition to protect patients from tissue heating. SAR (Specific Absorption Rate) is the primary metric.

64.1 Specific Absorption Rate (SAR)

SAR is the RF power absorbed per unit mass of tissue:

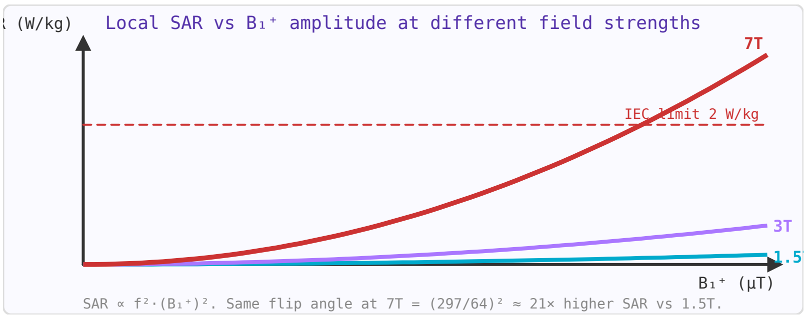

Figure 64.1: Local SAR grows with $B_1{}^+$ squared and with frequency squared. Managing SAR is the key challenge at ultra-high field (7T+).

$$SAR = \frac{\sigma |E|^2}{2\rho} \ (W/kg)$$

where σ is tissue conductivity, E the electric field amplitude, and ρ tissue density. IEC and FDA limits for clinical MRI:

- Whole-body average: 2 W/kg (normal), 4 W/kg (first level)
- Head: 3.2 W/kg; local (any 10 g): 10 W/kg

64.2 B1+ Field

The rotating component of the RF magnetic field that drives spin excitation is $B1^+$. At higher field strengths ($\geq$3 T), wavelength effects cause inhomogeneous $B1^+$ distribution in the body, leading to non-uniform flip angles. At 7 T (300 MHz), the wavelength in tissue (~12 cm) is comparable to body dimensions, causing significant $B1^+$ non-uniformity and hotspots in SAR.

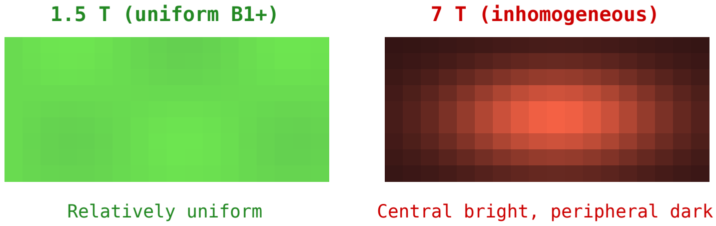

Figure 64.2: B1+ field maps at 1.5 T (uniform, left) and 7 T (inhomogeneous, right), illustrating how wavelength effects create SAR hotspots at high field.

64.3 Implant Safety

Conductive implants (pacemaker leads, orthopaedic implants, surgical clips) can act as RF antennas, concentrating the induced electric field and causing local tissue heating far exceeding whole-body SAR limits. MRI-conditional implants are tested and labelled with specific conditions (field strength, coil type, SAR limits) under which they are safe.

64.4 SAR Reduction Strategies

- Reducing flip angle (lower SAR $\propto$ flip^2)
- Increasing TR to allow pulse duty cycle reduction
- Parallel transmit (pTx) — using multiple transmit channels to shape B1$^+$ and reduce local SAR hotspots
- Lower field strength systems for implant patients